

CFA Level 2 Notes

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1 Quantitative Methods

1.1 Multiple regression

MLR requirements: (1) linearity, (2) homoskedasticity, (3) independence of errors (no autocorrelation), (4) normality (regression residuals are normal), (5) independence of independent variables - independent variables aren't exactly related (otherwise, collinearity). For (2)-(4), plot residuals versus predicted dependent values - for outliers in this chart, check for shocks not considered in

the proposed model. For (4), can also use a normal q-q plot of regression residuals. For (5), use a scatterplot matrix to view whether there is multicollinearity.

Regression residuals versus dependent variable - to look for heteroskedasticity, outliers.

Regression residuals versus an independent variable - shows whether the independent variable is causing an extreme value, AND whether the independent variable is influencing the residual (which would mean mis-specification, i.e. the model's mathematical form is wrong).

Normal q-q plot. Y-axis is empirical distribution of regression residuals (empirical quantiles); x-axis is theoretical normal distribution (take observed values, turn them into uniformly-spaced quantiles, then use inverse CDF of normal distribution - which correspond to quantiles - to get Z-values (theoretical quantiles of a normal distribution)). A q-q plot with a higher slope is leptokurtic and lower slope is platykurtic.

1.2 Regression fit and interpretation

Goodness of fit and overfitting. Coefficient of determination, R^2

$$R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST} = \frac{\sum(\hat{Y}_i - \bar{Y})^2}{\sum(Y_i - \bar{Y})^2}$$

SST is fixed. SSE can only fall with the addition of independent variables because otherwise, OLS isn't efficient. So, SSR can only increase.

Coefficient of determination cannot tell if model fit is good, particularly if many independent variables. To help address overfitting, use (1) adjusted R^2 , (2) AIC, or (3) BIC.

Adjusted R^2 doesn't automatically increase with more independent variables. Adjusted R^2 divides by DOFs:

$$\begin{aligned}\bar{R}^2 &= 1 - \frac{SSE/(n - k - 1)}{SST/(n - 1)} \\ \bar{R}^2 &= 1 - \frac{n - 1}{n - k - 1}(1 - R^2)\end{aligned}$$

k rises as more independent variables are added. Adjusted R^2 can be negative, unlike normal R^2 . If additional variable's coefficient t-statistic is greater than 1 or less than -1 (i.e. explains the dependent variable well), adjusted R^2 increases when the additional variable is added; otherwise, it decreases.

AIC. Use for selecting best PARSIMONIOUS + PREDICTIVE model:

$$AIC = n \ln \frac{SSE}{n} + 2(k + 1)$$

BIC. Use for selecting the most PARSIMONIOUS model:

$$AIC = n \ln \frac{SSE}{n} + \ln(n)(k + 1)$$

because $\ln(n) > 2$. For AIC and BIC, lower values = more parsimonious. AIC better for prediction (at the cost of increased, possibly unnecessary, complexity), BIC better for explanation or inference or causality (at the cost of predictive accuracy, by simplifying the model more).

Testing joint hypotheses. Can test the coefficients together in an F-test as in Level 1. While can test each coefficient, multiple coefficients might share explanatory power, so might not want to do this. ANOVA F-test:

$$F = \frac{MSR}{MSE} = \frac{SSR/k}{SSE/(n - k - 1)}$$

In the ANOVA table, if there are n samples, then the total DF is (n-1) to account for the average value (= DOF of intercept). Then the DOF of the regression = number of independent variables = k. DOF of residuals = n-k-1.

To test whether an unrestricted model with, e.g. 5 variables has better fit than a model with e.g. 3 of those 5 variables, can test whether b_4 and b_5 are jointly = 0.

$$F = \frac{(SSE_{restricted} - SSE_{unrestricted})/q}{SSE_{unrestricted}/(n - k - 1)}$$

q is the number of restrictions, and k is for the unrestricted model. Unrestricted model = **more regression coefficients**. Predictions using MLR models. The forecast error = model error + sampling error. The model error is residuals, the sampling error comes from how well the independent variables were forecasted!

Standard error in general = standard deviation of sampling distribution, e.g.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

and since we often don't know the true value of σ , the population standard deviation, we often just use the sample standard deviation $s = \sigma$.

Standard error = SE of regression = SE of estimate =

$$\sqrt{\frac{SSE}{n - k - 1}}$$

Standard error can therefore increase with more independent variables. SSE can fall but will never rise. Denominator will only fall with more independent variables. Used for prediction confidence intervals. Called 'standard error' even though more like standard deviation - though can think of it as one sample

regression out of the population, so the denominator = $\sqrt{1} = 1$ (as with any sample standard deviation).

Standard error of coefficient = use as the 'standard deviation' to calculate t-statistic. Note: can't get a 'real' standard deviation for the coefficient, because there's only one coefficient estimate for the entire sample! So use a 'standard error':

$$t = \frac{\hat{\beta} - 0}{SE(\hat{\beta})}$$

where

$$SE(\hat{\beta}) = \sqrt{\frac{MSE}{X'X}}$$

where $X'X$ is like the variance of the i-th variable. For a simple linear regression,

$$SE(\hat{\beta}) = \sqrt{\frac{\frac{1}{n-2} \sum (y_i - \hat{y}_i)^2}{\sum (x_i - \bar{x})^2}}$$

The $\sqrt{MSE}$ is the same for all variables in the regression, but if $X'X$ is large (e.g. $\sum (X_i - \bar{X})^2$ for single-variable regression), then a wide 'spread' of X leads to lower (dilutive) $SE(X)$. Basically, apportioning $\sqrt{MSE}$. The SE of each regression coefficient has the same DOF as the residual: $n - k - 1$.

1.3 Model misspecification

Functional forms. OVB, variable form, variable scaling, data pooling.

Heteroskedasticity. From OVB, incorrect functional form, others. Unconditional heteroskedasticity - error variance uncorrelated with independent variables; no statistical problems. Conditional heteroskedasticity - error variance is statistically problematic; usually (but not always) UNDERESTIMATED standard errors because SE has a square root term, leading to inflated t-statistics.

Conditional heteroskedasticity. Regress squared residuals (=error variance) against each of the independent variables. Or regress squared residuals against all independent variables together = Breusch-Pagan test, and calculate nR^2 , assuming that it's chi-square distributed.

$$\chi^2 = nR^2$$

Or use robust (White) standard errors, which take the residuals and scale them so that they are homoskedastic. White's standard errors give each observation its own unique "weight" in calculating the overall variance of the coefficients, specifically by using that observation's squared residual as a proxy for its error variance.

Serial correlation. Intuitively, like a version of heteroskedasticity. Usually underestimates standard errors of coefficients because the errors are serially 'linked', so the actual number of samples in the calculation is 'artificially' too high - it's as-if autocorrelation makes the n smaller in the denominator.

- If (1) independent variable is lagged value of dependent variable, then coefficient and SE are biased; (2) independent variable is NOT a lagged version of the dependent variable, then coefficient estimate is consistent but SE is biased. The coefficient in (1) is always biased because $y_t = a + by_{t-1} + e_t$ means $e_t = (y_t - by_{t-1}) - a$ and the bracketed term is autocorrelated - it's basically OVB, whereas (2) gives a chance that the problem is just low SE of coefficients. Intuitively, if something affects y_{t-1} , it affects y_t . Usually, SE will be underestimated.
- Positive versus negative serial correlation.
- Use Durbin-Watson test (only 1 lag; 'first order') or Breusch-Godfrey test (up to p order).
- DW test (only 1 lag is testable). Regress, take square of successive residuals and compare to square of lone residuals. If positive residual autocorrelation, very small test statistics. If negative residual autocorrelation, very high test statistics. If no residual autocorrelation, medium values. **DW test doesn't work if independent variable include lags of the dependent variable** - use Breusch-Godfrey test instead. (can e.g. look at residual autocorrelations instead). DW positive autocorrelation detection means coefficient SEs are biased downwards!
- BG test (can test up to p lags). Regress as normal. Then, regress residuals against independent variables AND lagged version of residual (which tests the autocorrelation), e.g. $u_t = a_0 + a_1X + bu_{t-1}$. Test statistic is $F_{n-p-k-1,p}$.
- Correct using Newey-West SE, robust SEs. Robust SEs use HAC estimators.

Multicollinearity. SEs are very large. So t-tests have little power. Often significant F-test, high R^2 , but individual coefficient t-tests fail. Multicollinearity may be present if adding an independent variable causes adjusted R^2 to not change much but SEs to rise. Variation inflation factors,

$$VIF = \frac{1}{1 - R_j^2}$$

which have minimum of 1, are high (e.g. more than 5) if there's multicollinearity. R_j^2 is from regression the j -th independent variable against the others. Correct MC by removing variables, using different proxies of the variable, increasing sample size.

1.4 Extensions of multiple regression

Influence analysis. 2 types of influential observations: (1) high-leverage points (x value is far from others) and (2) outliers (y value is far from predicted).

Leverage h_{ii} = x-value of i-th observation - mean x-value (of all n observations, including i-th observation). If

$$h_{ii} > 3 \left(\frac{k+1}{n} \right)$$

then it is potentially influential.

Studentized residuals. Calculate the residual between the observed y-value and the y-value when that observation is deleted and the regression is re-estimated on the remaining $n - 1$ observations:

$$e_i^* = Y_i - \hat{Y}_{i^*}$$
$$t_{i^*, n-k-2} = \frac{e_i^*}{s_{e^*}} = \frac{e_i}{\sqrt{MSE_{-i}(1 - h_{ii})}}$$

Dummy variables. Intercept and slope dummies, e.g.

$$Y_i = b_0 + d_0 D_1 + b_1 X_i + d_1 D_i X_i + \epsilon_i$$

Qualitative dependent variables. Can't use

$$Y_i = b_0 + b_1 X_{1,i} + b_2 X_{2,i} + \epsilon_i$$

as it is at most a linear probability model which is not shaped conveniently (e.g. it's hard to find the 'center' or recenter it so that we can establish binary distinctions). Instead, try to get cumulative logistic distribution shape - the odds $P/1 - P$ is exponential, and taking its log can make it symmetrical and therefore center-able. Get this by taking log odds

$$\ln \left(\frac{P}{1-P} \right) = b_0 + b_1 X_{1,i} + b_2 X_{2,i} + \epsilon_i$$

Solve for P:

$$P = \frac{1}{1 + \exp(-(b_0 + b_1 X_{1,i} + b_2 X_{2,i} + \epsilon_i))}$$

This is the logistic transformation. Can't just use

$$\frac{P}{1-P} = b_0 + b_1 X_{1,i} + b_2 X_{2,i} + \epsilon_i$$

or any other linear function because they're asymmetrical because they're not centered around 0 (i.e. below odds of 1, negative; above odds of 1, positive).

Use MLE to estimate logistic regression as OLS breaks down; use binomial distribution for the probability function and then maximize the joint probability using MLE. Use log-likelihood, which is χ^2 distributed as the test statistic. Can also use pseudo- R^2 .

Pseudo- R^2 can only be used to compare fit of different specifications of the same model, NOT models based on different datasets.

Interpretation. Non-linear, so marginal effect of each variable depends on the specific values (or levels) of EACH variable. So, plug in the average of each of the independent variables into the formula, then change the value of each independent variable in turn, e.g. by raising it 1% to see what the effect is on average P. That's the marginal effect.

Chi-squared test. Likelihood ratio (LR) test:

$$LR = -2 \times (\text{Log-likelihood restricted model} - \text{Log-likelihood unrestricted model})$$

The LL unrestricted model is the LL-null (intercept only) from the regression output.

1.5 Time series analysis

Linear trends - linear-linear models.

$$y_{t+1} = b_0 + b_1 t + \epsilon_i$$

Log-linear models:

$$y_{t+1} = e^{b_0 + b_1 t + \epsilon_i}$$

$$\ln(y_{t+1}) = b_0 + b_1 t + \epsilon_i$$

where the rate of growth of y is $y \times b_1$ (by differentiation, $\frac{dy}{dt}$).

However, trend models like linear-linear and log-linear models violate serial correlation, i.e. their residuals are serially correlated.

Autoregressive models. Can estimate AR models with OLS if (1) stationary, and (2) errors are uncorrelated. AR(1):

$$x_t = b_0 + b_1 x_{t-1} + \epsilon_t$$

AR(2):

$$x_t = b_0 + b_1 x_{t-1} + b_2 x_{t-2} + \epsilon_t$$

Covariance stationary (aka weakly stationary, stationarity; strict stationarity has little practical application). Mean must be constant and finite; covariance must be constant and finite:

$$Cov(y_t, y_{t-s}) = \lambda_s$$

If not covariance stationary, e.g. linear trend or random walk, then first differencing can sometimes make a non-stationary time series stationary. But if the AR equation has lags in it, the variable is already stationary, and it's just that the errors are autocorrelated, then first differencing will generally not resolve it.

Define: Autocorrelation (independent variable is lag of dependent variable; DW test doesn't work) - correlations of with its own k-th lag:

$$\rho_k = \frac{Cov(x_t, x_{t-k})}{\sigma_x^2} = \frac{E(x_t - \mu)(x_{t-k} - \mu)}{\sigma_x \sigma_{x-t}}$$

Define: Error autocorrelations (estimated by residual autocorrelations):

$$\rho_{\epsilon,k} = \frac{Cov(\epsilon_t, \epsilon_{t-k})}{\sigma_\epsilon^2}$$

Residual correlation's standard error is $\frac{1}{\sqrt{T}}$. This is the standard error of the RESIDUAL AUTOCORRELATION, it is not the standard error of the residual.

$$t = \frac{\rho_k}{\frac{1}{\sqrt{T}}}$$

To test if AR(p) model is valid, (1) estimate the model, (2) compute residual autocorrelations, (3) see if residual autocorrelations differ significantly from 0 using the t-statistic defined above. If residual autocorrelation is statistically significant, then try adding further AR lag orders.

AR models have independent variables that lag the dependent variable. So they should have autocorrelated errors. However, if the model is covariance stationary, and enough lags are added, then the autocorrelation can be captured in the variable itself and the residual will be mostly white noise. However, only some variables are amenable to AR modelling. Test the statistics on the AR regression coefficients as with linear regression, i.e. standard errors of coefficients.

In AR models, the error is white noise or almost white noise. White noise means uncorrelated errors but it's not the iid normal errors usually assumed in linear regression. However, white noise errors lead to AR OLS estimators that are consistent and asymptotically normal estimators.

Mean reversion: $x_t = x_{t+1}$. Assume constant x-values, then mean-reverting level is:

$$x_t = b_0 + b_1 x_t$$

$$x_t = \frac{b_0}{1 - b_1}$$

Forecast errors. To compare the accuracy of two models, for their in-sample forecast errors, compare their standard errors. For their out-of-sample forecast

errors, compare their RMSE, which is taken by the average of (actual - forecast value) over the number of these values (or test period), and square root of that.

If the covariance stationarity changes between two time periods, it is a regime change, so estimate equations for them separately.

$$RMSE = \sqrt{\frac{\sum (x_{actual} - x_{forecast})^2}{N_{forecasts}}}$$

where N is the number of forecasts, not the number of total observations used in the model.

Random walks. Has a unit root. Special case of AR(1) with $b_0 = 0$ and $b_1 = 1$.

$$x_t = x_{t-1} + \epsilon_t$$

$$E(\epsilon_t) = 0$$

$$E(\epsilon_t^2) = \sigma^2$$

$$Cov(\epsilon_t, \epsilon_s) = E(\epsilon_t \epsilon_s) = 0$$

Can add drift:

$$x_t = b_0 + x_{t-1} + \epsilon_t$$

For $x_t = b_0 + b_1 x_{t-1}$, mean reversion is $x_t = \frac{b_0}{1-b_1}$ but in random walk this is $=0/0$, an undefined mean. Further,

$$Var(x_t) = \sigma^2$$

$$Var(x_{t+1} = x_t + \epsilon_{t+1}) = \sigma^2 + \sigma^2 = 2\sigma^2$$

and so on. So variance is undefined too. So, standard regression techniques don't work, though we will get high R^2 and very statistically significant regression coefficient on lagged variable, but the coefficient will be close to or equal to 1 using standard regression techniques. But this is wrong; the standard error is wrong.

Note: **error is covariance stationary after first-differencing:**

$$x_t - x_{t-1} = \epsilon_t$$

and this is mean-reverting (a finite mean) and has finite variance.

So, regress

$$x_t - x_{t-1} = y_t = b_0 + b_1 y_{t-1} + \epsilon_t$$

which should yield $b_0 = b_1 = 0$ and an error term which is not serially correlated = random walk. Other ways to determine if random walk: (1) errors have no autocorrelation or drop quickly, (2) unit root by Dickey-Fuller.

Dickey-Fuller.

$$\begin{aligned}x_t &= b_0 + b_1 x_{t-1} + \epsilon_t \\x_t - x_{t-1} &= b_0 + (b_1 - 1)x_{t-1} + \epsilon_t \\x_t - x_{t-1} &= b_0 + g_1 x_{t-1} + \epsilon_t\end{aligned}$$

If unit root, then $g_1 = 0$. Use modified t-distribution from DF (different from t-distribution because errors are not iid).

Moving averages. Just like ARs, MAs require (1) covariance stationarity, and (2) no autocorrelation of errors. Test the statistics in the MA regression coefficients as with linear regression, i.e. standard errors of coefficients. Smoothing MAs for past values, but slow to react. Predictive MAs, e.g. MA(1):

$$x_t = \epsilon_t + \theta \epsilon_{t-1}$$

Look at error autocorrelations to see if MA and if MA, which order p of MA(p). MAs have error autocorrelations and then suddenly go to 0. ARs have error autocorrelations that gradually decline.

Seasonality. Try AR(1) with the lag, or AR(1) differenced with lag, e.g. for quarterly seasonality, x_t is influenced by x_{t-4} , so:

$$x_t = b_0 + b_1 x_{t-1} + b_2 x_{t-4} + \epsilon_t$$

$$\ln(x_t) - \ln(x_{t-1}) = b_0 + b_1(\ln(x_{t-1}) - \ln(x_{t-2})) + b_2(\ln(x_{t-4}) - \ln(x_{t-5})) + \epsilon_t$$

If many of the error autocorrelations are statistically significant but one of them is much higher, try just adding that one to the AR model - the other error autocorrelations might shrink automatically.

ARMA(p,q) models. Like ARs and MAs, require covariance stationarity and no autocorrelation of errors. With lots of data, can provide better fit. But often fitting is more art than science and doesn't forecast well. Coefficient estimates can be very unstable.

ARCH. Heteroskedasticity can be modelled, e.g. ARCH(1):

$$\epsilon_t^2 = a_0 + a_1 \epsilon_{t-1}^2 + u_t$$

to predict the next period's variance:

$$\sigma_{t+1}^2 = a_0 + a_1 \epsilon_t^2$$

which can be generalized to more complex forms with GARCH. So if suspect heteroskedasticity, can test for ARCH(n). As with other heteroskedasticity, if ARCH is present, then standard errors are invalid. Need to use GLS or other methods.

More than one variable. Take each variable's time series and look for unit roots. If none have unit roots, can combine in regressions. If only some have unit roots, stop. If all have unit roots, see if cointegrate by combining in regression and test whether error term has unit root (must use Engle-Granger critical values under the DF test).

Time series procedure. Look for linear, exponential (logged) trends and test residuals for stationarity, and then for serial correlation. If there is serial correlation, use more complex model like AR(1) and if (1) linear trend then first-difference, (2) exponential, then log first-difference, (3) regime changes, then need separate regressions. If errors are serially correlated, add lags, and then graph for seasonality and ARCH/GARCH.

Covariance stationarity versus random walk (and unit root). If a time series is covariance stationary, then it cannot be a random walk and therefore doesn't have a unit root. However, a time series which is not a random walk is not necessarily stationary, e.g. it might have a changing mean and/or variance.

Analyzing time series - after looking at the errors re serial correlation + ARCH + seasonality, check the coefficient estimates to see if close to 1 (i.e. check for unit root to see if stationary).

1.6 Machine learning

Statistical approaches - explicit models of structure, assumptions (e.g. underlying probability distributions). ML doesn't have these - learn the pattern, apply the pattern. ML: (1) supervised learning, (2) unsupervised learning, (3) deep learning/reinforcement learning (deep learning built on NN; RL is about interacting with itself or mirror image).

Supervised learning. Requires labelled dataset (x = features, y = target(s)). Dimensionality = number of features. Supervised learning: (1) regression (target is continuous), (2) classification (target is ordinal or binary; then x = classifier).

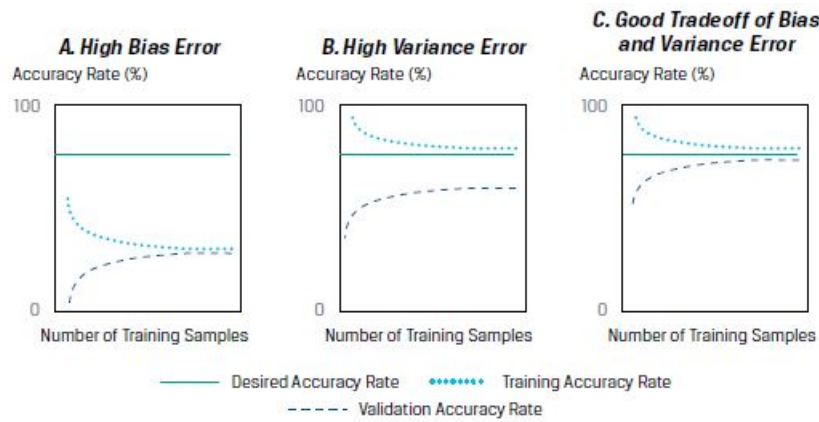
Unsupervised learning. No targets, just features. Good for (1) dimension reduction, and (2) clustering.

Generalization and overfitting. If ML algorithm incorporates too much noise, can't generalize because noise is different in new situations, reducing predictive value. Use prediction error to evaluate goodness of fit. Overfitting increases with complexity of the model.

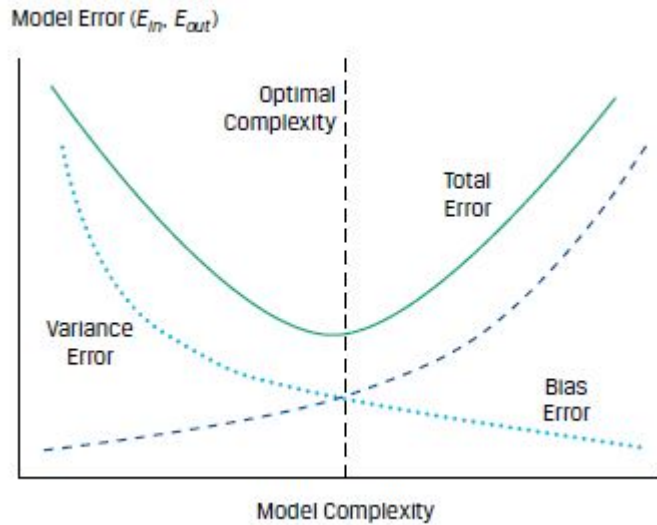
In-sample errors (training data set), out-of-sample errors (validation or test data set). Part of bias-variance tradeoff. Types of errors: (1) bias error: low variance but high bias (high in-sample error), typically from underfitting, (2)

variance error: low bias but high variance (high out-of-sample error), typically from overfitting, (3) base error: irreducible error due to randomness of data. A perfect model would have zero bias and zero variance, but it would still be subject to the base error inherent in the data.

Learning curves. Training learning curves decline in accuracy with increasing training sample size as small sample size = just memorize pattern; larger sizes = have to learn patterns. Validation accuracy rates rise with increasing training sample size. Notice in B below that the model fits the training data accurately but the out-of-sample accuracy (validation rate) is low.



Fitting curves. Cost (vertical axis; model error) versus complexity (horizontal axis). Bias error falls with increasing model complexity; variance rises with increasing model complexity. The addition of the costs (or model errors) is at a optimal low in the middle.



To prevent overfitting, use (1) **regularization** (an overfitting penalty if there are too many features; based on Occam's razor), and (2) **cross-validation**, e.g. k-fold cross-validation, time-series cross-validation. In cross-validation, train k models on various hyperparameters, then take hyperparameters with best results (e.g. accuracy, ROC-AUC) and train on entire dataset; this becomes the model.

Supervised methods. Penalized regression (regularization). LASSO:

$$\sum (y_i - \hat{y}_i)^2 + \lambda \sum |\hat{b}_k|$$

where the b_k are regression coefficients. λ is a hyperparameter. The penalty incentivizes parsimony; helpful to reduce multicollinearity. Can be used for non-linear models.

Support vector machine (SVM). Linear classifier. Divides points using hyperplanes with margins. Want wide margins for the max boundary between regions or zones. Cost parameter is a hyperparameter which is a penalty for misclassifications; higher cost means thinner margins, lower cost means thicker margins. Can have non-linear boundaries, but more complex and prone to overfitting.

K-nearest neighbors. Pick $k=1, 5$, etc. Used e.g. for credit modelling. Note: In KNN, targets are already labelled (this is a supervised ML after all); the clusters are already there; KNN finds the nearest neighbor of a new, unlabelled point.

Classification and regression tree (CART). Supervised. Provide labelled (target) data and it will make a decision tree with the features by using cutoff values

from the features as hyperparameters - which can be tuned. This will classify or partition the data (partition space) into different areas, maximizing the separation of the data to minimize classification error. To reduce overfitting, add regularization, e.g. limit tree size, minimum population of tree; also, can prune the decision trees.



Ensemble learning and ensemble method. Combine multiple algorithms. Typically works better out-of-sample than non-ensemble methods, and more robust to noise. Can use **majority-vote classifiers** to pick the final result. Can use **bootstrap aggregating (bagging)** where the same ML algorithm is used but with different training data.

Random forest classifier. Ensemble method. E.g. bootstrapped CART, which produces n CART trees (one for each sample generated by bootstrapping). Can reduce the number of features to reduce overfitting.

Oversampling - increasing proportional representation of rare data in the training set to increase model precision (e.g. ratio of correctly predicted fraudulent cases to all predicted fraudulent cases) and recall (ratio of correctly predicted fraudulent cases to all actual fraudulent cases).

Evaluation of **supervised** models. False positives (FP) are Type I errors. False negatives (FN) are Type II errors. **Accuracy** is percentage of correctly predicted classes out of total predictions. Low type I and II errors. **F1 score** is weighted average (harmonic average) of precision and recall - precision is ratio of correctly predicted positive classes to all predicted positive classes (useful where cost of Type I error is high); recall (sensitivity) is ratio of correctly pre-

dicted positive classes to all actual positive classes (useful where cost of Type II error is high). Information gain quantifies information that feature holds about response, like a non-linear strength of correlation between X and Y. It's about how relevant a variable is.

$$F1 = \frac{2 \times P \times R}{P + R}$$

This means:

$$F1 = \frac{2}{\frac{TP}{TP+FN} + \frac{TP}{TP+FP}} = \frac{2TP}{(TP+FN) + (TP+FP)} = \frac{2TP}{2TP + (FN+FP)} = \frac{2TP}{2TP + 2(F_{average})} = \frac{TP}{TP + F_{average}}$$

So, it measures: 'how good is the TP'?

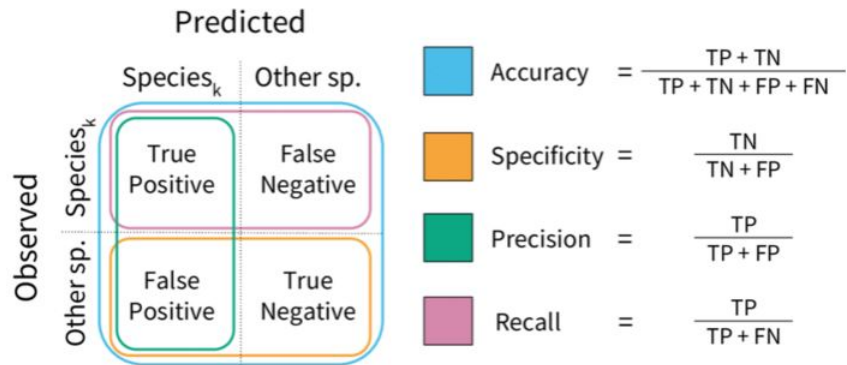
True negative rate = specificity. False positive rate (FPR) = 1- specificity.

True positive rate = recall.

$$FPR = 1 - \frac{TN}{TN + FP} = \frac{FP}{FP + TN}$$

So, the FPR is about: what proportion of the actual negatives were falsely labelled as positives?

Confusion matrix:

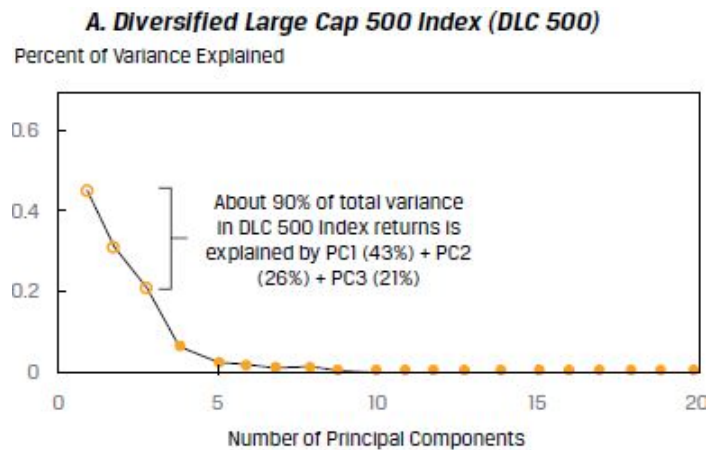


A model that hardly finds any true positives (low recall) but is very precise about the few it finds, is useless. Harmonic averages generally filter out large members. The F1 only allows high numbers if both recall and precision are high.

Unsupervised learning. Dimensionality reduction and clustering.

PCA. Dimensionality reduction. Transform highly correlated features of data into a few main, uncorrelated composite variables. Transforms covariance matrix of features and involves eigenvectors and eigenvalues (determines how much

of variance of data is explained by the eigenvector). Orient eigenvectors by minimizing projection distance errors but maximize the spread (data points parallel to the eigenvector to draw 'long' eigenvectors with high eigenvalues). Orders eigenvectors from highest eigenvalues to lowest. Use a Scree plot here - example shown below.



PCA is often used for EDA. ML models are quicker to train, tend to reduce overfitting, and easier to interpret with lower-dimensional datasets.

Clustering. Can specify number of clusters, distances (e.g. R^2). K-means clustering: select k (a hyperparameter) initial centroids randomly and divide into clusters depending on distances, then move centroid to center of cluster and repeat. Randomize initial centroid positions.

Hierarchical clustering includes (1) agglomerative (bottom-up) clustering, and (2) divisional (top-down) clustering. Agglomerative algorithms are faster. **Dendrograms** depict clusters along x-axis, and y-axis showing distances between clusters.

Neural networks. Can often handle better if multicollinearity because learns complex relationships instead of trying to perfectly isolate correlations. Input, hidden layer(s), output layer. Hidden layer nodes often comprise input values fed into a summation operator (linear transformation) and then fed into an activation function (often logit function with dimmer switch to determine activation threshold). This is then forward-propagated. The output is then compared to the actual and penalized, and then backpropagated to adjust the weights: new weight - old weight - learning rate $\times$ partial derivative of total error wrt old weight. The **learning rate** is a hyperparameter.

Number of neurons is a hyperparameter. Can add regularization, e.g. LASSO, which will increase in-sample MSE but reduced out-of-sample MSE, i.e. improve

bias-variance tradeoff.

Deep neural networks (DNN). 2-20 hidden layers.

Reinforcement learning.

1.7 Big data

ML modelling steps: (1) Problem conceptualization, (2) Data collection, (3) Data preparation and wrangling: (a) data cleansing (address errors, blanks, duplicates, non-uniform data, out of range invalidity, data inconsistencies), (b) data preprocessing (outliers, scaling data, transforming data, extracting e.g. age from birthdate, aggregating variables, filter for appropriate use and range, select the features, data conversion), (4) EDA, feature selection and engineering, (5) Model training.

Data preparation - errors

- Incompleteness
- Invalidity
- Inaccuracy
- Inconsistency
- Non-uniformity
- Duplication

Data wrangling (preprocessing) transformations:

- Extraction (of new variables from existing ones)
- Aggregation
- Filtration
- Selection
- Conversion (of data types, e.g. nominal, ordinal, continuous, categorical)

Data wrangling normalization and standardization. Normalization is to scale the range to 0 to 1 - but note that it's sensitive to outliers:

$$X_{i,normalized} = \frac{X_i - X_{min}}{X_{max} - X_{min}}$$

Standardization - note data has to be normally distributed:

$$X_{i,standardized} = \frac{X_i - \mu}{\sigma}$$

Unstructured data. Text preparation and wrangling transforms into structured data, a **document term matrix (DTM)**. Note: a document can be a sentence or other larger unit.

Text preparation: remove html tags, punctuation, numbers (won't work well in a DTM), white spaces.

Text wrangling: tokenization (split text into individual tokens - often individual words but can also be n-grams) by normalization: (1) lowercasing, (2) stop words (the, in, no), (3) stemming (e.g. turn 'happiness' into 'happi'), (4) lemmatization (e.g. turn 'ran' into 'run'). Latter two reduce data sparsity (by removing these useless words as dimensions). Remove duplicates. This gives a 'bag of words' (BOW).

Data (usually numerical, ordinal, categorical) exploration: (1) **EDA** with histograms, scatterplots, boxplots, etc, (2) feature selection, (3) feature engineering.

Feature selection: remove irrelevant features and basic diagnostics: homoskedasticity, multi-collinearity, univariate chi-square and correlation tests, information-gain measures (R-squared). Dimensionality reduction.

Feature engineering: combine, decompose variables. Transform variables. One-hot encoding: transform categorical variable into binary forms (like dummy variables), but this increases sparsity. The process of decomposing a single categorical feature with multiple values, e.g. individual features by type (e.g., isdrizzle, israin, issnow), and recording the results in binary form (0 or 1).

Apply EDA, feature selection, and feature engineering for text:

Text exploration: text classification (supervised ML), topic modelling (unsupervised ML), sentiment (supervised, unsupervised, supervised + unsupervised ML). Text EDA: look at term frequency of words, wordclouds.

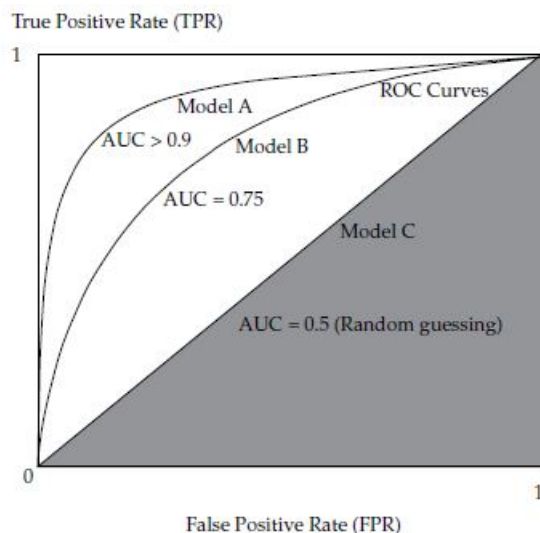
Text analysis feature selection: frequent tokens lead to underfitting, rare tokens to overfitting. **Document frequency (DF)** discards tokens present in all or most documents; the DF of a token is the number of documents containing the token divided by the total number of documents. DF, chi-squared, mutual information (high MI if token usually only appears in specific texts).

Text analysis feature engineering. Using specific numbers or classes of numbers, N-grams, NER (name entity recognition; classifies tokens into classes using a dictionary), parts of speech (POS) which classifies tokens into grammatical categories.

ML model training. More art than science. Model fitting (overfitting, underfit-

ting) caused by dataset size and number of features. Note: class imbalance can be reduced by undersampling and oversampling.

Performance evaluation. Label data TP, TN, FP, FN and build confusion matrix to get accuracy, precision, recall, true positive rate, F1. Then use accuracy, ROC, or RMS to evaluate performance. Receiver operating characteristic (ROC): TPR v FPR; AUC (area under curve) of 1 is perfect prediction rate, 0.5 is guess or random. Note: ROC requires testing every threshold 0-1 for the model output probability to get the data points to plot the ROC.



TPR: What proportion of the actual positives were identified as positive? FPR: What proportion of the actual negatives were identified as positive?

Smaller RMSE is potentially better model performance:

$$RMSE = \sqrt{\sum \frac{(Predicted_i - Actual_i)^2}{n}}$$

Tuning. **Parameters** depend on training data. Parameters are the internal variables of the model that are learned from the training data during the training process. These are the values that the model adjusts itself to fit the data. They are an intrinsic part of the model's learned function.

Hyperparameters don't depend on training data and depend on the researcher. For parameters, select the model parameters with highest accuracy. For hyperparameters, use grid search or ceiling analysis for optimum bias-variance trade-off.

DF = document frequency = sentence count with the word / total number

of sentences. Inverse DF = IDF = $\log(1/DF)$, indicates how unique the term is; higher value means more unique. Term frequency = TF, the number of times the term appears in the document. **TF-IDF** = TF(sentence) x IDF, which indicates importance of the token. A word that appears very often in one document (high TF) AND is rare in the overall collection (high IDF) will get a high TF-IDF score. This makes it a strong candidate for representing the core topic of that document.

Token frequency (collection level) = total word count / total number of words.
Token frequency (sentence level) = word count in sentence / total word count of sentence.

Feature selection reduces overfitting; feature engineering reduces underfitting.

Bias and variance used for tuning, not performance evaluation. Performance evaluation to measure goodness of fit: (1) error analysis, (2) ROC, (3) RMSE.

2 Economics

2.1 Currency exchange rates

Notation. FX quoted as price/base currency and foreign/domestic. Spot is for settlement on the second business day after trade (T+2 settlement, except CAD/USD which is T+1; if longer than T+2, it is a forward contract). **Two-sided price**, e.g. bid-offer of USD/EUR of 1.1648/1.1652 means dealers will pay USD 1.1648 for 1EUR and sell one EU for USD 1.1652. The **interbank spreads** are NARROWER than the dealer-client spreads; this allows the dealer to offset the client transaction with the interbank market, earning an instant profit. The pips unit - 0.0001 except JPY which is 0.01.

Bid-offer spreads. Some currency pairs are more liquid. Size of spread also depends on time of day and when and which liquid markets are open. Also depends on market volatility, and if the trade size is large harder to offset risk so spread is wider.

Most currency pairs most liquid during 8-11am NY time, which is London afternoon.

Cross-rate bid/offer spreads.

$$\left(\frac{GBP}{USD}\right)_{bid}$$

means pay 1 USD to get x GBP. So the currency paid into is the denominator.

$$\left(\frac{GBP}{USD}\right)_{offer}$$

means pay 1 GBP to get x USD. So the currency paid into is the numerator.

Mark-to-market. For an investor to close out a forward position early, use the 'all-in' FX (spot + forward pips). Then find the investor's counterparty's position under the forward contract (in other words, it's the opposite buy-sell side of the investor's contract, and it's in the opposite currency). Then find the CURRENT condition for the counterparty, and discount this back. Example: if forward sale of 10 NZD in 3 months to hedge US FX, then BUY 10 NZD in USD for the calculation; if forward sale of 10 USD in 3 months, then BUY 10 USD in NZD for the calculation.

International parity conditions. Help form long-run equilibrium views:

- Covered (hedged) interest rate parity
- Uncovered (unhedged) interest rate parity
- Forward rate parity
- PPP (relative or ex-ante)
- International Fisher effect

Note: Uncovered interest rate parity says that the expected **future exchange rate** follows interest rate parity, and covered interest rate parity says that **forward rates** follow interest rate parity.

Covered interest rate parity. Investment in a foreign money market instrument which is completely hedged against FX = return on domestic money market investment. Arbitrage makes the CIP equation hold.

Uncovered interest rate parity. Change in spot FX rate over investment horizon should on average = differential in interest rates. UIP equation (which just replaces CIP's forward FX with the expected spot FX rate) doesn't hold well because of risk aversion, so the expected spot rate changes as perceived risk of sovereign bond rates changes - so the expected spot FX rate just floats around and changes with those perceived risk changes. So, UIP would hold if investors were risk neutral. If FX spot rates follow a random walk, then current FX spot rate = future FX spot rate.

In the short term, only the covered interest rate parity generally holds because of arbitrage. In the short and medium term, uncovered interest rate parity doesn't hold well because there is no arbitrage there to remove disequilibria. It sometimes doesn't hold well because of risk-aversion (UIP assumes risk-neutrality). It holds better over the long run.

$$\% \Delta S_{f/d}^e = i_f - i_d$$

Forward rate parity. Forward FX rate is an unbiased predictor of future FX rate. Sometimes higher, sometimes lower, but unbiased predictor.

Purchasing power parity. Relationship between interest rate and inflation. Law of one price: foreign prices should equal domestic prices adjusted by FX rate. **Absolute version** of PPP - law of one price for broad price level of foreign and domestic countries, and equilibrium exchange determined entirely by this ratio (same exact weights of goods and service bundles and that everything is tradeable):

$$S_{f/d} = \frac{P_f}{P_d}$$

But if transaction costs are high, then absolute version of PPP not valid. Instead perhaps changes in FX will hold - the **relative version** of PPP - the change in the exchange rate between two currencies over a period should approximately offset the difference in inflation rates between the two countries during that same period:

$$\begin{aligned}\% \Delta S_{f/d} &= \frac{\% \Delta P_f}{\% \Delta P_d} \\ \Delta S_{f/d} / S_{f/d} &= \frac{\Delta P_f / P_f}{\Delta P_d / P_d} \\ 1 + S_{f/d} &= \frac{1 + \pi_f}{1 + \pi_d} \\ \% \Delta S_{f/d} &\approx \pi_f - \pi_d\end{aligned}$$

The 'ex ante' version of this, i.e. with every term being an expected term, is the long-run expected FX rate. So, **ex-ante PPP** comes from relative PPP:

$$\% \Delta S_{f/d}^e \approx \pi_f^e - \pi_d^e$$

Fisher effect:

$$i = r + \pi^e$$

where r is the real interest rate. Then,

$$\begin{aligned}i_f - i_d &= (r_f - r_d) + (\pi_f^e - \pi_d^e) \\ (r_f - r_d) &= (i_f - i_d) - (\pi_f^e - \pi_d^e)\end{aligned}$$

Input UIP and **ex-ante PPP** (difference in expected inflation rates = expected change in the FX rate). results below into the above:

$$\begin{aligned}\% \Delta S_{f/d}^e &= i_f - i_d \\ \% \Delta S_{f/d}^e &= \pi_f^e - \pi_d^e\end{aligned}$$

to get the **real interest rate parity**:

$$r_f - r_d = \% \Delta S_{f/d}^e - \% \Delta S_{f/d}^e = 0$$

If r 's are the same, **International Fisher effect**:

$$i_f - i_d = \pi_f^e - \pi_d^e$$

Note there could be a risk premium on top of this, e.g. if country has high debt. Might be reflected in real interest rate.

Can combine UIP, ex-ante PPP, real interest rate parity, international Fisher effect:

$$\begin{aligned}\% \Delta S_{f/d}^e &= i_f - i_d = (r_f - r_d) + (\pi_f^e - \pi_d^e) \\ \% \Delta S_{f/d}^e - \% \Delta S_{f/d}^e &= r_f - r_d = (i_f - i_d) - (\pi_f^e - \pi_d^e)\end{aligned}$$

Carry trade. Studies show UIP doesn't hold in the short and medium term. So, can use FX carry trade by borrowing and then shorting low-yield (low i) currencies and long high-yield currencies. These are leveraged trades. However, note high-yield currencies' yields can be very volatile - so the return distribution of carry trades is negatively skewed with fat tails. Also, if everyone unwinds carry trades in a flight to quality, this exacerbates losses.

Balance of payments. BOP = current account + capital account + financial account. Here, just group latter two to capital account (financing/investing).

Current accounts and BOP flows. (1) Trade (goods and services): Export surplus leads to currency appreciation and vice versa, which should lead to correction as export competitiveness declines and imports become cheaper for the exporter. However, takes a long time and often a change in FX only leads to partial change in net trade as exporters adjust prices to keep market share. (2) Portfolio: portfolio managers and asset managers also make decisions on what to do with currencies which affect FX rates. (3) Debt: If current account deficit is persistent and large and debt is used to fund imports and this is unsustainable, market will begin to think major depreciation is coming.

Current accounts over the long term, i.e. BOP flows.

- The portfolio balance channel assumes that current account imbalances shift financial wealth from deficit nations to surplus nations. Countries with trade deficits will finance their trade with increased borrowing. This behavior may lead to shifts in global asset preferences, which in turn could influence the path of exchange rates.
- The debt sustainability channel as a means to assess the long-run equilibrium value of exchange rates. According to this mechanism, there should be some upper limit on the ability of countries to run persistently large current account deficits. If a country runs a large and persistent current account deficit over time, eventually it will experience an untenable rise in debt owed to foreign investors. If such investors believe that the deficit country's external debt is rising to unsustainable levels, they are likely

to reason that a major depreciation of the deficit country's currency will be required at some point to ensure that the current account deficit narrows significantly and that the external debt stabilizes at a level deemed sustainable.

- The flow supply/demand channel mechanism focuses on the fact that purchases and sales of internationally traded goods and services require the exchange of domestic and foreign currencies in order to arrange payment for those goods and services. Such shifts in currency demand should exert upward pressure on the value of the surplus nation's currency and downward pressure on the value of the deficit nation's currency.

Capital account and flows. Excessive emerging market capital inflows lead to major external debt, asset bubbles, consumption binges and wasteful over-investments, and then a crash, so often capital controls are imposed. If there's a crash, usually the currency doesn't appreciate; instead, the risk premium for that country rises. Correlations between equity returns and FX rates are unstable in the ST and tend towards 0 in the LT.

The **flow supply/demand channel** assumes that trade imbalances will be corrected as the deficit country's currency depreciates, causing its exports to become more competitive and its imports to become more expensive. Studies indicate that there can be long lags between exchange rate changes, changes in the prices of traded goods, and changes in the trade balance. In the short run, exchange rates tend to be more responsive to investment and financing decisions.

Mundell-Fleming approach to FX rates. P and inflation play no role. As monetary and fiscal policies change, i and GDP change, but depending on capital controls and FX rate flexibility, this changes capital flows + FX. Example: fiscal expansion + monetary tightening raises i and with high capital mobility, currency appreciates; with low capital mobility, indeterminate effect on GDP and therefore imports and therefore currency. Note: fiscal expansion raises i as more borrowing needed.

Monetary approach to FX rates. In contrast to Mundell-Fleming, GDP is fixed and monetary policy affects FX through P and inflation rate. Monetary expansion increase P , and assuming PPP holds, depreciation. However, PPP doesn't hold in ST and MT; to address this, use Dornbusch: P fixed in ST but fully flexible in long run.

Dornbusch. If the domestic price level is assumed to be inflexible in the short run, however, the model implies that the exchange rate is likely to overshoot its long-run PPP path in the short run. With inflexible domestic prices in the short run, any increase in the nominal money supply results in a decline in the domestic interest rate. Assuming that capital is highly mobile, the decline in domestic interest rates will precipitate a capital outflow, which in the short run will cause the domestic currency to depreciate below its new long-run equilibrium level. In

the long run, once domestic nominal interest rates rise, the currency will appreciate and move into line with the path predicted by the conventional monetary approach.

Reconciling UIP and Mundell-Fleming and monetary approaches. Mundell-Fleming and monetary approaches are about short-term. UIP is about what would have to happen in the long-term if the interest rate differentials were to hold.

Note: even if there are capital controls, an tightening in monetary policy will still lead to FX effects. As i rises, there will still be some leakage and therefore appreciation. There will also be reduced GDP and therefore less imports and therefore appreciation.

Portfolio balance approach. Focus on investors with assets, including bonds. Unlike Mundell-Fleming and Monetary approach, this looks more at LT. Example: a growing government budget deficit leads to higher i and therefore appreciation in the short term. But this may require in the LT more domestic bonds, which requires higher expected returns because the debt is growing - so need (1) higher interest rates or premium and/or (2) IMMEDIATE currency depreciation to get a future currency appreciation gain on the bonds. Governments running long-run deficits and growing debt will lead to LT currency depreciation - e.g. need to stop and therefore lead to tighter fiscal policies causing i to decline and/or central bank buys bonds by monetary expansion.

High fiscal spending leads to high debt, and therefore high interest rates and currency depreciation.

Capital controls. Many developing countries intervene and use capital controls to reduce FX volatility and to try to prevent too much inflow, speculative attacks (by unwinding carry trades which are shorting one currency and long on another), asset bubble formation (by too much capital inflow - hot money), etc. Most developed country central banks hold negligible FX reserves compared to daily transactions and don't intervene in FX markets much; developing countries have more FX reserves and intervene more.

Currency crises. From unwinding carry trades, having to pay US-denominated debt, flight to safe haven currencies. Some currency crises might be predicted in advance with various changes to market liberalization, changes in capital inflows, banking crises, currency rate changes, trade balance changes, money growth, inflation, etc.

A high level of FX reserves reduces probability of currency crisis.

Use the UIP to calculate the 'missing' forward FX rate. Use the covered interest rate parity to find the 'missing' MRR.

2.2 Economic growth

Differences in GDP per capita (PPP): K, L, technology, and institutions (financial markets and intermediaries, rule of law and property rights, education, health, taxes, fewer regulations, unrestricted trade and capital flows).

Problem for developing countries. Low K/worker. But to increase K, need savings. But low savings if salaries are low. So need to attract foreign investment. Financial intermediaries - a good banking system can help by making good investments and attract savings, and attract savings to fund investments. FDI can be direct (e.g. in PPE) and indirect (e.g. in bonds and equities).

Economic growth and security returns. Earnings growth cannot exceed GDP growth forever - for this to happen, the ratio of profits to GDP must trend upward, but this means capital owners get all income and stagnant labor income - workers won't work and undermine demand.

Grinold-Kroner decomposition.

$$\begin{aligned}
 ROE &= dy + \text{expected capital gain} \\
 &= dy + \text{expected repricing} + \text{earnings growth per share} \\
 &= dy + \Delta(P/E) + \pi + g - \Delta S \\
 &= \frac{D}{P} + \% \frac{\Delta P}{E} + \pi + g - \% \Delta S \\
 &= \frac{D}{P} + \% \frac{\Delta P}{E} + \% \Delta E - \% \Delta S
 \end{aligned}$$

where E is nominal earnings, dy is dividend yield, ΔS is the increase in shares outstanding, π is inflation rate and g is real GDP growth rate. Expected capital gains = $\% \frac{\Delta P}{E} + \% \Delta E = \% \frac{\Delta P}{E} + r + g$. $\% \Delta E - \% \Delta S$ is the nominal earnings growth per share. $\% \Delta S < 0$ means EPS grows faster than total nominal earnings. In the long run, the P/E goes to 0, $\% \Delta S$ goes to 0, and $\% \Delta E$ goes to nominal GDP. **Expected repricing - increase in shares outstanding = 0** over the long run; anything else might not be sustainable.

$$\Delta S = -nbb + rd$$

where nbb is net buybacks and rd is relative dynamism of the economy - economic growth from SMEs which will issue shares and increase S. Privatizations generally lead to $nbb < 0$ and delistings generally lead to $nbb > 0$.

Bonds. If actual GDP growth above potential GDP growth, inflation rises, increasing nominal interest rates (monetary policy affected by difference between actual and potential GDP - monetary policy easing if actual GDP below potential GDP, which will reduce interest rates and therefore a good idea to buy bonds now) and reducing bond prices. Faster potential GDP raises real asset

prices, real interest rates, which induce savings for investments.

Production function. $Y = \text{Potential GDP}$

$$Y = AF(K, L)$$

A is total factor productivity. A is technology, but so is the function F as it can change K and L productivity. However, 'technology' in standard terminology refers to just TFP. For concreteness, use Cobb-Douglas production function

$$F(K, L) = K^\alpha L^{1-\alpha}$$

$$Y = AK^\alpha L^{1-\alpha}$$

It has constant returns to scale. It also has diminishing marginal returns wrt to both L and K because α and $1 - \alpha$ are less than 1.

Labor productivity:

$$y = \frac{Y}{L} = A \left(\frac{K}{L} \right)^\alpha \left(\frac{L}{L} \right)^{1-\alpha} = Ak^\alpha$$

The rental price of capital is

$$\frac{dY}{dK} = MPK = \alpha AK^{\alpha-1} L^{1-\alpha} = \alpha A \frac{K^\alpha}{K} L^{1-\alpha} = \alpha \frac{Y}{K} = r$$

So,

$$\alpha \frac{Y}{K} = r$$

$$\alpha Y = rK$$

which means that share of GDP paid out to capital suppliers is α , and therefore the share of GDP paid out to labor is $1 - \alpha$. Also, a 1% increase in K leads to a $\alpha\%$ increase in Y, so α is the elasticity of Y wrt K.

Growth accounting. Solow's model:

$$\Delta Y/Y = \Delta A/A + \alpha \Delta K/K + (1 - \alpha) \Delta L/L$$

Derivation:

$$Y = AK^\alpha L^{1-\alpha}$$

$$\ln(Y) = \ln(A) + \alpha \ln(K) + (1 - \alpha) \ln(L)$$

$$\frac{d}{dt} \ln(Y) = \frac{d}{dt} (\ln(A) + \alpha \ln(K) + (1 - \alpha) \ln(L))$$

$$\frac{1}{Y} \frac{dY}{dt} = \dots$$

which is approximated in discrete time by:

$$\frac{\Delta Y}{Y} = \dots$$

Can estimate α and then back out TFP growth rates. Can simplify growth accounting to **labor productivity growth accounting**:

$$\Delta Y/Y = \Delta LF/LF + \Delta LP/LP$$

where Y = **Potential GDP**, LF is labor force and LP is labor productivity. This can be rewritten as

$$\Delta Y/Y = n + \frac{\theta}{1 - \alpha}$$

since labor productivity = $y = Y/L$, and in the steady state,

$$\Delta y/y = \frac{\theta}{1 - \alpha}$$

Note that

$$\frac{d}{dt}y = \frac{\Delta y}{y} = \frac{d}{dt}\frac{Y}{L} = \frac{\Delta Y}{Y} - \frac{\Delta L}{L} = \frac{\Delta Y}{Y} - n$$

So in the steady state,

$$\frac{\Delta y}{y} = \frac{\theta}{1 - \alpha}$$

becomes

$$\frac{\Delta Y}{Y} - n = \frac{\theta}{1 - \alpha} \rightarrow \frac{\Delta Y}{Y} = n + \frac{\theta}{1 - \alpha}$$

Capital deepening. With more investment, K/L increases but diminishing marginal returns and investors will only invest in K to MPK = marginal cost of $K = r$. Thereafter, can only increase by a rise in TFP (e.g. university research investment). However, once TFP increases, then Y rises, so MPK rises again, so K/L can increase again.

Natural resources. Only need access to natural resources, don't have to own according to observations. Also, if have lots of natural resources, there's a 'resource curse' if (1) don't have institutions to extract them and spread their gains through society, or (2) Dutch disease.

Labor supply. Labor force = working age population (employed + unemployed) available for work, depends on population growth, labor force participation, net migration, average hours worked. Human capital quality enhances growth. Labor force participation rate - risen in developed countries with women joining.

Physical capital and investment. Investment-to-GDP helps determine growth rate. For physical capital, ICT gives a bigger boost because of network externalities and because it's a 'general purpose technology' that affects many other

areas. Increasing technology is important for developed economies as this is the only way they can grow, whereas developing countries can grow cheaper because technological copying or imitation is cheaper. Note however that increasing technology will contribute to growth but has the disadvantage that it will displace workers - the double-edged nature of Schumpeterian creative destruction.

Theories of growth. Classical, neoclassical, endogenous.

Classical: Malthusian growth. Y depends on labor and fixed land. As $y=Y/L$ grows, population growth ACCELERATES after income per capita rises above subsistence, leading to diminishing labor returns, and therefore y returns back to subsistence. So, technological progress and land expansion will lead to higher population but not per capita wealth. Failed because (1) as y rose, population growth slowed, and (2) y grew as TFP rose faster and more than offset diminishing marginal returns.

Neoclassical. Solow model. Potential GDP is:

$$Y = AK^\alpha L^{1-\alpha}$$

$$y = Y/L = A(K/L)^\alpha = Ak^\alpha$$

Steady-state growth when $y=Y/L$ and $k=K/L$ grow at a steady rate. Take the rate of the above:

$$\Delta y/y = \Delta A/A + \alpha \Delta k/k$$

Decompose the definitions:

$$y = \frac{Y}{L}$$

$$\ln(y) = \ln(Y) - \ln(L)$$

$$\frac{d}{dt} \ln(y) = \frac{d}{dt} \ln(Y) - \frac{d}{dt} \ln(L)$$

$$\Delta y/y = \Delta Y/Y - \Delta L/L$$

Similarly, for k :

$$k = \frac{K}{L}$$

$$\ln(k) = \ln(K) - \ln(L)$$

$$\frac{d}{dt} \ln(k) = \frac{d}{dt} \ln(K) - \frac{d}{dt} \ln(L)$$

$$\Delta k/k = \Delta K/K - \Delta L/L$$

Since physical capital stock changes:

$$\Delta K = sY - \delta K$$

where δ is capital depreciation. Then plug into the above:

$$\Delta k/k = sY/K - \delta - \Delta L/L = sY/K - \delta - n$$

In the steady state, we must have (see the graph two graphs below):

$$\Delta k/k = \Delta y/y$$

Combine with $\Delta y/y = \Delta A/A + \alpha \Delta k/k$ above:

$$\Delta k/k = \Delta y/y = \Delta A/A + \alpha \Delta k/k$$

Rearrange:

$$\Delta A/A = \Delta k/k - \alpha \Delta k/k = \Delta k/k(1 - \alpha)$$

$$\Delta y/y = \Delta k/k = \frac{\Delta A/A}{1 - \alpha} = \frac{\theta}{1 - \alpha}$$

Growth rate of output per capita:

$$\Delta y/y = \Delta Y/Y - \Delta L/L = \Delta Y/Y - n = \frac{\theta}{1 - \alpha}$$

Growth rate of output:

$$\frac{\Delta Y}{Y} = \frac{\theta}{1 - \alpha} + n = \frac{\Delta K}{K}$$

Recall growth rate of potential output (GDP) calculated using the labor productivity growth equation; so, compare results with that. Combine some equations above:

$$\Delta y/y = \Delta k/k = sY/K - \delta - n$$

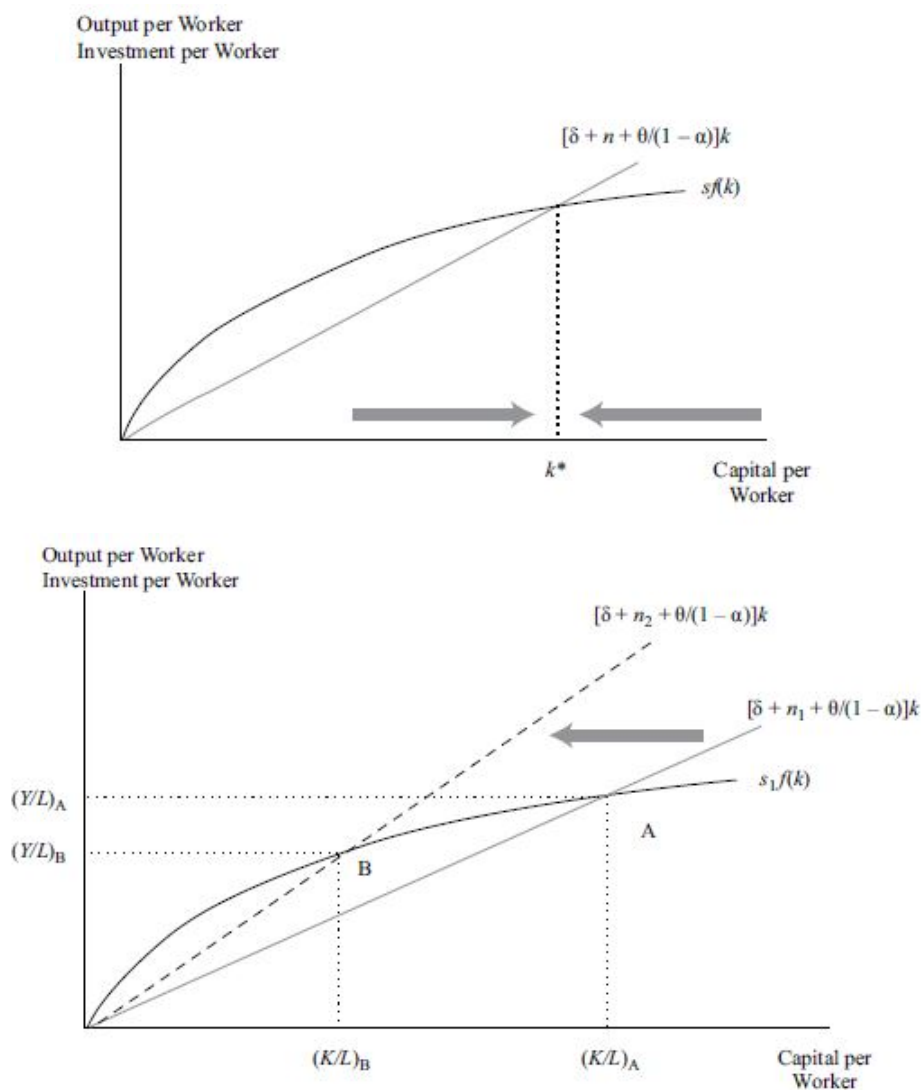
$$\Delta y/y = \Delta k/k = \frac{\theta}{1 - \alpha}$$

to get

$$\frac{Y}{K} = \frac{1}{s} \left[\left(\frac{\theta}{1 - \alpha} \right) + \delta + n \right] = \Psi$$

$$sy = \left[\left(\frac{\theta}{1 - \alpha} \right) + \delta + n \right] k$$

The output function $sf(k)$ intersects the investment line $\left[\delta + n + \frac{\theta}{1 - \alpha} \right] k$:



Both the $s f(k)$ and k^* are growing at rate $\frac{\theta}{1-\alpha}$, because that's what $\frac{\Delta k}{k} = \frac{\Delta y}{y}$ means. If s rises, then function shifts up, so y and k levels rise. If n , δ , θ rise, the investment line shifts left.

If $\text{TFP} = \theta$ rises, investment line shifts left so y and k FALLs, but this isn't what is actually happening: the entire graph is recalibrated for a given LEVEL of TFP. So, the previous y is now the same y on the new recalibrated graph, so that it looks like y falls but in fact it doesn't. **Growth rises unlike if δ and/or n rise.**

Note in the steady state, Y and K grow at the same rate g . This is because of constant returns to scale of the neoclassical production function.

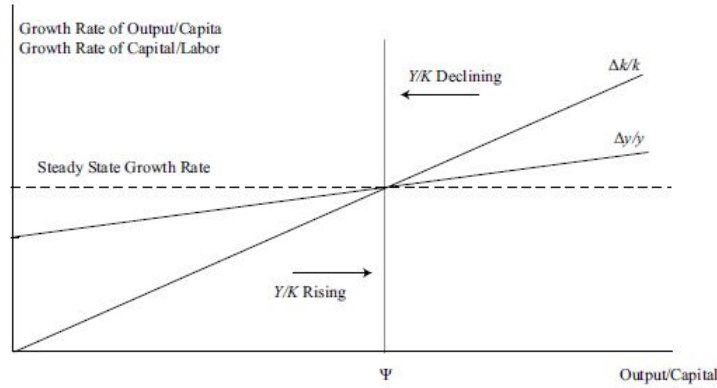
$$Y = AL^{1-\alpha} K^\alpha$$

Combine previous equations to get the NON steady state:

$$\frac{\Delta y}{y} = \left(\frac{\theta}{1-\alpha} \right) + \alpha s \left(\frac{Y}{K} - \Psi \right) = \left(\frac{\theta}{1-\alpha} \right) + \alpha s \left(\frac{y}{k} - \Psi \right) = g + \alpha s \left(\frac{y}{k} - \Psi \right)$$

$$\frac{\Delta k}{k} = \left(\frac{\theta}{1-\alpha} \right) + s \left(\frac{Y}{K} - \Psi \right) = \left(\frac{\theta}{1-\alpha} \right) + s \left(\frac{y}{k} - \Psi \right) = g + s \left(\frac{y}{k} - \Psi \right)$$

which corresponds to the steady state graph below. If $sy > Xk$, investment is very high (or TFP just rose), Y/K is high, so we are on the right-side of the steady state chart below. With $\Delta k/k > \Delta y/y$, capital grows faster than output, so Y/K falls, and we move left to the equilibrium at Ψ and steady state growth rate $\frac{\theta}{1-\alpha}$. The difference between the slopes is due to $\alpha < 1$.



$$Y = AK^\alpha L^{1-\alpha}$$

$$\frac{Y}{K} = AK^{\alpha-1} L^{1-\alpha}$$

$$\left(\frac{Y}{K} \right)^{\frac{1}{\alpha-1}} = A^{\frac{1}{\alpha-1}} \frac{K}{L} = A^{\frac{1}{\alpha-1}} k$$

$$\frac{k_{new}}{k_{old}} = \left[\frac{(Y/K)_{new}}{(Y/K)_{old}} \right]^{\frac{1}{\alpha-1}}$$

and

$$y = Ak^\alpha$$

$$\frac{y_{new}}{y_{old}} = \left(\frac{k_{new}}{k_{old}} \right)^\alpha$$

Note: **In the steady state, Y-K ratio is constant, but even in steady state, k and y are not constant because of TFP (= growth).** Neoclassical growth model has TFP being determined exogenously, and has diminishing returns to capital per worker. Augmented Solow model - incorporate human capital, research and development, infrastructure. But still diminishing returns.

Endogenous growth model. No diminishing marginal returns because e.g. R&D has positive externalities.

$$y = f(k) = ck$$

where c is the constant marginal product of capital, $c = y/k$. Plug into

$$\Delta k/k = sY/K - \delta - n$$

from above, to get:

$$\Delta y/y = \Delta k/k = sc - \delta - n$$

The RHS is constant, so this is both ST and LT equilibrium and growth rates. A higher s means permanently higher growth rate.

Neoclassical growth versus Endogenous growth models. If all private investment is subsidized, this just increases capital deepening. Subsidizing private research that provides positive externalities would convert the neoclassical model into the endogenous model.

In the endogenous growth model, there is no neoclassical

$$sy = \left(\frac{\theta}{1 - \alpha} + \delta + n \right) k$$

because the neoclassical

$$\frac{\Delta k}{k} = \frac{\Delta y}{y} = \frac{\theta}{1 - \alpha}$$

which comes from

$$\frac{\Delta y}{y} = \theta + \alpha \frac{\Delta k}{k}$$

which in turn comes from

$$y = Ak^\alpha$$

doesn't exist. Instead, the

$$y = ck$$

leads to

$$\ln(y) = \ln(c) + \ln(k)$$

$$\frac{\Delta y}{y} = \frac{\Delta k}{k}$$

So, instead of the neoclassical

$$\frac{\Delta y}{y} = \frac{\Delta k}{k} = sY/K - \delta - n$$

there is just

$$\Delta y/y = \Delta k/k = sc - \delta - n$$

Absolute convergence. Same per capita output regardless of characteristics. However, neoclassical model assumes same technology A, same per capita output GROWTH RATES converge at $\frac{\theta}{1-\alpha}$, but not their LEVELS.

Conditional convergence. Same s, n, f will lead to same per capita output levels under neoclassical growth. However, if these variables differ, neoclassical model says only convergence in growth rate at $\frac{\theta}{1-\alpha}$.

Club convergence. Only members of club (middle-income and rich) converge, these countries have good institutions. Lowest y/l members of club grow fastest. If not, they fall into non-convergence trap.

Trade. More competition, larger markets, more knowledge and information flows.

Neoclassical growth. With more open trade policy, generally developing countries will have capital inflows as they lack capital, and developed countries will have capital outflows. So developing countries will run trade deficits and grow faster than steady state while developed countries will run trade surpluses and grow slower than steady state.

Endogenous growth. Open trade policy will permanently raise growth rate: more competition, larger markets, more knowledge and information flows.

Trade and industrial policies. Inward policies (high tariffs, domestic substitution) work worse than outward policies (export-oriented and which also attracts FDI).

Note: (1) Growth in labor productivity is $\Delta y/y$, (2) Capital deepening is not just $\Delta k/k$; it's $\alpha \Delta k/k$, (3) Higher potential GDP growth, lower sovereign credit risk (and therefore higher rating), (4) Capital deepening occurs when increase in K-L ratio, (5) Compare potential GDP Y growth to steady state growth rate g.

3 Derivatives

3.1 Pricing and valuation of forward commitments

Carry arbitrage models - no-arbitrage where underlying instrument is bought or sold along with establishing a forward position. S_t is spot, F_t is forward, f_t is futures price (of new contracts at time t ; so a contract determined at $t=0$ is e.g. F_0), V_t and v_t are values of forward and futures contracts ($v_t = 0$ after mark-to-market settlement each day).

$$F_0 = FV(S_0)$$

Carry costs and convergence. If carry costs are higher than carry benefits, future and forward prices are higher than spot. They converge at maturation. Spot and futures prices differ because of basis and carry (benefit and costs) values.

Carry arbitrage, $F_0 > FV(S_0)$. If future or forward price is higher than carry-arbitrage model (which is just the predicted future or forward price being higher at the risk-free rate), then (1) sell forward, (2) borrow funds, (3) buy underlying. Arbitrageur pays interest cost.

Reverse carry arbitrage, $F_0 < FV(S_0)$. If future or forward is lower than carry-arbitrage model, then (1) buy forward, (2) sell underlying short, and (3) invest shorted funds at riskfree rate. Arbitrageur receives interest cost by lending.

Carry benefits CB_t and costs CC_t . CB_0 , CC_0 are PVs and CB_T , CC_T are FVs.

$$F_0 = FV(S_0 + CC_0 - CB_0)$$

With dividends

$$F_0 = FV(S_0 - PV(D))$$

For stock indices, many dividends, so there's a dividend index point which measures the amount of dividends. Use dividend yield which is often continuously compounded annually in

$$F_0 = S_0 e^{(r+CC-CB)t}$$

Note that r , CC , CB are continuously compounded RATES.

Forward rate agreements (FRA). 3 x 9 FRA is FRA settled in 3 months based on $T = 9$ maturity date. Advanced set, advanced settled. Initiation date $t = 0$, FRA expiration at $t = h$, underlying maturity date (for calculation) is $t = T = h + m$. So, to value at $t = h$, must discount back from $t = T = h + m$; need the discount rate for the FRA settlement, the MRR at $t = h$, and the predicted MRR at $t = h$ at $t = 0$ (i.e. the fixed FRA rate). Note the MRR at $t = h$ at $t = 0$ is an implied forward rate. Note2 the FRA value before

expiration at $t = h$ is calculated BACK to that point before expiration!

Settlement for advanced set, advanced settled. Settlement at $t = h$ for receive-floating (long on FRA contract for interest rates):

$$FRA_h = NA \times \frac{(L_m - FRA_0)t_m}{1 + D_m t_m}$$

where L_m is MRR spot rate (set at time = 0) for an m-day deposit. NA = notional amount. t_m is total accrual period, fraction of year for an m-day deposit. Since this is for receive-floating, FRA_0 is paid (outflow). D is discount factor.

To calculate the FRAs, e.g. based on MRRs for $t = h + m$ and $t = h$, this is just a forward rate:

$$FRA_0 = \frac{1 + MRR_{h+m}}{1 + MRR_h} - 1$$

and make sure to 'scale up' FRA_0 , e.g. if $t = m$ is 3 months, then scale up FRA_0 by 4x.

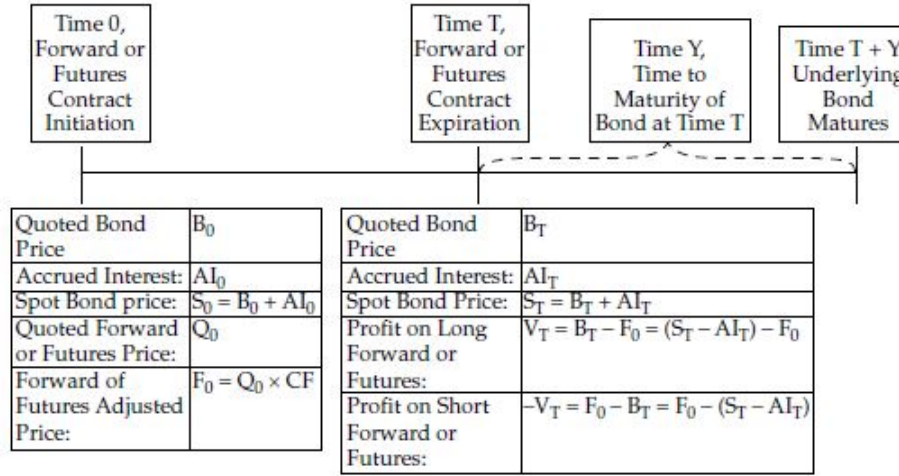
Note: when calculating forward rates, e.g. FRA 6 x 9, MRRs are quoted for annual, so need to divide to get 3 months, and then discount that back.

Fixed-income futures. Although it depends, generally the bond future value is clean, e.g. 150. If at bond future expiration, the underlying bond price is 150 (= dirty price), then the long position receives the underlying bond and pays the futures price of $150 + AI$. This is if the conversion factor = 1.

In a more sophisticated example, using CME bond futures. If a bond future value is 150, at expiration the short position (bond future seller) chooses the cheapest-to-deliver bond from a basket of eligible bonds based on rules of bond eligibility at the CME. The bond seller looks at the invoice price:

$$\text{Invoice price} = \text{Futures price} \times CF + AI$$

for each eligible bond, and selects the bond where the seller gains the most or loses the least. So, the futures price in the invoice price formula is a clean price, as AI is added. The futures price is always, regardless of bond, the futures price agreed in the contract. The buyer pays the invoice price to the seller. Though low coupons are associated with low CFs, the seller won't automatically deliver the bond with the lowest CF; they will also look at the market price as they need to buy the bond in the first place.



If bonds are quoted with the accrued interest,

$$F_0 = FV(S_0 + CC_0 - CB_0)$$

If bonds quoted without accrued interest,

$$S_0 = B_0 + AI_0$$

where B_0 is the quoted bond price. In either case, because any bonds can be used to meet the futures contract in theory - and a variety of bonds should be used to improve liquidity as individually specified bonds can have low liquidity - each bond in the exchange is standardized to, e.g. 6% YTM, using a conversion factor (CF) though this is not precise. This leads to using the cheapest bond to meet the futures bond is often used by the futures seller. The CF is then used to derive the quoted futures price based on the bond price at maturity, F_0 , i.e. $Q_0 = F_0/CF$. Note that this quoted price can be with or without AI depending on convention.

To value the fixed-income future, short at time $t = T$ if bought future at $t = 0$ and vice versa.

$$Q_0 = \frac{1}{CF}(FV(B_0 + AI_{0,t}) - AI_{0,T} - FV(CI))$$

where FV is future value, CI is coupon interest, and the AI is accrued interest. Since Q_0 , the quoted futures contract price, subtracts AI_T , it is a 'clean' price. AI_T 's T is the time from the last interest payment (even if it was before the current time) to futures contract expiry. The AI_0 's 0 refers to the time until now from the time of last coupon or interest payment.

When calculating, we often just use quoted bond prices (i.e. without AI) and

forward and quoted prices (i.e. without AI) and then make adjustments for AI later.

Swap contracts. Swap rate is the fixed rate determined at the beginning of the contract.

Interest-rate swaps. Use bonds to replicate, though can use FRAs, options (e.g. long call and short put) and other ways. Can use swap to turn a fixed (floating) bond synthetically into a floating (fixed) bond. The value of such a floating rate bond is equal to par 1 which will be set to 1 because the MRR (coupon) will be the same as the discount rate for each period:

$$V_{flt} = par = 1$$

The fixed bond with par value 1 and coupon rate discounted against par 1 by c :

$$V_{fix} = c \sum_{i=1}^n PV_i(1) + PV_n(1)$$

and since

$$V_{fix} = V_{flt} = 1$$

and

$$c = r_{fix} \times AP$$

where the AP is the accrual period. So,

$$V_{fix} = c \sum_{i=1}^n PV_i(1) + PV_n(1) = 1$$

$$r_{fix} = \frac{1 - PV_n(1)}{\sum PV_i(1)} \times \frac{1}{AP}$$

To value a swap, buy/sell fixed/floating to reverse the original swap. The swap value is

$$V_{swap} = NA \times AP \times (r_{fix} - r_{flt}) \times \sum PV_i$$

where PV is calculated from $t = t$.

Currency swaps. There is a notional amount (NA) swapped at the beginning, then each leg is in different currencies, with each leg being fixed or floating. But for calculations, assume fixed-on-fixed currency swaps. The NA is first swapped:

$$V_a = S_0 V_b$$

So that the value of the currency swap is

$$V_{CS} = V_a - S_0 V_b$$

which is = 0 at initiation. Then use the r_{fix} from above to calculate r_a, r_b separately based on their own PV schedules.

Calculating PV factors. When calculating PV factors for anything other than a year, calculate based on the appropriate AP. For days:

Days to Maturity	A\$ Spot Interest Rates (%)	Present Value (A\$1)	US\$ Spot Interest Rates (%)	Present Value (US\$1)
90	2.50	0.993789 ^a	0.10	0.999750
180	2.60	0.987167	0.15	0.999251 ^b
270	2.70	0.980152	0.20	0.998502
360	2.80	0.972763	0.25	0.997506
	Sum:	3.933870	Sum:	3.995009

^a $A\$0.993789 = 1/[1 + 0.0250(90/360)]$.
^b $US\$0.999251 = 1/[1 + 0.00150(180/360)]$.

For half-years:

Date (Years)	Present Value Factors (PV)	Fixed Cash Flow	PV (Fixed Cash Flow)
0.5	0.994036	150,000	149,105
1.5*	0.982318	150,000	147,348
2.5	0.970874	150,000	145,631
3.5	0.959693	150,000	143,954
4.5	0.948767	10,150,000	9,629,981
		Total:	10,216,019

* Answers may differ due to rounding: $PV(1.5) = 1/(1 + 3 \times (0.012/2)) = 0.982318$.

Currency swap valuation.

$$V_{CS} = V_a - S_t V_b$$

$$V_{CS} = (NA_a(AP_0 \times r_{a,0} \sum PV_i(1) + PV_n(1)) - S_t NA_b(AP_0 \times r_{b,0} \sum PV_i(1) + PV_n(1)))$$

where AP and r are as at **initiation** $t = 0$, and PV's are **new schedules at** $t = t$. To get floating-on-fixed currency swaps, combine fixed-on-fixed currency swap + interest-rate swap.

Notes re currency swaps. (1) The dealer gets the currency that it paid NA originally in. (2) Even if the FX rate changes, the original NAs in both currencies are returned at the end without accounting for FX rate changes. (3) Fixed

currency swap rates - calculate independently of other fixed or floating rates.

Equity swaps. Types:

- Equity-fixed
- Equity-floating
- Equity-another equity, which is like adding (1) receive-equity, pay-fixed + (2) pay-equity, receive-fixed interest rate swap

Equity returns can be with or without dividends. For simplicity, assume all dividends are re-invested. Equity swaps can use equity, equity indices, or equity portfolio.

Valuation. Receive-fixed, pay-equity swap V_{EQ} . Replicate by (1) buying NA of equity, and then sell it at the end; (2) short fixed-rate bond V_{FIX} at par. The net cash flow is

$$V_{EQ} = V_{FIX} - NA - PV(Par - NA)$$

because intermediate fixed and equity cash flows cancel out. Assuming

$$V_{EQ} = 0$$

$$NA = Par = 1$$

then

$$r_{FIX} = \frac{1 - PV_n(1)}{\sum PV_i(1)} \times \frac{1}{AP}$$

as for fixed-floating swaps. To value at $t = t$,

$$V_{EQ,t} = V_{FIX,t} - \frac{S_t}{S_{t-1}} NA - PV(Par - NA)$$

where $V_{FIX,t}$ is based on a new PV schedule for time t , **using the old coupon rate but new discount spot rates**. The S_t/S_{t-1} is because have to pay back more if the equity price rises. Generally assume bond par value = NA.

For V_{FIX} , calculate PV factors, fixed cash flow (= original fixed rate x NA), and multiply PV factors x fixed cash flow to get PV and total the PVs.

Date (in years)	PV Factors	Fixed Cash Flow	PV (fixed cash flow)
0.5	0.998004 or $1/[1 + 0.0040(180/360)]$	\$400,000	\$399,202
1.5	0.985222 or $1/[1 + 0.0100(540/360)]$	\$400,000	\$394,089
2.5	0.970874 or $1/[1 + 0.0120(900/360)]$	\$400,000	\$388,350
3.5	0.934579 or $1/[1 + 0.0200(1,260/360)]$	\$400,000	\$373,832
4.5	0.895255 or $1/[1 + 0.0260(1,620/360)]$	\$20,400,000	\$18,263,205
Total			\$19,818,678

There are 5 concepts in this chapter:

- Carry arbitrage.
- Quoted futures Q_0 and quoted bond futures B_0 , with $Q_0 = \frac{1}{CF}((1 + r)^t(B_0 + AI_0) - AI_T - FVCI)$, s.t. $FV(A_0) \neq A_T$ though it should be. $A_T = \frac{T}{\text{Time to next payment even if after futures expiration}} \times \text{Payment}$. Note F_0 is the forward or futures price, $Q_0 = F_0/CF$ is the QUOTED contract forward or futures price; B_0 is the quoted bond (clean) price. When calculating arbitrage profit, find the differences in the forward or futures price, not the quoted price.
- Calculating the swap rate $r_{FIX} = \frac{1 - PV_T(1)}{\sum PV_i(1)} \times \frac{1}{AP}$. This comes from normalizing the coupons = par = 1.
- Note that forward or futures prices as par, e.g. 130 (130 percent of par=100), where par is based on the NA.
- Currency swap values $V_{CS} = NA_a(AP \times r_{FIX,a,0} \sum PV_i(1) + PV_T(1)) - S_t NA_b(AP \times r_{FIX,b,0} \sum PV_i(1) + PV_T(1))$.
- Equity swap values $V_{EQ,t} = V_{FIX,0} - (S_t/S_{t-1})NA - PV(Par - NA)$ where $V_{FIX,0}$ is the value with (1) old coupon rate, (2) new discount rate, and (3) new PV schedule at $t=t$: $V_{FIX,0} = NA(r_0 \sum_t^T PV_i(1) + PV_T(i))$.

3.2 Valuation of contingent claims

$$u = \frac{S^+}{S}$$

$$d = \frac{S^-}{S}$$

One-period binomial model: European options. In a **no-arbitrage approach**,

$$\begin{aligned} -c^+ + hS^+ &= -c^- + hS^- \\ h &= \frac{c^+ - c^-}{S^+ - S^-} \\ +c - hS - PV(-hS^- + c^-) &= 0 \\ +c - hS - PV(-hS^+ + c^+) &= 0 \end{aligned}$$

where c is a call option value and h is the hedge ratio. The h is essentially a delta: it is a measure of sensitivity of the call option price to the underlying asset price. If selling a call option, hedge by buying the underlying asset. If buying a call option, hedge by selling the underlying. These should equal each other; otherwise, arbitrage. Moreover, their PVs will be $+c - hS$, again otherwise there is arbitrage. So, can buy hS and partially finance by borrowing $PV(..)$ for a long call:

$$c = hS + PV(-hS^- + c^-)$$

If the call option is overpriced, sell (write) the call, buy S and borrow the $PV()$ for an arbitrage opportunity:

$$\begin{aligned} c &> hS + PV(-c^- + hS^-) \\ c - hS - PV(hS^- - c^-) &> 0 \end{aligned}$$

where c is positive, so it's a sell, S is negative so it's a buy, and PV is negative so it's buying in the future i.e. borrow.

For a long put, sell short hS and lend $PV(...)$:

$$\begin{aligned} h &= \frac{p^+ - p^-}{S^+ - S^-} < 0 \\ p &= hS + PV(-hS^- + p^-) \end{aligned}$$

Note

$$h_p = \frac{p^+ - p^-}{S^+ - S^-} \neq h_c = \frac{c^+ - c^-}{S^+ - S^-}$$

Instead of the no-arbitrage approach, can use the **expectations approach**: based on **risk-neutral probability**.

$$S_0 = S = \frac{S^+ \pi + S^- (1 - \pi)}{1 + r}$$

Re-arranging,

$$\begin{aligned} \pi &= \frac{(1 + r) - d}{u - d} \\ c &= \frac{\pi c^+ + (1 - \pi) c^-}{1 + r} \end{aligned}$$

$$p = \frac{\pi p^+ + (1 - \pi)p^-}{1 + r}$$

Or use put-call parity:

$$p + S = \frac{X}{1 + r} + c$$

Note that if $S = \frac{X}{1+r}$, then $p = c$.

Two-period binomial model. European options. To build the two-period binomial tree, use u and d to determine the final 3 underlying asset values. Then find $c^{++}, c^{+-} = c^{-+}, c^{--}$ and then work backward through the tree to find c^+, c^- and finally, c . Note the intermediate value, $c^+ > uS - X$ and

$$c^{++} = \max(0, S^{++} - X) = \max(0, u^2 S - X)$$

$$c^{+-} = \max(0, udS - X)$$

Assume recombining binomial lattice: $S^{+-} = S^{-+}$.

To calculate c, first calculate risk-neutral probability for up move THROUGHOUT the binomial tree:

$$1. \pi = \frac{(1+r)-d}{u-d}$$

$$2. c^+ = \frac{\pi c^{++} + (1-\pi)c^{+-}}{1+r}$$

$$3. c = \frac{\pi c^+ + (1-\pi)c^-}{1+r}$$

Moreover,

$$h^+ = \frac{c^{++} - c^{+-}}{S^{++} - S^{+-}}$$

$$h = \frac{c^+ - c^-}{S^+ - S^-}$$

Using the expectations approach: use repeated substitutions and assume recombining binomial lattice to get the following.

$$c = \frac{\pi^2 c^{++} + 2\pi(1-\pi)c^{+-} + (1-\pi)^2 c^{--}}{(1+r)^2}$$

$$p = \frac{\pi^2 p^{++} + 2\pi(1-\pi)p^{+-} + (1-\pi)^2 p^{--}}{(1+r)^2}$$

and therefore that

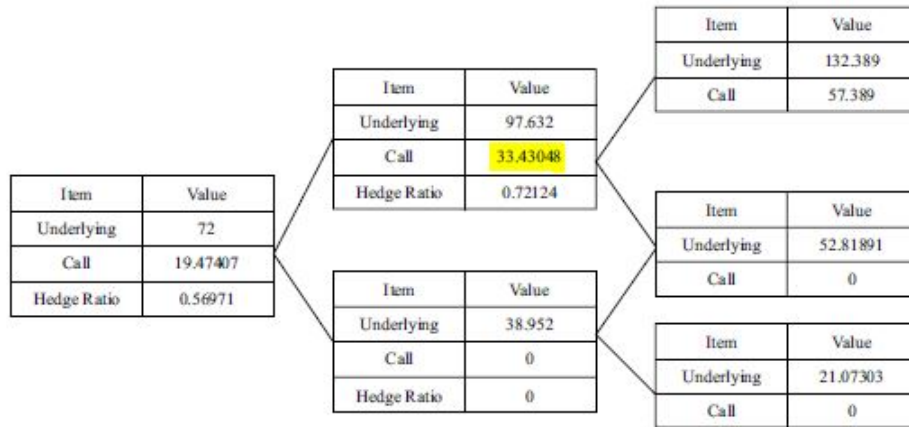
$$E(c_2) = \pi^2 c^{++} + 2\pi(1-\pi)c^{+-} + (1-\pi)^2 c^{--}$$

$$E(p_2) = \pi^2 p^{++} + 2\pi(1-\pi)p^{+-} + (1-\pi)^2 p^{--}$$

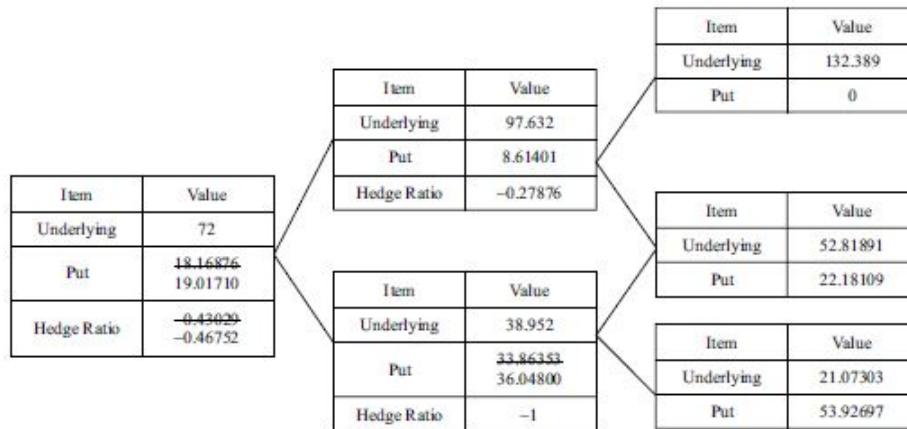
In the two-period binomial model, put-call parity is over two periods:

$$p + S = c + \frac{X}{(1+r)^2}$$

Two-period model: American options. Note that in the intermediate step, in the 'up' position, S^{++} and S^{+-} are much 'higher' than the strike price X , so $c^+ > uS - X$ and therefore a long American call option wouldn't be exercised - it's worth more than the exercise value $uS - X$. For instance, with $X = 75$, the call option value highlighted below is higher than the exercise price ($97.632 - X$). So call option value is not affected by changing from European to American options.



However, in the 'down' position of the intermediate step, S^{--} , S^{-+} are much 'lower' than the strike price X , so $p^- < X - uS$ so a long American put would be exercised. With the new p^- , recalculate p and the hedge ratio h . See put option example below. So, put option values are affected and need to be adjusted when changing from European to American options.



Intuitively, a call option has unlimited upside so it's better to hold it, while a put option has limited upside so it is often better to sell it and make interest off of the proceeds.

Note in American options that the binomial table requires some BSM modelling of values.

Dividends in the binomial model. Escrow method: dividends perfectly predictable and reduce the value of option. Security price without dividends:

$$\hat{S} = S - \gamma$$

and at expiration

$$\hat{S}_T = S_T$$

Most option contracts don't provide protection against dividends. With dividend stocks, often advantageous to exercise BEFORE the ex-dividend date for American options. That's one reason why American options can be worth more than European options. Create the binomial model as before, and then adjust for dividends, particularly look for **early exercise opportunities for American options** where the stock **still includes the dividends**, i.e. before the dividend is issued.

Interest rate options. From the bond value, get the interest rate. (1) The strike (exercise) price is based on the interest rate (not the bond price) and (2) the discount rates are based on the current spot interest rate. This is different from standard 2-period binomial model. Calculate $c^+ = PV(\pi c^{++} + (1-\pi)c^{+-})$, etc. where the PV is based on the BOND VALUE, e.g. 0.96, for c^+ .

BSM model. Extend multiperiod model to infinitesimally continuous time. Bachelier uses normally-distributed stock price returns, but that gives negative stock prices too; so, assume stock price returns are lognormal. This leads to

prices also being lognormal.

$$R = \ln \left(\frac{S_T}{S_t} \right)$$

$$S_T = S_t e^R$$

Since R is normally distributed, and e^R is an exponential function of a normally distributed variable, it means that e^R is lognormally distributed. Since S_T is simply S_t multiplied by a lognormally distributed variable, S_T is also lognormal.

Use geometric brownian motion as the stochastic process. Then, growth rate of stock price μ = annual geometrically compounded rate. Moreover, volatility, σ = standard deviation of log returns = annualized standard deviation of continuously compounded percentage changes in the underlying price. High volatilities for higher prices. BSM model assumes constant μ, σ .

BSM model: European options. No dividends. At expiration, the expected values are:

$$E(c_T) = S e^{rT} N(d_1) - X N(d_2)$$

$$E(p_T) = X N(-d_2) - S e^{rT} N(-d_1)$$

The present values then are just divided by e^{rT} :

$$c = S N(d_1) - e^{-rT} X N(d_2)$$

$$p = e^{-rT} X N(-d_2) - S N(-d_1)$$

where

$$d_1 = \frac{\ln(S/X) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

The d_1 term is basically $= \sigma\sqrt{T}$ + idiosyncratic drift. The d_2 isolates the idiosyncratic drift. So, $N(d_2)$ **is the idiosyncratic drift, and it is also the probability that the call option expires in-the-money**, and $1 - N(d_2) = N(-d_2)$ **is the probability the put expires in-the-money** (also, it must be that $N(-d_2)$ is the probability that the put option expires in-the-money because of the standard normal CDF's rotational symmetry). $N()$ is the standard normal CDF.

Further, d_2 is a Z-score for the strike price at expiration. It essentially measures how many standard deviations the strike price is away from the expected future stock price $S_0 e^{rT}$ in a risk-neutral world, after accounting for the volatility. $N(d_2)$ then converts that into a probability. If you exercise a call, you pay X . So, $X N(d_2)$ is the expected value of paying the strike price.

Option replication strategy prices are based on the security price and bond price, i.e. leveraged security purchase or lending by security short-sales:

$$\text{Replication cost} = n_S S + n_B B$$

where $n_S = N(d_1) > 0$ for calls and number of bonds is $n_B = -N(d_2) < 0$ for calls. Note as S increases, $N(d_1)$ increases too because d_1 increases. So, replicating (for call options) means continuously buying shares in a rising market and continuously selling in a falling market.

Options typically cost more than BSM theory: (1) transaction costs, and (2) discontinuities introduce more volatility, (3) volatility not known in advance.

BSM model: Carry benefits and costs. Increase in carry benefit reduces call option value and increases put option value, and vice versa. Carry benefit (e.g. dividends and bond coupons, foreign interest rates; carry costs = negative carry benefits = storage and insurance costs) is made continuous to continuous yield γ . Just replace S with $Se^{-\gamma T}$ which means $E(c) = Se^{rT}e^{-\gamma T} = S^{(r-\gamma)T}$ so replace r with $r - \gamma$:

$$\begin{aligned} c &= Se^{-\gamma T} N(d_1) - e^{-rT} X N(d_2) \\ p &= e^{-rT} X N(-d_2) - Se^{-\gamma T} N(-d_1) \end{aligned}$$

where

$$\begin{aligned} d_1 &= \frac{\ln(S/X) + (r - \gamma + \sigma^2/2)T}{\sigma\sqrt{T}} \\ d_2 &= d_1 - \sigma\sqrt{T} \\ \gamma &= \delta \end{aligned}$$

where δ is the continuously compounded dividend yield. Put-call parity in continuous time with carry:

$$p + Se^{-\gamma T} = c + e^{-rT} X$$

So, dividends influence the dynamically managed portfolio by lowering the number of shares to buy for calls and raising the number of shares to short sell for puts.

BSM model: FX options. S = spot FX rate (e.g. 135JPY/EUR), X = an FX rate (JPY/EUR). Risk-free rate is of HOME country, e.g. JPY here, and carry rate is the FOREIGN rate, e.g. EUR here, continuously compounded. The currencies in the equations are in the domestic currency (e.g. JPY in the example). The volatility σ is volatility of log return of FX spot rate. BSM model for call options = foreign exchange component minus bond component and reverse for put options. Just replace γ above with r^f :

$$c = e^{-r^f T} S N(d_1) - e^{-r^d T} X N(d_2)$$

$$p = e^{-r^d T} X N(-d_2) - e^{-r^f T} S N(-d_1)$$

$$d_1 = \frac{\ln(S/X) + (r - r^f + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Above, use the home currency. So, FX rate = (home currency) / (foreign currency).

BSM model: Futures options. Ignore margin requirements and mark-to-market. Just replace S with $e^{-rT} F_0(T)$ and γ above with another r :

$$c = e^{-rT} F_0(T) N(d_1) - e^{-rT} X N(d_2)$$

$$p = e^{-rT} X N(-d_2) - e^{-rT} F_0(T) N(-d_1)$$

where

$$d_1 = \frac{\ln(F_0(T)/X) + (0(=r-r) + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Note d_1 is missing interest rate r because futures don't need to account for the cost of carry, unlike spot security prices (i.e. long securities). Put-call parity:

$$c = e^{-rT} (F_0(T) - X) + p$$

Interest rate options. The underlying is an interest rate based on FRA. Though advanced set (t_{j-1}) , settled in arrears $(t_{j-1} + t_m)$, so must discount back from $t_{j-1} + t_m$. FRA is a rate, e.g. $FRA(0, 0.25, 0.5) = 2\%$ means 2% set a $T = 0$ paid in $0.25 + 0.5 = 0.75$ years. For notional amount $NA = 1$:

$$c = (AP)e^{-r(t_{j-1}+t_m)} (FRA(0, t_{j-1}, t_m) N(d_1) - R_X N(d_2))$$

$$p = (AP)e^{-r(t_{j-1}+t_m)} (R_X N(-d_2) - FRA(0, t_{j-1}, t_m) N(-d_1))$$

$$d_1 = \frac{\ln(FRA(0, t_{j-1}, t_m)/R_X) + (\sigma^2/2)t_{j-1}}{\sigma\sqrt{t_{j-1}}}$$

$$d_2 = d_1 - \sigma\sqrt{t_{j-1}}$$

Notes.

- R_X is the exercise rate. This is an option, after all.
- If $X = \text{FRA rate}$, long an interest rate call + short an interest rate put = receive-floating, pay-fixed FRA. The X makes it fixed-paying. Conversely, long an interest rate put + short an interest rate call = received-fixed, pay-floating FRA.
- Interest-rate cap = portfolio of interest rate call options (caplets). Use this to hedge floating-rate payments.

- Interest-rate floor = portfolio of interest rate puts (floorlets), analogous to interest-rate caps.
- Apply put-call parity. Long an interest rate cap + put interest rate floor with X = swap rate = receive-floating, pay-fixed swap. Converse is true.

Swaptions. Option for a swap - holder can enter at swap at pre-agreed exercise swap rate. Payer swaption buyer = swaption to pay fixed, receive floating. Receiver swaption buyer = receive fixed, pay floating.

Swaption Black model. For swaps. Since swaps determined periodically, the PVAs (PV of annuities) are in discrete, not continuous, time (which is the discount factor summed; it is summed because unlike the previous options, swaptions are a series of payments). The payer swaption buyer wants R_{FIX} to rise above R_X so that they can exercise the option, and once they do, they can enter an offsetting swap where they receive fixed (now at a higher R_{FIX}) and pay floating. The difference is then an annuity:

$$PVA = \sum_{j=1}^n PV_{0,t_j}(1)$$

For a payer swaption, for $NA = 1$:

$$PAY_{SWN} = (AP)PVA(R_{FIX}N(d_1) - R_XN(d_2))$$

$$REC_{SWN} = (AP)PVA(R_XN(-d_2) - R_{FIX}N(-d_1))$$

where

$$d_1 = \frac{\ln(R_{FIX}/R_X) + (\sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

where there is a swap component and a bond component. Notes

- Long receiver swaption + short payer swaption at same X = receive-floating, pay-fixed forward swap.
- Long payer swaption + short receiver swaption at same X = receive-fixed, pay-floating forward swap.

Option Greeks. Delta. Dividend yield (continuous) = δ . Change in call or put price for a small change in value of underlying.

$$Delta_c = e^{-\delta T}N(d_1) < 1$$

$$Delta_p = -e^{-\delta T}N(-d_1) > -1$$

The more of the share of the underlying N , the higher the Delta because a \$1 change in the underlying leads to a greater change in the call and put values. However, $Delta_c < 1$ because it reflects the 'zero-downside' of a call option: a

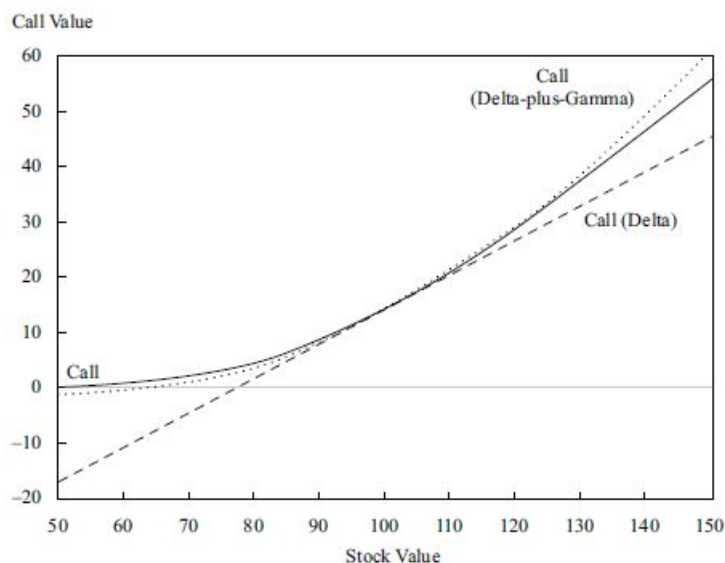
rise of \$1 in the underlying doesn't lead to a +1 value in the call option ('the delta of a stock is always 1'). As N changes through time, Deltas are also changing through time.

Deltas interpretation. Delta of long one stock is +1, short one stock is -1. This means if the stock price rises by +1, the Delta **on the stock, not the call option** = +1. Deltas are also approximate changes in call and put values based on changes in underlying, where the relationship is linear:

$$\hat{c} \approx c + Delta_c(\hat{S} - S)$$

$$\hat{p} \approx p + Delta_p(\hat{S} - S)$$

However, there is a 'gamma short' because of the call and put price convexity. Below is for call values and Delta estimated call values $S = 100 = X, r = 0.05, \sigma = 0.3, \delta = 0$.



Delta hedging. Make sum of option deltas = 0 (delta neutral), so that the portfolio position is immune to small changes in the underlying price using N_H units of a hedging instrument at Δ_H :

$$PortfolioDelta + N_H Delta_H = 0$$

$$N_H = -\frac{PortfolioDelta}{Delta_H}$$

Note: Delta of stock itself is 1, and gamma of stock itself is 0. However, delta of call option on stock is not 1, and neither is the gamma.

Gamma. Change in call or put delta for a change in underlying value. Curvature of call or put value when underlying price changes. **Gamma of long or short one stock is simply 0** because the stock always moves 1-for-1 with itself (delta is always +1 for long position and -1 for short position). Gamma values highest near the money (even though distance from Delta prediction is far when deeply in or out of the money).

$$Gamma_c = Gamma_p = \frac{e^{-\gamma T} n(d_1)}{S\sigma\sqrt{T}} > 0$$

where n = standard normal PDF (not CDF). Gamma of put and call are equal because of put-call parity:

$$c - p = S_0 - e^{-rT}X$$

since neither S_0 nor $e^{-rT}X$ depend on delta, the right hand side of put-call parity above has a delta = 1.

$$\hat{c} \approx c + Delta_c(\hat{S} - S) + \frac{Gamma_c}{2}(\hat{S} - S)^2$$

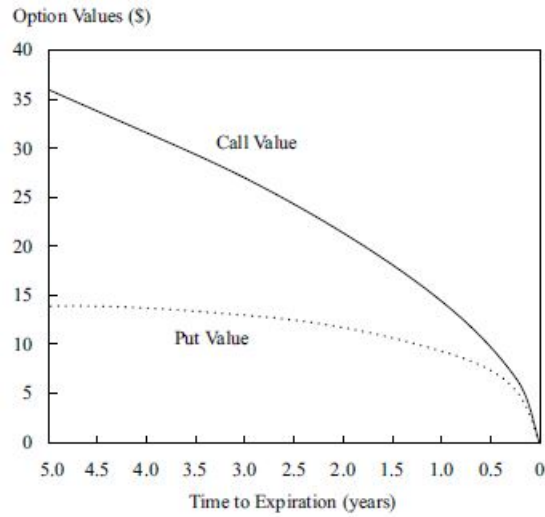
$$\hat{p} \approx p + Delta_p(\hat{S} - S) + \frac{Gamma_c}{2}(\hat{S} - S)^2$$

Delta + gamma approximation is biased low for a fall in the underlying price but biased up for underlying price rise.

Gamma neutrality. Portfolio gamma = 0. (1) Reduce gamma, (2) neutralize delta. Neutralize gamma first because delta can be neutralized by simply using long or short stocks, because **stocks have 0 gamma**.

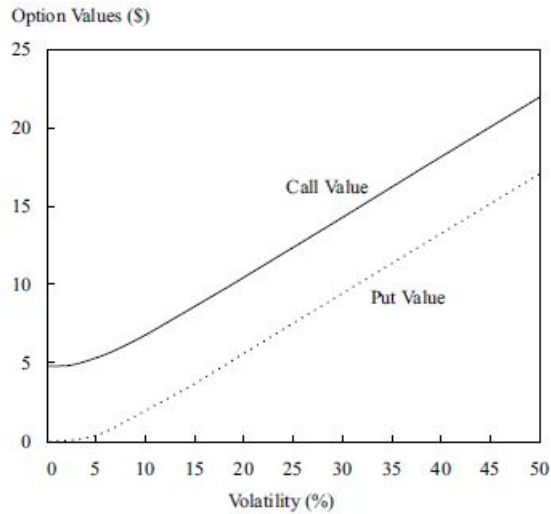
Gamma risk. Sometimes stock moves are discontinuous jumps, which leads to 'gamma risk', i.e. gamma higher than theory.

Theta. Change in option value per calendar time. Only declines with time (time decay), so option theta < 0. Call and put thetas decline faster closer to expiration. Stocks don't expire so their theta = 0, and Theta can't be adjusted with stock trades.



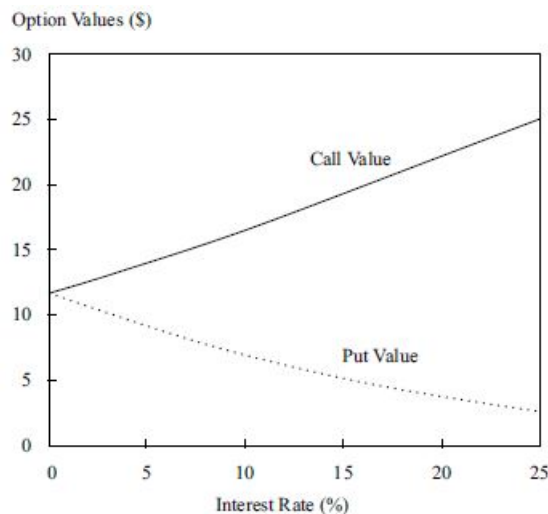
Above is option values and time to expiration $S = 100 = X, r = 0.05, \sigma = 0.3, \delta = 0$.

Vega. Change in option value given small change in σ . $\text{Vega} > 0$. A rise in Vega leads to a rise in option value. Though historical volatility can be estimated, future volatilities (which are assumed constant in BSM) require guesses. Vega is very sensitive and influential in portfolios. Vega of call and put are same because RHS ($S_0 - e^{-rT}X$ of put-call parity is not volatile (so, RHS Vega = 0).



Above option values and volatility $S = 100 = X, r = 0.05, T = 1, \delta = 0$.

Rho. Change in option value per change in r_f . Rho is positive for a call. Rho is negative for a put; investment in put option delays opportunity to earn interest on proceeds from sale.



Above option values and interest rates, $S = 100 = X, r = 0.05, T = 1, \delta = 0$.

Implied volatility. Invert BSM and use observed prices to infer implied volatility. BSM itself assumes constant volatility, but prices change, changing implied volatilities. Implied volatilities can also change across time (**term structure of volatility**) and X (**volatility smile or skew**), which can result in volatility surfaces. Often, compare implied volatilities rather than call or put options which depend on time, X , interest rate, etc.

4 Fixed Income

4.1 The term structure and interest rate dynamics

Forward rate, $f_{A,B-A}$. Forward price is the bond price at time $t = A$. Forward rate is

$$\left(\frac{1 + z_B}{1 + z_A} \right)^{\frac{A}{B-A}} (1 + z_B) = 1 + f_{A,B-A}$$

Forward rate is above spot rate if spot rate rises.

Par curve. YTM on coupon-paying government bonds priced at par. The coupon rate to achieve par (usually = 1). Use **bootstrapping** to calculate.

Bootstrapping: derive spot rates from par rates. Example: 1y par rate = 0.05,

2y par rate = 0.0597, 3y par rate = 0.0691. 1y spot is 0.05. 2y spot is derived from:

$$1 = \frac{0.0597}{1.05} + \frac{1 + 0.0597}{(1 + z_2)^2}$$

and so forth.

Rising yield curve. Rising or at least stable future inflation and risk premiums for longer time.

Falling yield curve. Expectation of declining future inflation rates, economic slowdown.

YTM. Do Example 5 on page 14.

If bond trader expects forward curve to hold (i.e. spot rate in future will follow the forward curve), then will get a standard rate, e.g. if flat yield curve of 8% with 3y bond with 8% annual coupon, then trader gets 8% return. If trader expects forward curve (i.e. future spot rates) will be higher than current forward curve, then will get lower than 8%; if trader expects lower than market, then will get higher than 8% because of discounting.

Active bond portfolio management. **Rolling down the yield curve.** If the yield curve is upward sloping (so the par rates are rising), then can benefit from buying longer maturities and selling them early - because (1) the par rate (i.e. coupon rate) will be higher, and (2) capital gain will be higher (because coupon rates will be higher than discount rates, because of the upward-sloping yield curve - note the yield curve is 'constantly moving', e.g. 2y rate will be the 2y rate in 5 yrs). In other words, valued at higher prices but with lower discount rates. Example: buying a 10y bond with 10% coupon and selling it at 2y is better than buying a 2y bond with 2% coupon to maturity. To calculating rolling or riding the yield curve: Return = coupon + **reinvested coupons** + capital gain.

Maturity spread carry trade. Borrow short-term and lend long-term by buying bonds (rolling down yield curve) - popular before GFC when yield curves were rising upward.

Swap rate. Based on par swaps, where the contract value is 0 at initiation. Swaps can be used to convert fixed to and from floating.

See example 8. Remember the last DF needs to be considered twice - once for the final coupon, and once for the terminal FV.

Swap spread is premium over spot yield - spread paid by fixed-rate payer in interest rate swap over government rate of same maturity. Swap spread is for short-term risky debt. I-spread is bond rate over swap rate. Swap rate is higher than government spot yield because of higher default risk. After GFC, higher

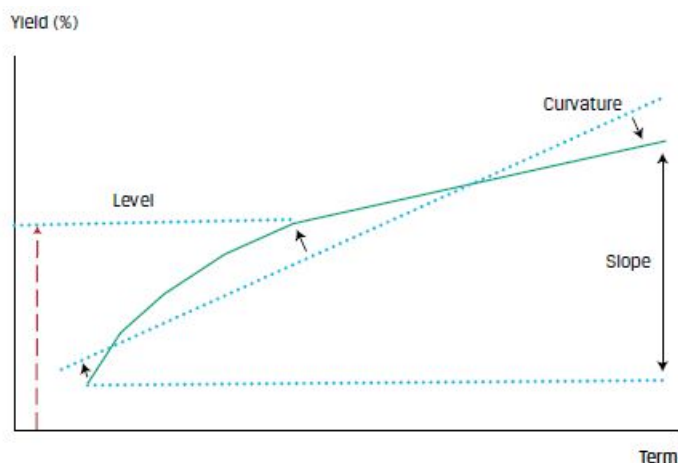
capital requirements for banks and swap dealers (e.g. Basel III) have led to banks having less capital available for swaps, so this has resulted in negative swap spreads (they won't arbitrage negative swap spreads - this arbitrage requires buying government bonds and then entering pay-fixed (low because negative swap rates) and receive-floating (high), but this consumes lots of capital).

Term structure theories.

- Expectations theory (assumes risk neutrality) - unbiased/pure expectations theory that the forward rate is an unbiased predictor of the future spot rate; local expectations theory - short-term returns are r_f . Expectations theory require risk-neutrality and ignore other factors.
- Liquidity preference theory. Long-term rates are higher because of liquidity issues.
- Segmented markets theory. Lenders and borrowers determine supply and demand of maturities and therefore term structure.
- Preferred habitats theory. Segmented theory but lenders and borrowers will move out of habitats if rates are good.

In segmented market theory, bond market participants are limited to purchase of maturities that match the timing of their liabilities. In the preferred habitat theory, participants have a preferred maturity for asset purchases, but they may deviate from it if they feel returns in other maturities offer sufficient compensation for leaving their preferred maturity segment.

Yield curve factor models. Shaping risk - risk that yield curve will change. Level, steepness, curvature are the variables in the model. Level explains most, then steepness, then curvature.



Yield volatility. **Interest rate volatility.** Lognormal interest rate:

$$\sigma = \frac{\sigma(\Delta r/r)}{\sqrt{\Delta t}}$$

$$\Delta t = \frac{1}{\text{No. of months}}$$

where t is now, T is maturity. The Δt is an annualizing factor. Short-term volatility from monetary policy; long-term volatility from real economy and inflation.

Managing yield curve risks using key rate durations. Note the negative below - an increase in the interest rate z reduces the price.

$$KeyDur = \% \Delta P \approx -KeyDur_a \Delta z_a - KeyDur_b \Delta z_b - KeyDur_c \Delta z_c$$

where a, b, c are different duration bonds. To calculate KeyDur, move the rate by e.g. 100 bps and see what the sensitivity shift is for the portfolio.

$$KeyDur \approx -KeyDur_L \Delta x_L - KeyDur_S \Delta x_S - KeyDur_C \Delta x_C$$

for the level (L), steepness (S), and curvature (C) factors. Note implicitly $\Delta = 100bps$

$$KeyDur_L = KeyDur_a + KeyDur_b + KeyDur_c$$

$$KeyDur_S = -KeyDur_a + KeyDur_c$$

$$KeyDur_C = KeyDur_a + KeyDur_c$$

Example. There are individual bond durations used to get the key durations (each year is composed of the individual bond durations), and then key durations are used with the shape factors to calculate the key durations for yield curve shaping changes. The below shows how the the yield curve (not the bond price) moves UP for positive numbers. It DEFINES what a parallel, steepness and curvature are.

Year	5	10	30
Parallel	1	1	1
Steepness	1	0.5	-1
Curvature	0.5	0	1
Key rate curations	1.8	3.6	8.7

The portfolio's effective duration is the **weighted sum of the effective duration of each bond position**, e.g. if there is 100 USD invested in each year, then $0.333(5 + 10 + 30) = 15$. The year 1 key duration is $KeyDur_{5y} = 1/((100 + 100 + 100) \times 0.01)$ in response to a standard 100 bps change.

$$KeyDur_{portfolio} = KeyDur_{5y} + KeyDur_{10y} + KeyDur_{30y}$$

or alternatively

$$KeyDur = \frac{Dur_{5y} + Dur_{10y} + Dur_{30y}}{(100bps) \times (USD_{5y} + USD_{10y} + USD_{30y})}$$

which is

$$KeyDur = \% \Delta P \approx -KeyDur_5 \Delta z_5 - KeyDur_{10} \Delta z_{10} - KeyDur_{30} \Delta z_{30}$$

So, for instance,

$$KeyDur_{5y} = \frac{Dur_{5y}}{100bps \times (USD_{5y} + USD_{10y} + USD_{30y})}$$

For the level shift,

$$KeyDur_L = \frac{L_1 Dur_1 + L_2 Dur_2 + L_3 Dur_3}{TI \times 100bps} = L_1 KeyDur_1 + L_2 KeyDur_2 + L_3 KeyDur_3$$

where TI = total investment, and L = 1 as in the table above for each year in the example. For steepness,

$$KeyDur_S = \frac{-(S_1 \times Dur_1) + (S_3 \times Dur_3)}{TI \times 100bps} = -S_1 KeyDur_1 + S_3 KeyDur_3$$

$$KeyDur_C = \frac{(S_1 \times Dur_1) + (S_3 \times Dur_3)}{TI \times 100bps} = S_1 KeyDur_1 + S_3 KeyDur_3$$

Key rates: multiply change in interest rate by shape sensitivity factor by key rate duration and by weight (e.g. 0.333) and make it negative. That is the price change from the interest rate change.

Bond risk premium = expected excess return of a default-free long-term bond less equivalent short-term bond = term (duration) premium. Captures uncertainty of default-free rates.

Yield curve movements. Bullish is when the yield curve translates lower; bearish when it rises. Flattening versus steepening. Buying a single maturity = bullet portfolio; buying more than one maturity = barbell portfolio.

Use swap rate where government bond market is not liquid; often have more swap data points to construct swap rate curve.

TED spread between T-bill and MRR (between banks), so TED spread measures credit and liquidity risk in the banking sector. SOFR (secured overnight financing rate) is a US treasury overnight repo market rate.

Bullish: lower interest rates. Bearish: higher interest rates. Flattening and steepening. Interest rates to fight inflation in economic growth: bearish flattening. Flight to quality usually results in bullish flattening (where investors GO, not leave) because investors usually flock to long-term bonds.

4.2 The arbitrage-free valuation framework

Arbitrage types: value additivity and dominance.

From YTM (par rate), take year 1 spot rate, then bootstrap to get other spot rates. From spot rates, derive forward curve.

Benchmark bonds. Benchmark par curve. Bonds are priced at par so that the YTM and coupon rates (annual) are the same, i.e. bond price is 100.

Binomial interest rate tree. Need interest rates to populate the branches of the tree. Model interest rates as lognormal random walks. Lognormality used by assuming (1) interest rates > 0 , (2) higher volatility at higher interest rates. Lognormality implies - which is backed empirically - that the rate of change of an interest rate (or price) is proportional to the interest rate (or price). If the differential equation looks like this:

$$\frac{di}{dt} = a \times i$$

then the solution (which can be verified by plugging it into the above DE) is

$$i = e^{ai}$$

which is also why if the lognormal interest rate is logged, then the result is the rate of change of the interest rate (or price). If the 'a' factor is normally distributed, then the interest rate (or price) is lognormal.

Implied forward rate from yield curve is average, i_H is high implied forward rate 1 standard deviation above average, i_L is low implied forward rate 1 standard deviation below average. With lognormality,

$$i_H = i_L e^{2\sigma}$$

$$\ln(i_H) = \ln(i_L) + 2\sigma$$

$$\sigma = \frac{1}{2}(\ln(i_H) - \ln(i_L))$$

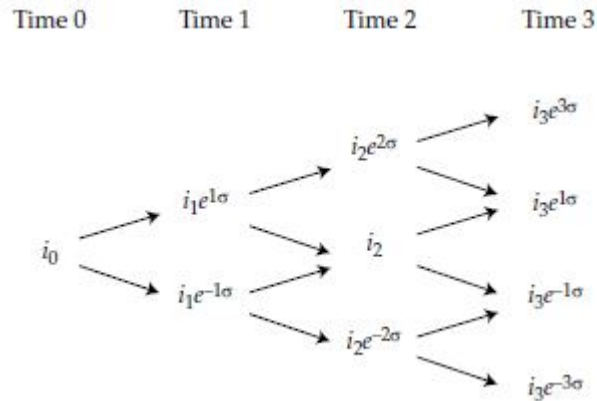
σ is the interest rate volatility for ONE YEAR or PERIOD. In the binomial interest rate tree, assume lattice recombination.

$$i_{HH} = i_{LL}(e^{4\sigma})$$

$$i_{HL} = i_{LL}(e^{2\sigma})$$

$$i_{HHH} = i_{LLL}(e^{6\sigma})$$

So with one up movement in a binomial tree and one down movement, up is $ie^{1\sigma}$ and down is $ie^{-1\sigma}$.



The standard deviation of interest rate i is $i\sigma$, because

$$e^\sigma \approx 1 + \sigma$$

for small σ . Then,

$$ie^\sigma = i(1 + \sigma) = i + i\sigma$$

which means the standard deviation is $i\sigma$. But this is also intuitive.

Binomial interest rate tree use (calculating BOND PRICES, not calculating interest rates). Start at last branches. Take the par (e.g. 100) + coupon rate and discount using the last node interest rates to get the final bond prices. Work backwards, assuming weight of each branch is 0.5, and remember to add the coupon rates:

$$\text{Bond value at node} = \frac{\text{coupon} + (0.5VH + 0.5VL)}{1 + i}$$

Binomial interest rate tree construction. Corresponds to a spot/forward/YTM curve.

- Get the benchmark par curve, and extract spot and IFRs from it. Here, we have the $t = 0$ benchmark par bond price (=100) used below, as well as the par rates = coupons.
- Assume a volatility σ , e.g. 0.15 for 15%.
- Unlike with call and put options, the branch probabilities are always 0.5. This is a binomial tree.
- Start with time = 0 root of tree using the benchmark par rate for a bond maturing at $t = 1$.
- Get the $t = 1$ coupon rate (= 1y par rate) and $t = 2$ values (= 100 + 1y par-rate) from the first step.

- For the first branch, take the implied forward rate $f_{1,1}$ as a base and take a slightly lower interest rate (not bond price). Use $i_H = i_L e^{2\sigma}$ to get the higher-branch interest rate, which should be slightly higher than the implied forward rate. The simple arithmetic average of the high and low interest rates should be slightly higher than the IFR because of log-normality. Note that if $\sigma = 0$, the binomial tree collapses to the IFR curve.
- Calculate the bond value at the first branch nodes. If $t = 0$ bond price (from above) is too low, adjust i_L lower. Repeat. Use goalseek or other iterative process.
- Add branches, using coupon rates and values from the first step. Note: use the $t = 2$ coupon rate as the $t = 1, 2, 3$ coupon rates now (i.e. retroactively change the $t = 1$ coupon rate). Keep the previous branch **interest rates** once set, but note that the bond values and coupons will retroactively change depending on the added branches.

Note that interest rate paths through a binomial tree are like IMPLIED FORWARD RATES, not spot rates.

Using a binomial tree calibrated to a spot/forward/YTM curve to calculate an option-free bond price gives same answer as (1) just using that curve, (2) pathwise valuation - calculate PV for each path and averaging, weighting each path the same amount, (3) Monte Carlo.

Monte Carlo. Often used when cash flows are path dependent. So many paths, so assume a probability distribution and volatility assumption, and then sample paths with same initial par and spot curves. Add as in pathwise valuation, and then adjust with a constant (drift adjustment).

Term structure models. Quantitatively describe interest rates. They can fit binomial and lattice models. Interest rate process:

$$dr = \theta_t dt + \sigma_t dZ$$

The first term is a drift term - can be constant or mean-reverting. The second term is a volatility term. Z is a Wiener process distributed normally. Two types of term structure models: (1) equilibrium models which are meant to derive equilibrium interest rates through, e.g. mean-reversion, and have only a few or only one variable, e.g. r_t , and (2) no-arbitrage models which are bootstrapped to the existing, real term structure, and have many parameters, each for different $time = t$.

Cox-Ingersoll-Ross (CIR) model. Equilibrium model.

$$dr_t = k(\theta - r_t)dt + \sigma\sqrt{r_t}dZ$$

θ is the long-term mean rate, k is the speed of mean-reversion. Volatility term depends on size of r_t , preventing negative interest rates.

Vasicek model. Equilibrium model.

$$dr_t = k(\theta - r_t)dt + \sigma dZ$$

Constant volatility. Theoretically possible for interest rate to be negative.

Ho-Lee model. No-arbitrage model.

$$dr_t = \theta_t dt + \sigma dZ$$

There is a θ_t for each time step.

Kalotay-Williams-Fabozzi model. No-arbitrage model.

$$\ln(dr_t) = \theta_t dt + \sigma dZ$$

4.3 Valuation and analysis of bonds with embedded options

Bonds with embedded options like callable, puttable, convertible bonds. European, American, and Bermudan options.

Sinking bond funds (sinker). Issuer sets aside funds over time to retire bond, reducing credit risk. Can be callable (call option where X = above par) with other options - acceleration provision to buy a percentage of outstanding bonds (issuer has option to repurchase at par, and then use call option to buy rest), delivery option (issuer can pay sinking fund payment by delivering bonds instead of cash; works if interest rates rise and bond is liquid). Acceleration is like call option in addition to the extant call option, while the delivery option is like a put. Delivery option + call option = long straddle.

Valuation of embedded options. From the investor's (i.e. market's) perspective:

- Callable bond value = straight bond value - (issuer) call option.
- Puttable bond value = straight bond value + (investor) put option.

Example: Bonds with embedded Bermudan options at $X = 100$. Work backwards from final cashflow ($100 + c$), discount using implied forward rates, and then if call or put, then call or put and set bond value to strike price, and discount back to $t = 0$.

	Today	Year 1	Year 2	Year 3
Cash flow		4.250	4.250	104.250
Discount rate		2.500%	3.518%	4.564%
Value of the call-able bond	$\frac{100 + 4.250}{1.02500} = 101.707$	$\frac{99.700 + 4.250}{1.03518} = 100.417$ Called at 100	$\frac{104.250}{1.04564} = 99.700$ Not called	

	Today	Year 1	Year 2	Year 3
Cash flow		4.250	4.250	104.250
Discount rate		2.500%	3.518%	4.564%
Value of the puttable bond	$\frac{100.707 + 4.250}{1.02500} = 102.397$	$\frac{100 + 4.250}{1.03518} = 100.707$ Not put	$\frac{104.250}{1.04564} = 99.700$ Put at 100	

If term structure might change, have to strategize when and if to call for embedded Bermudan and American options. For European options, there is only one exercise date so it is straightforward.

Embedded option bond valuation with interest rate volatility and other term structure changes.

- Interest rate volatility. Call option value (just the option, not the entire bond) to the issuer rises with rising volatility, while the straight bond price remains the same - this means the callable bond price falls with increasing volatility as per the callable bond value equation above. Put option rises with increasing rate volatility; this is added to the straight bond, leading to higher puttable bond values.
- Yield curve rises (flat curve horizontal increase). If the interest rate rises, straight bond price declines, and the call option also declines. So the callable bond value declines. If interest rate rises, straight bond value declines again, and put option rises in value. But overall, puttable bond value declines.
- Yield curve slope. If upward sloping, call option has lower value (and callable bond therefore has higher value); if downward sloping, call option has higher value (and callable bond therefore has lower value). Upward sloping yield curve = put option rises, so puttable bond value rises. Downward sloping yield curve - put option value falls, so puttable bond value is lower.

Note: Option Greek Rho on call option is positive, because if interest rate rises, stock price probably rises. However, if interest rate rises, issuer embedded call is lower in value because the embedded call option is about the **bond price, not an underlying stock price**

Valuation of DEFAULT-FREE callable and puttable bonds with interest rate volatility. Generate the usual binomial interest rate tree, using

$$R_u = R_d \times e^{2\sigma\sqrt{t}}$$

And then manually adjust for calls or puts.

Valuation of RISKY callable and puttable bonds. First method is to simply increase the **discount rates**. Second method is getting issuer-specific rate curves, but this is expensive. The third method involves adding uniform Z-spreads or OASs.

- Add **Z-spread** (zero volatility spread) to the **1y forward rates** (not spot rates) in the standard, single-line PV calculation. Z-spread calculated by looking at bonds of similar credit quality. This is used in the one-line NPV calculations, not the binomial trees (it's 'zero-volatility').
- **Option-adjusted spread (OAS)** which is **uniformly** applied to interest rates (**1y forward rates**) in the binomial tree, until the binomial tree bond price = market price. Note: this is a trial-and-error process because as the OAS is added, the calls and puts in the bond will change. OAS falls as σ rises almost linearly because there's a 'total' amount of riskiness, and this can be overwhelmed by increase in the call option as volatility rises.
- Z-spread for option-free bond = OAS at zero volatility.
- Z-spread can be used to value the theoretical bond price and compare with market bond price. OAS is extracted from the market bond price, so compare the extracted OAS with rates on similar bonds.
- Low OAS compared to comparable bonds = bond is overpriced; high OAS = bond is underpriced.
- As σ rises, binomial tree bond price falls (more likely to be called), indicating a rise in equivalent interest rate, 'crowding out' OAS. There is more certainty in the callable bond price, so there is less role for OAS.

Note actual situations can be complicated, e.g. 5-year bond but investment horizon is shorter. Then if interest rates fall, bond might be called, and both principal + coupons have to be reinvested. This creates lots of uncertainty.

Effective duration. Sensitivity of bond prices to interest rate changes. Use full prices in the formula.

$$EffDur = \frac{PV_- - PV_+}{2\Delta Curve PV_0}$$

where $\Delta Curve$ is the magnitude of the parallel shift in the yield curve. To calculate:

- Given PV_0 , calculate the IMPLIED OAS using a benchmark yield curve, assuming a σ .
- Shift the benchmark yield curve down $\Delta Curve$, generate a NEW binomial interest rate tree, apply the implied OAS from above, apply call/puts and get PV_- .

- Shift the benchmark yield curve up ΔCurve , generate another NEW binomial interest rate tree, apply the implied OAS from above, apply calls/puts to get PV_+ .
- Calculate the EffDur.

As callable (and puttable) bond prices are less sensitive to interest rates (because they 'buffer' interest rates changes partially through the embedded call option), effective duration of straight bond is higher than corresponding callable bond.

One-sided duration. Call and put options limit the movement of $PV_- - PV_+$. Can look at one-sided durations in such cases - e.g. the up-side duration will be larger than the down-side.

Key rate durations (partial durations). Sensitivity of bond prices to specific maturities. Note: though key rate durations are sensitivities to changes in the spot yield at particular points in time, changing the spot rate at a non-maturity point (i.e. before bond maturity) leads to changes in interest rates after that time to preserve par bond prices - this is where the spot yield curve is bootstrapped from. So if the interest rate at non-maturity rises, it falls after that. Note also that a rise in interest rates at maturity = fall in prices = positive key rate duration.

Option-free bonds. Concentrated at maturity date - that's where most of the weight of the duration lies. For the par bond, YTM = coupon rate, and key durations for anything less than maturity is 0 because if the short-duration rates change, rest of the rates change to keep the bond priced at par. As coupon rate declines, the duration is even more concentrated in the final year. For zero- or low-coupon bonds, key rates for durations before the final year can be negative because if their interest rate rises, adjusting the rest of the rates in the same way as the par coupon bond leads to an INCREASE in the bond price, which is a negative key rate duration.

Coupon (%)	Price (% of par)	Key Rate Durations				
		Total	2-Year	3-Year	5-Year	10-Year
0	67.30	9.81	-0.07	-0.34	-0.93	11.15
2	83.65	8.83	-0.03	-0.13	-0.37	9.37
4	100.00	8.18	0.00	0.00	0.00	8.18
6	116.35	7.71	0.02	0.10	0.27	7.32
8	132.70	7.35	0.04	0.17	0.47	6.68
10	149.05	7.07	0.05	0.22	0.62	6.18

Callable bonds. At low coupon rates, the bond price is low, so unlikely to be called. So, key rate durations are concentrated in the final years. As coupon rates increase, the bond is more likely to be called, so key rate durations shrink.

Coupon (%)	Price (% of par)	Key Rate Durations					
		Total	2-Year	3-Year	5-Year	10-Year	30-Year
2	64.99	19.73	-0.02	-0.08	-0.21	-1.97	22.01
4	94.03	13.18	0.00	0.02	0.05	3.57	9.54
6	114.67	9.11	0.02	0.10	0.29	6.00	2.70
8	132.27	7.74	0.04	0.17	0.48	6.40	0.66
10	148.95	7.14	0.05	0.22	0.62	6.06	0.19

Puttable bonds. Opposite of callable bonds. At low coupon rates, bond prices are so likely to be called, so key rate durations are high at earlier maturities. As coupon rates increase, key rate durations concentrate in the final year.

Coupon (%)	Price (% of par)	Key Rate Durations					
		Total	2-Year	3-Year	5-Year	10-Year	30-Year
2	83.89	9.24	-0.03	-0.14	-0.38	8.98	0.81
4	105.97	12.44	0.00	-0.01	-0.05	4.53	7.97
6	136.44	14.75	0.01	0.03	0.08	2.27	12.37
8	169.96	14.90	0.01	0.06	0.16	2.12	12.56
10	204.38	14.65	0.02	0.07	0.21	2.39	11.96

Effective convexity. Sensitivity of duration to interest rate changes. Use full prices in the formula.

$$EffCon = \frac{PV_- + PV_+ - 2PV_0}{(\Delta Curve)^2 PV_0}$$

Callable bond convexity becomes negative when interest rates fall because downside duration (PV_-) gets limited by the call. Puttable bonds always have positive convexity, but it has higher positive convexity than straight bond when interest rates rise, as upside duration (PV_+) gets limited.

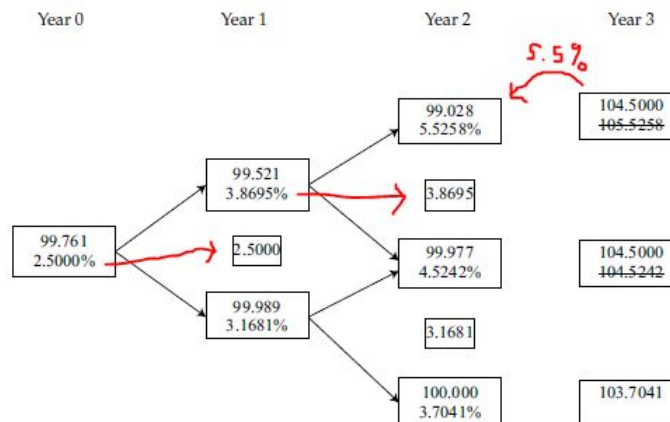
Capped and floored FRNs. These are FRN options which are exercised automatically depending on interest rates. If coupon rate rises or falls below threshold, cap or floor automatically applies. Similar to callable and puttable bonds.

Capped floater is favorable to issuer because if coupon rate rises too high, capped. Floored floater is favorable to investor.

- Value of capped floater = straight bond value - value of embedded cap.
- Value of floored floater = straight bond value + value of embedded floor.

FRN binomial tree. Take the pre-made binomial tree (based on prevailing par curve). Without a cap, the value is 100 in the tree everywhere because coupon

rate = discount rate (so, if coupon rate is capped, bond price should be lower than the straight bond). Add floating rate coupons, e.g. if year 1 is 2.5%, then coupon after the first branch has coupons of 2.5. After the final branch is the 100 + coupon rate value. Discount back in each branch as for other binomial trees, even if bond values are capped.



Capped and floored floaters not applicable to fixed-rate bonds. Floaters apply to the COUPONS, while calls and puts apply to the BOND VALUES. Can have calls and puts on FRNs, but less usual (usually for credit changes).

Convertible bonds. Issuer pays lower coupon, investor buys with an option. However, issuer faces share dilution if converted. If not converted, issuer has to pay the rest of the debt and investor is stuck with low-coupon bond. Conversion ratio and conversion price essentially provide the same information.

Convertible bonds dividends. If share pays dividends while convertible bond still unconverted, (1) may be no compensation to bondholder, (2) may be compensated by **adjusting conversion price lower**, (3) in between with a **threshold dividend**.

Convertible bonds: issuer changes control. Bond prospectus governs what happens, e.g. put option with full redemption of nominal bond value, **adjust conversion price lower**.

Convertible bonds can include, apart from convertibility, put options. Hard put (classic put). Soft put - issuer choose how payment made, e.g. cash, stock, etc.

Convertible bonds can include, apart from convertibility, call options. Issuer option. Issuer might call if (1) interest rates fall and/or credit rating rises and bond price rises, (2) if share price rises or believes the share price will rise above conversion price (**'forced conversion'**) and the bondholder has not converted.

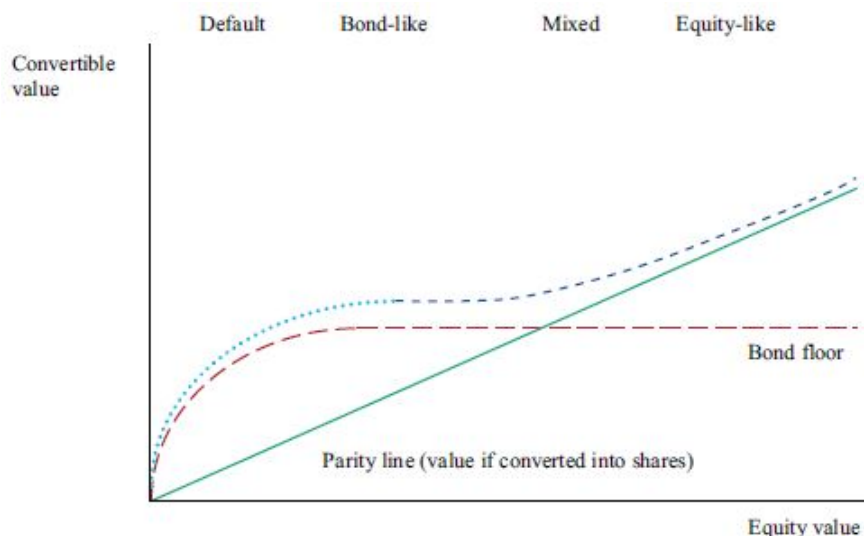
Advantageous for issuer to call callable convertible bond when share price is high because otherwise, bondholders will convert to get high share prices. So, bondholders will pre-empt the issuer calling the convertible bond by converting. So the issuer has effectively forced a sale - they could sell the shares at the higher market price, but this would time take, underwriting fees, and could depress the price if sold at large volumes. Moreover, the forced conversion means no more debt, possibly higher credit rating, issuer doesn't pay the call price, prevents having to pay future coupons and the principal, and reduces leverage in their capital structure. If the share price is high, existing shareholders might resist dilution.

Convertible bonds. Effectively, a convertible bond is like a call option, with the floor being the straight bond.

- Conversion value = current share price x conversion ratio. This is constantly changing and is the 'call option' part.
- Minimum value of a convertible bond = max(conversion value, straight bond). Moving floor price.
- Market conversion price = convertible bond price / conversion ratio. Break-even price; price investors pay for shares if they buy the convertible bond and convert it to shares.
- Market conversion premium per share = market conversion price - share price.
- Market conversion premium ratio = market conversion premium per share / share price.
- Premium over straight value = convertible price / straight price - 1.

Convertible bond value.

- Classic. CB value = straight bond (SB) value + call option on ISSUER'S stock value
- Callable CB value = SB value + call option value - issuer call option value.
- Callable puttable CB value = SB value + call option on issuer stock - issuer call option + investor put option



Left side of above = 'busted convertible'.

Contingent convertibles (CoCos). High coupons but deeply subordinated and converted to equity or face principal write-downs if regulatory capital ratios breached. Convertible CoCos (CoCoCos) - upside if share price rises.

4.4 Credit analysis models

G-spread = credit spread. Single year YTM spread, unlike Z-spread which is the entire curve.

Risk-neutral default probability of a bond. Example 1-y maturity, priced currently at 100, 5% coupon at par 100, recovery rate = 40%, $r_f = 3\%$. Calculated using the current price, the expected receipt at maturity with no default (that is, $100 + 5 = 105$), the expected receipt at maturity in the event of a default (that is, $0.40 \times 105 = 42$), and the risk-free rate of interest (0.03):

$$\text{Bond price, } 100 = \frac{105(1 - p) + 105(0.4)p}{1.03}$$

and solve for p .

To calculate the CVA (**credit valuation adjustment**), calculate (1) expected exposure to default - which is basically the bond value at each year discounted BACKWARDS using r_f , (2) $\text{LGD} = \text{EE} \times (1 - \text{RR})$, (3) POD - risk-neutral probabilities (to allow discounting using r_f and determined using structural or reduced-form credit models), not historical - calculated like hazard rates, (4) expected loss $\text{EL} = \text{LGD} \times \text{POD}$, (5) discounted the ELs, (6) sum the discounted

ELs = CVA. Take the straight bond - CVA = fair value, which is compared to the current bond market price. From this, calculate the IRR of the fair value, and then IRR of fair value minus IRR of straight bond = credit spread.

To calculate returns if issuer defaults in a given year, take the recovery as the final year and calculate the IRR. Note: if there is no default, the realized rate of return = IRR of fair value!

Given credit spread and fair value of bond, by trial-and-error get POD and other variables.

Credit scores (personal) and credit ratings (companies). Notching from senior unsecured debt. **Credit rating transition matrices** show probability of changing from one credit rating to another; for each change in credit rating, use modified duration x (change in credit spread) x probability from transition matrix; then add these up to get the CVA. Note that these expect changes are skewed to the downside because credit spreads are larger for downgrades and more of the changes are skewed to the downgrades.

From/To	AAA	AA	A	BBB	BB	B	CCC, CC, C	D
AAA	90.00	9.00	0.60	0.15	0.10	0.10	0.05	0.00
AA	1.50	88.00	9.50	0.75	0.15	0.05	0.03	0.02
A	0.05	2.50	87.50	8.40	0.75	0.60	0.12	0.08
BBB	0.02	0.30	4.80	85.50	6.95	1.75	0.45	0.23
BB	0.01	0.06	0.30	7.75	79.50	8.75	2.38	1.25
B	0.00	0.05	0.15	1.40	9.15	76.60	8.45	4.20
CCC, CC, C	0.00	0.01	0.12	0.87	1.65	18.50	49.25	29.60
Credit Spread	0.60%	0.90%	1.10%	1.50%	3.40%	6.50%	9.50%	

From A to AAA:	$-7.2 \times (0.60\% - 1.10\%) = +3.60\%$
From A to AA:	$-7.2 \times (0.90\% - 1.10\%) = +1.44\%$
From A to BBB:	$-7.2 \times (1.50\% - 1.10\%) = -2.88\%$
From A to BB:	$-7.2 \times (3.40\% - 1.10\%) = -16.56\%$
From A to B:	$-7.2 \times (6.50\% - 1.10\%) = -38.88\%$
From A to CCC, CC, or C:	$-7.2 \times (9.50\% - 1.10\%) = -60.48\%$

If investment-grade bond, then typically don't use the D rating in the table above. However, if junk bond, then need to consider - which means using CVA = VND minus recovery from the discussion above for defaults.

Credit models: (1) structural and (2) reduced-form. These help determine the probability that a certain credit change will occur - i.e. describes in detail the corporate transition matrix probabilities. Or, can be used to calculate probability of default (POD) for use in credit spread calculations. Structural. requires company-specific variables like stock price. Based on options pricing theory, so it's BSM-like. Company defaults on debt if asset values falls below

liabilities and probability of that even has features of an option; provides explanation of why company defaults. Structural models require stock information, but if stock not traded, then use reduced-form models - default is based on probability distribution (Poisson) below the 'default barrier'. Options theory. Let $A(T)$ be random asset value at time T , $D(T)$ debt and equity $E(T)$:

$$D(T) + E(T) = A(T)$$

$$E(T) = \max(A(T) - K, 0)$$

$$D(T) = A(T) - \max(A(T) - K, 0)$$

Reduced form. Contrary to structural models, default is no longer tied to the firm's assets falling below a threshold level under reduced-form models. Instead, it occurs according to some exogenous hazard rate process. Reduced-form model use company-specific variables (leverage ratio, return-to-assets ratio, etc.) and macro variables; reduced-form models don't explain why defaults.

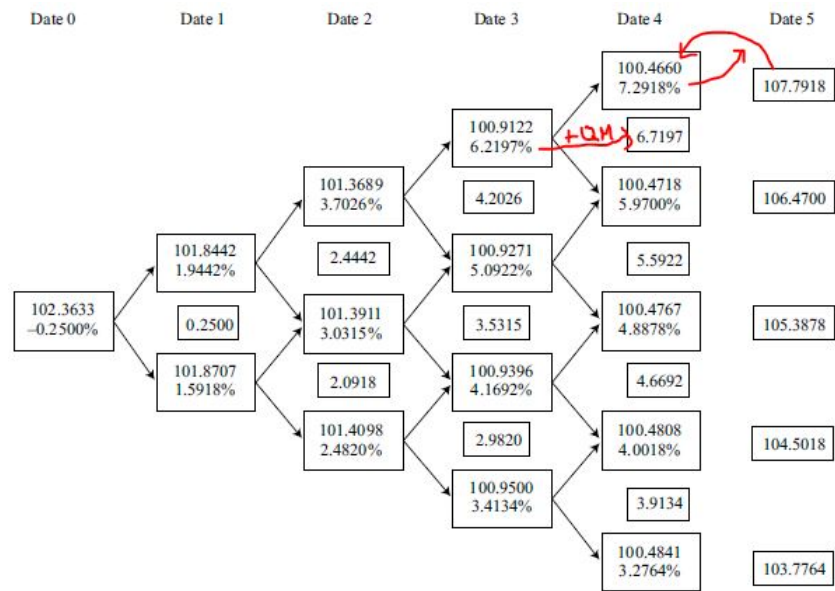
Valuing risky bonds with binomial trees. From the binomial tree, for each time (set of branches), get the expected exposure = multiply bond prices by the probability (binomial probability coefficients), and **add the coupon ONCE**, and then sum. Then convert to LGD. $POD = POD_{annual}x(1-POD_{annual})^{t-1}$. Then discount and sum to get CVA.

Date	Expected Exposure	LGD	POD	Discount Factor	CVA per Year
0					
1	103.2862	61.9717	1.2500%	1.002506	0.7766
2	101.5481	60.9289	1.2344%	0.985093	0.7409
3	101.0433	60.6260	1.2189%	0.955848	0.7064
4	102.0931	61.2559	1.2037%	0.913225	0.6734
5	103.5000	62.1000	1.1887%	0.870016	0.6422
			6.0957%	CVA =	3.5394

The CVA is subtracted from the non-defaulting bond price. Then the YTM of that now-risky bond can be calculated and compared to the YTM without the CVA.

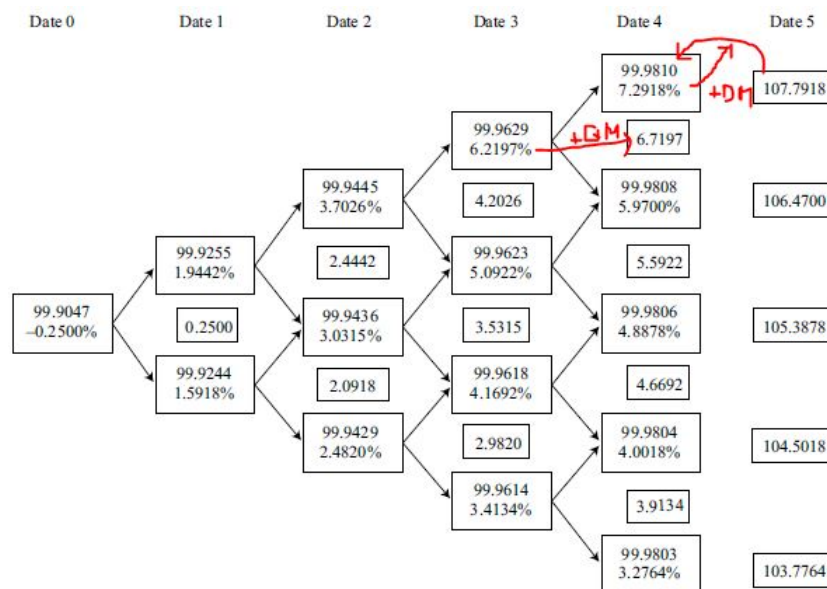
Changing the volatility of the binomial tree should theoretically not change the CVA, but in practice it does because of the asymmetry of the lognormality assumption of tree branches.

Valuing FRN (MRR + quoted margin) risky bonds with binomial trees. **Add MRR + quoted margin to the COUPONS in the NEXT PERIOD of the binomial tree.** But the binomial tree interest rates remains otherwise intact.



$$\begin{aligned}
 & \left[(0.0625 \times 100.4660) + (0.25 \times 100.4718) + (0.375 \times 100.4767) \right] \\
 & + (0.25 \times 100.4808) + (0.0625 \times 100.4841) \\
 & + [(0.125 \times 6.7197) + (0.375 \times 5.5922) + (0.375 \times 4.6692) + (0.125 \times 3.9134)] \\
 & = 105.6535.
 \end{aligned}$$

Then add each date's branches to get the expected exposures (multiplying by their respective binomial probabilities) including coupons (where **coupons are multiplied by their respective binomial probabilities and summed**). If fair value is below par (this is after all an FRN), it must mean the QM is larger than the DM.



DM calculation. QM = quoted, DM = discounted margin which is what investors actually see in the market. Use binomial tree interest rates intact, but add Dm to each of the discounted rates - see Exhibit 19. Use Excel goalseek to solve for DM which equalizes the bond value at Date = 0 to the fair value (= current market value - CVA).

Credit spreads XVAs. Risk premium on bond = taxation + liquidity + credit risk. Hard to often separate. XVA = CVA + TVA (taxation value adjustment) + LVA (liquidity value adjustment) + other risks. XVA engines (i.e. models) generally use Monte Carlo simulations for thousands of paths of interest rates - way beyond binomial interest rate trees manually calculated.

Term structure of credit spreads. Credit curves = yield curves + credit spreads. Typically for unsecured bonds. Generally, credit spreads slope up as longer maturities mean more risk and economic and business uncertainty. However, if the business is expected to improve (e.g. economy will improve, the company was bought out), the credit spreads can invert.

Credit spreads - factors affecting. Strong economy = high government yields and low credit spreads. Market supply and demand for the corporate bonds.

Credit spreads - default. VND - recovery rate = CVA.

4.5 Credit default swaps

Credit derivatives: underlying is somebody's credit quality. Types: (1) total return swaps, (2) credit spread options, (3) credit-linked notes, (4) credit default swaps. CDS most liquid. Credit protection buyer (short position) makes regular payments in a CDS and gets big payment if credit event from credit protection seller. For buyer, it's like a long put: put security value less expected recovery value. CDSs can be useful if bond markets become illiquid.

CDS types: (1) single-name CDS, (2) index CDS, (3) tranche CDS. Single-name is for one reference entity's reference obligation - the minimum obligation, usually senior unsecured debt. Even if the CDS reference obligation is 5 years, as long as it's senior unsecured (or higher in credit rating), it can be 2 years, etc. - and the CDS buyer (upon a credit event) delivers to the CDS seller the CHEAPEST-TO-DELIVER bond of these, in exchange for the par value of that bond (essentially, the CDS seller gives the par minus recovery value to the CDS buyer upon a credit event). Index CDSs are groups of credit risk positions in companies; they are more riskier if there's significant credit correlation. Note: index CDS credit protection buyer is strangely the LONG position. Tranche CDSs - cover up to a specified credit risk for a combination of borrowers.

CDS contracts. Notional amount (can be higher than total debt of the reference entity). CDS coupon - the payment by the credit protection buyer, often = credit spread. Often, credit coupon rates are standardized, like 1% and 5%, so use upfront payments/premiums to balance. If the credit quality of the reference obligation declines, CDS coupon rate remains but the credit protection buyer is now holding a CDS worth more and could sell it for a gain.

CDS credit events. ISDA-standard contracts have detailed definitions. Can have bespoke CDS terms. If ISDA, and unclear, there are committees that vote based on studying laws and event whether it is a credit event.

Credit events and settlement preferences. If credit event, right not obligation to exercise CDS. Credit protection buyer can (1) physical settlement - put all bonds currently owned (if any) to the CDS seller and get the notional amount, or (2) cash settlement - get the notional minus the recovered amount (which the credit protection buyer owns or doesn't own). See Example 2.

Credit curves for different bonds = benchmark + CDS spreads. CDS market is very efficient. Usually, credit curve is upward sloping but can be flattened if hazard rate is constant, though benchmark curve and discounting are also factors in shape of credit curve. If downward sloping credit curve, usually there's an acute stress in the short term.

CDS pricing. If default can ONLY occur at the end of the maturity period of

the CDS, CDS price is roughly:

$$\text{CDS spread} \approx (1 - RR) \times POD$$

Here however, the POD = probability of SURVIVAL, which is calculated by the **hazard rate**. The hazard rate is the conditional probability of survival, given it has survived, e.g. a constant 10% per year or 2.5% per quarter. The conditional probability of survival is then 97.5% per quarter.

Credit spread = bond YTM minus MRR. Compare this to CDS implied credit spread, for arbitrage opportunities.

If there is an upfront, balancing, payment, then:

$$\text{Upfront payment} = PV(\text{Protection leg}) - PV(\text{Premium leg})$$

$$\text{Upfront payment} = PV(\text{Credit spread}) - PV(\text{Fixed coupon})$$

where the coupon is the fixed payment in the CDS. Above is equivalent to:

$$\text{Upfront premium (buyer to seller) \%} \approx (\text{Credit spread} - \text{Fixed coupon rate}) \times \text{Dur}$$

where approximate Dur with EffDur. Moreover:

$$\text{CDS price per 100 par} = 100 - \text{Upfront premium (buyer to seller) \%}$$

where the LHS is the price paid by the buyer to protect 100. If greater than 100, then the CDS price is high, but the buyer is likely being compensated by an upfront premium.

If a credit spread changes:

$$\text{Profit for credit protection buyer} = \Delta \text{Spread} \times \text{EffDur} \times \text{NA}$$

Alternatively,

$$\% \text{ change in CDS price} = \Delta \text{spread} \times \text{Dur}$$

CDS uses. Transfers credit risk, can separate credit risk from rest of bond cheaply.

CDS trades.

- Long/short credit trade - sell protection on company credit expected to improve and buy protection on company credit expected to worsen.
- Curve trade - buy credit protection at one maturity and sell at another maturity.
- Basis trade - difference between CDS spread and bond market spread. If bond market spread higher than CDS, negative basis. Reverse is positive basis.

- Component arbitrage trade. If index CDS components mispriced, buy/sell single-name CDS.
- Others. E.g. combinations of CDS with equity. If there is an LBO, then likely large leverage, so credit spread will rise on the taken-over entity, so buy entity stock (potential gain) + buy CDS on it (potential gain), where the potential gains are offsetting or move in the same direction.

5 Equity valuation

5.1 Equity valuation: Applications and processes

What is value? Mostly concerned here with intrinsic value, which is where market efficiency should take the market price. Guessing intrinsic value and being right = abnormal returns, alpha, due to mispricing. Mispricing has two components:

$$V_{estimated} - P = (V - P) + (V_{estimated} - V)$$

where P is market price and V is intrinsic value. If mispricing exists, often need a 'catalyst' to get the market to revalue the asset.

Often value companies as **going-concern**, but if not then estimate its **liquidation value** (note: if assets are liquidated orderly, then 'orderly liquidation value'). Fair market value pretty close to fair value - market transaction, no compulsion, parties informed of material aspects of transaction. If value is worth more than intrinsic value to investor, then there's **investment value**.

Absolute valuation model - NPV models, DCF, DDM models. FCFE, FCFF or 'residual income' (= accrual accounting earnings in excess of opportunity cost of generating those earnings) used in numerator. Use sensitivity analysis to analyze.

Relative valuation model - P-to-E, MC-to-EV, etc. Note: undervalued, overvalued are used to describe relative to intrinsic value; relatively undervalued and relatively overvalued used to compare to a given market benchmark. Note: *Relative value investing* such as 'pairs trading' by longing undervalued and shorting overvalued stocks, is often employed.

Types of valuation: sum-of-parts (break-up or private market value), conglomerates. But, conglomerate discount (low total company EV compared to sum of segment or division EVs) - arising possibly from inefficiency, poorly performing companies tend to expand into unrelated areas, research measurement errors. Note control premiums, lack of marketability discounts (non-publicly traded stocks), illiquidity discounts (and blockage factors).

5.2 Discounted dividend valuation

Cashflows = dividends, FCF, residual income. DDM most appropriate when:

- Company pays dividends
- Company dividend policy is related to profitability
- Not taking control perspective

Mature companies often have these qualities.

FCF. FCFF - CF generated by company's operations which can be withdrawn without economically impairing the company (so, good if have control over dividends or acquiree may want to view from control perspective if about to acquire). FCFE - FCFF and net debt payments. FCFE used to estimate equity. FCFF easier to use if leverage changes a lot. FCF most appropriate when:

- Company's actual dividends are much lower and/or higher than FCF
- FCF aligns with company profitability in the investor's forecast horizon
- Taking control perspective

Residual income. Earnings in excess of return on beginning-of-year investment (common stockholder equity), i.e. earnings in excess of opportunity costs. Opportunity cost = shareholder initial investment x required rate of return. Residual income model = book value per share + PV of expected future residual income (a restatement of DDM). Book value per share = common shareholder equity divided by number of shares; common shareholder equity = $BV_t + earnings_t - dividends_t$. If FCF is negative, use residual income approach. Residual income good when:

- Company not paying dividends
- Negative FCF over investor horizon

Dividend cover = EPS/DPS. See if steady to determine which DCF model to use. D divided by E is dividend payout ratio DPR.

Forward dividend yield = dividend yield (D divided by P) with future dividend.

Gordon growth model.

$$V_0 = \frac{D_0(1+g)}{r-g}$$

Note: D is dividend per share; E is earnings per share in the GG model. D_0 is the **just past** dividend payment, while $D_0(1+g) = D_1$ is the dividend payment **in one year's time**. Example: if there are 3 years of no dividend payment, then dividend payment in the 4-th year, then can calculate sum of dividends by

starting in the **third** year (when there's 0 dividend) and using D_4 in the GG model.

Terminal value is the value at the point of the last dividend payment. E.g. if there are 8 dividends in the first stage of a 2-stage DDM, then the terminal value is the UNDISCOUNTED value at $T = 8$, NOT $T = 9$. So the TV is discounted from $T = 8$.

Often get g from comparing with GDP growth. Good if steady growth, like mature industries or broad equity indices. **Sustainable growth rate** is g :

$$g = ROE \times (1 - DPR) = ROE \times b$$

Proof.

$$\begin{aligned} DPR &= \frac{D}{NI} \rightarrow D = NI \times DPR \\ RE &= NI - D = NI - DPR \times NI = NI(1 - DPR) \\ g &= (1 - DPR)ROE = \frac{NI - D}{NI} \frac{NI}{E} = \frac{NI - D}{E} \end{aligned}$$

but since retained earnings are:

$$RE = NI(1 - DPR)$$

and

$$g = \frac{RE}{E} = \frac{NI(1 - DPR)}{E} = \frac{NI - D}{E}$$

If DPR ('payout ratio') is constant, then $g = EGR$ where EGR is earnings growth rate. Note this **assumes constant capital structure**; changing leverage can change this.

$$g = EGR$$

Proof.

$$EGR = \frac{NI_2 - NI_1}{NI_1} = \frac{\Delta NI}{NI_1} = \frac{RE \times ROE}{NI_1} = \frac{(NI_1(1 - DPR)) \times ROE}{NI_1} = (1 - DPR) \times ROE = g$$

Required rate of return on equity = from CAPM (note R^2 of β often very low), or from 'pretax debt cost' + risk-premium.

Share repurchases (buybacks). Timing is opportunistic but over long periods is related to earnings. Neutral effect on shareholders. Need to incorporate this together with dividends - e.g. 'total distributions to shareholders' = dividends + repurchases.

Implied dividend growth rate. Given price, dividends, required rate of return, back out g .

PVGO (PV of growth opportunities) = value of growth = NPV of future projects. Only if NPV positive, then project value is worth more than opportunity cost; this is equivalent to ROE (equity = market cap, not book value) is greater than r (required rate of return). So, mature companies without good NPV projects pay dividends and growth companies don't.

Real options - e.g. to modify projects. PVGO companies usually have flexibility.

No-growth company = no profitable growth = no PVGO, and for these companies, earnings = ROE x equity and ROE is constant and equity is constant because no RE is added (beyond opportunity cost). No-growth value per share = E_1/r :

$$V_0 = \frac{E_1}{r} + PVGO$$

E_1 is future EPS. Assuming $g = 0$,

$$V_0 = P_0 = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r}$$

$$\frac{D_1}{E_1} = 1 - b \Rightarrow D_1 = E_1(1 - b)$$

and assuming $b = 0$, we get

$$V_0 = \frac{E_1}{r}$$

without PVGO. Above original formula can be re-arranged to

$$\frac{P}{E} = \frac{V_0}{E_1} = \frac{1}{r} + \frac{PVGO}{E_1}$$

$$\frac{P}{E} = \frac{P_0}{E_1} = \frac{1}{r} + \frac{PVGO}{E_1}$$

$\frac{1}{r}$ is the **P-E of a no-growth company**. The second term is the P-E related to growth opportunities.

Justified (fundamental) P-E. Based on Gordon growth model. Leading justified P-E:

$$\frac{P_0}{E_1} = \frac{D_1/E_1}{r-g} = \frac{1-b}{r-g}$$

where b = retention rate of funds, $1-b$ = dividend payout ratio = DPS divided by EPS = $\frac{D_t}{E_t}$. Trailing justified P-E:

$$\frac{P_0}{E_0} = \frac{D_0(1+g)/E_0}{r-g} = \frac{(1-b)(1+g)}{r-g}$$

The expression

$$P_0 = \frac{D_0(1+g)}{r-g}$$

can be solved for r:

$$r = \frac{D_0(1+g)}{P_0} + g$$

which means the total return = dividend yield (first term) + capital gains yield g. The term r = IRR.

Note: Dividend yield has share price in the denominator, *not* earnings (D-E is just dividend payout ratio):

$$\text{Dividend yield} = \frac{\text{DPS}}{P_0}$$

Multistage DDMs. Growth, transition, mature phase. Different g's for different phases. Sometimes use different r's.

2-stage DDMs. Use (1) Gordon growth model for sudden transitions, or (2) H-model for gradual

$$V_0 = \frac{D_0(1+g_L) + D_0H(g_S - g_L)}{r - g_L}$$

where H = half-life of high-growth period (2H = high-growth period), g_S = short-term (i.e. 'growth phase' from S to L growth rates) and g_L = long-term (i.e. 'long-term growth rate'). The V_0 is the whole value (it's not just the value of the transition time). Intuition for H: expected excess growth rate in high-growth period is $(g_S - g_L)/2$ and in 2H periods of high-growth, total excess dividends is $2HD_0(g_S - g_L)/2 = HD_0(g_S - g_L)$ - this is the upward adjustment to the first dividend term.

If $P_0 = 41.7$, $D_0 = 1.77$, initial dividend growth rate $g_S = 7\%$, declining linearly over 10 years $H = 5$, to perpetual growth of $g_L = 4\%$, assuming $r = 8\%$, then $V_0 = 52.75$.

3-stage DDMs. Can have (1) constant growth rates g_1, g_2, g_3 in a Gordon growth model, or (2) g_1 and then H-model.

Calculating g

- $g = b \times ROE$ (dividend displacement of earnings), with ROE: (1) DuPont, (2) set $r = ROE$ (mature phase companies can't do more than opportunity cost of capital), (3) ROE in mature phase equal to median industry ROE.
- Relate to g to macroeconomic, industrial growth.

Mature phase exceptions. $g = (b \text{ in } \textit{mature phase}) \times ROE$, and in mature phase, $r = ROE$.

If above $r = ROE$, then

$$\frac{P_0}{E_1} = \frac{D_1/E_1}{ROE - bROE} = \frac{1 - b}{ROE(1 - b)} = \frac{1}{ROE}$$

$$ROE = \frac{E_1}{P_0} = r$$

which is just $b = 0, g = 0$ in the above

$$r = \frac{D_0(1 + g)}{P_0} + g$$

to get

$$r = \frac{E_1}{P_0}$$

For the H-model, re-arrange:

$$r = \left(\frac{D_0}{P_0} \right) ((1 + g_L) + H(g_S - g_L)) + g_L$$

May need mathematical computation (e.g. goalseek) to find r for complex dividend patterns.

PRAT (profit margin P , retention rate R , asset turnover A and financial leverage T), which takes DuPont decomposition further and shows what makes up growth rate. It is $g = b \times ROE$:

$$g = \frac{NI - D}{NI} \times \frac{NI}{R} \times \frac{R}{A} \times \frac{A}{E} = \left(\frac{NI - D}{NI} \times \frac{A}{E} \right) \times ROA$$

RA determine ROA (return on assets), PT are company financial policies. Theoretically, should use beginning-of-year E for DuPont and PRAT but often just use average.

High-growth. If very high-growth ROE, then use DuPont decomposition and make 'more normal' anything unusual or unsustainable.

Price of a stock under DDM increases at the same rate as dividends:

$$P_t = \frac{D_t(1 + g)}{r - g}$$

$$P_{t+1} = \frac{D_{t+1}(1 + g)(1 + g)}{r - g}$$

5.3 FCF valuation

Grossman-Stiglitz paradox. If market is efficient, then nobody will analyze, but if nobody will analyze, market won't be efficient.

FCFF or FCFE. EBIT and EBITDA don't subtract taxes, and they may or may not subtract off preferred dividends (i.e. different capital structures) for FCFE (which is for *common* stockholders) or to add back interest for FCFF (which is available to bondholders; again, this is about different capital structures).

Use FCFE instead of FCFF for equityholders when:

- Stable capital structure

However, use FCFF over FCFE when:

- Levered company with negative FCFE
- Levered company with changing capital structure (net borrowing can change quickly, required return can change quickly if more debt)

Adjusted present value. Firm value = unlevered firm value + NPV of debt (including risk of financial distress). This is from the topic of capital budgeting.

Required rates of return. FCFF use WACC, FCFE use required rate of return on equity.

$$WACC = \frac{D}{D+E}r_d(1-t) + \frac{E}{D+E}r$$

where D and E are *current* market values, not book values. FCFF and constant-growth version.

$$FCFF = \sum \frac{FCFF_t}{(1+WACC)^t}$$
$$FCFF = \frac{FCFF_0(1+g)}{WACC-g}$$

FCFE and constant-growth version.

$$FCFE = \sum \frac{FCFE_t}{(1+r)^t}$$
$$FCFE = \frac{FCFE_0(1+g)}{r-g}$$

FCFF from net income (like indirect method of estimating CFO from NI).

$$FCFF = NI + NCE + I(1-t) - FCInv - WCInv$$

$$FCFF = CFO + I(1-t) - FCInv$$

FCInv and WCInv are net changes.

FCFE from NI.

$$FCFE = NI + NCE - FCInv - WCInv + NB$$

where NB = changes in : notes payable + long-term debt. FCFF and FCFE from CFO:

$$FCFF = CFO + I(1 - t) - FCInv$$

$$FCFE = CFO - FCInv + NB$$

Fixed capital investment is subtracted off because it's not subtracted in NI nor in CFO (it's an investing cashflow).

Interest is always in the FCFF for GAAP as interest is always CFO, but under IFRS can be CFO or CFF. If under IFRS interest is CFF, then don't add back to CFO (but still add back to NI).

	IFRS	US GAAP
Interest received	Operating or investing	Operating
Interest paid	Operating or financing	Operating
Dividends received	Operating or investing	Operating
Dividends paid	Operating or financing	Financing

NCEs. Adjustment to NI if treated as an expense on IS but not an equivalent CFO in the same reporting period. PNL gains and losses - e.g. on selling PPE is an NCE to be adjusted. If DTAs/DTLs are short-term and small, ignore but if longer than NCE to be adjusted. Share-based compensation (shares, RSUs, options) need NCE adjustment, as they're subtracted from NI and then classified (mostly or all) as CFF. Non-recurring items might also be NCE.

Noncash Item	Adjustment to NI to Arrive at FCFF
Depreciation expense	Added back
Amortization expense and impairment of intangibles	Added back
Restructuring charges (expense)	Added back
Restructuring charges (income resulting from reversal)	Subtracted
Amortization of long-term bond discounts	Added back
Amortization of long-term bond premiums	Subtracted
Losses on non-operating activity	Added back
Gains on non-operating activity	Subtracted
Deferred taxes	Added back but calls for special attention

Interest is added back after the tax shield because we want the *after-tax* interest in the cashflow. CFO subtracts off $I(1 - t)$ so we need to add it back for bondholders.

- Revenue: 100, all cash. Cash expenses: 70. Interest: 10. Tax 30%.
- $NI = (100 - 70)(1 - 30\%) - 10(1 - 30\%) = (100 - 70 - 9) - 10 + 3 = 14$
- $CFO = 100 - 70 - 10 - 6 = 14 = (100 - 70 - 9) - 10 + 3$. This is the 'true' cash flow - what the company has after subtracting expenses. The amount of cash that actually left the company is $I = 10$. However, **given** $CFO = 24$, the company effectively paid bondholders $10 - 3$ **in the past**.
- $NI = 14 = CFO$
- So, add $10 - 3$ to both sides: $14 + (10 - 3) = 21$

The cost of debt in the WACC equation is post-tax debt rate. For consistency, use post-tax interest. It can also be thought of as (1) the 'net' amount of cash that left the company (**not** the amount received by bondholders) by paying interest (which is after the tax shield effect) (so, company pays 100 in interest to bondholders and gets tax shield of 20 if $t = 20\%$, but to make the company whole again, the bondholders only need to give back 80), or (2) what is available to be paid to bondholders **before** any actual payment (and therefore, we should not get the tax shield benefit).

Depreciation and amortization (DA). DA also has a tax shield effect, like interest. However, the entire DA is added back. Example:

- Revenue: 100, all cash. Cash expenses: 70. Depreciation: 10. Tax 30%.
- $NI = (100 - 70)(1 - 30\%) - 10(1 - 30\%) = 14$
- $CFO = 100 - 70 - 6 = 24 = NI + D = 14 + 10$. CFO is about how much cash actually left the company. The $D = 10$ did not leave the company. We are adding back the full depreciation to NI to 'get back' to CFO.
- $NI + 10 = 14 + 10 = 24 = CFO$. So, $NI + Depreciation = CFO$

Working capital investment (= change). WCInv is *current* assets and liabilities like WC is AR, AP, inventories, but **excludes** cash and short-term debt (e.g. notes payable and current portion of long-term debt). **WCInv = current assets - current liabilities**, so an increase in AR is a rise in WCInv, so that's why it's subtracted from FCF.

- Exclude cash. We start with NI, CFO, etc. for cash. This has all residual cash. We want to find the non-cash elements that should be converted to cash, like WC. But if we include cash in WC, then we're adjusting the NI, CFO, etc. which was fine to begin with.

- Exclude short-term debt. They are financing items, not operating items. Moreover, they have interest costs which are already addressed in FCFF, FCFE. Goes into NB anyways.

FCFF is computed on an *after-tax* basis: after-tax cost of interest, and after-tax cost of capital. Alternatively, calculate FCFF using pretax cost of interest and pretax cost of capital. Just be consistent.

Note: add back preferred dividends for FCFF (firm = whole firm, including preferred shareholders), if they're subtracted.

Note: interest is the cost on the principal and found in the income statement; borrowing is the principal + interest and found in the balance sheet.

FCFF and FCFE.

$$FCFE = FCFF - I(1 - t) + NB$$

NB = net borrowing (e.g. = long-term debt + notes payable). Alternatively, can take the debt ratio of ratio of D and E to get the FCFE from FCFF.

$$FCFE = FCFF \times \frac{E}{D + E}$$

From EBIT and EBITDA. Assume below the only NCE = depreciation = D.
From EBIT:

$$FCFF = NI + D + I(1 - t) - FCInv - WCInv$$

$$NI = (EBIT - I)(1 - t) = EBIT(1 - t) - I(1 - t)$$

$$FCFF = EBIT(1 - t) + D - FCInv - WCInv$$

From EBITDA:

$$NI = (EBITDA - D - I)(1 - t) = EBITDA(1 - t) - D(1 - t) - I(1 - t)$$

$$FCFF = EBITDA(1 - t) + D(t) - FCInv - WCInv$$

For FCFE:

$$FCFE = EBIT(1 - t) + D - I(1 - t) - FCInv - WCInv + NB$$

$$FCFE = EBITDA(1 - t) + D(t) - I(1 - t) - FCInv - WCInv + NB$$

Since many NCEs other than DA are subtracted *after* EBIT and EBITDA, they don't have to be added back (i.e. they are already part of EBIT and EBITDA).

FCF based on cashflow **uses** instead of sources as above. Uses: retain or draw-down cash + cash equivalents, pay down more debt or borrow more debt, pay equityholders more (e.g. dividends, net share repurchases over issuances) or

less. Example FCFF: after-tax interest payments + new borrowing + dividend payments + net share repurchases. Example FCFE: dividend payments + net share repurchases.

Forecasting FCFF and FCFE. Can define 'incremental FC' = $FCInv - D$, **assume only NCE is D**, and that both 'increment FC' and incremental WC (= change in WC, same as before) grow at the same rate as sales. FCFF:

$$FCFF = EBIT(1 - t) - FCInc - WCInc$$

FCFE. Assume FCInc and WCInc depend on DR.

$$FCFE = NI - (FCInv - D) - WCInv + NB = NI - FCInc - WCInc + NB$$

$$NB = DR(FCInc) + DR(WCInc)$$

where DR = debt ratio (D divided by D + E).

$$FCFE = NI - FCInc - WCInc + DR(FCInc) + DR(WCInc)$$

$$FCFE = NI - (1 - DR)FCInc - (1 - DR)WCInc$$

Other issues with FCFF and FCFE. Sometimes, CFO doesn't match contents in BS; could be (1) acquisitions and divestitures and discontinued operations - e.g. inventory discrepancy from M&A; (2) nondomestic subsidies - e.g. currency translations.

Leverage changes, e.g. more debt increase interest tax shield (reduce taxes) and reduce cash flow to equity. Doesn't change FCFF (because there's nothing in the FCFF formula to change; intuitively, if the company borrows, it doesn't have any more cash to give to equityholders + debtholders because debtholders have to pay it back) but changes FCFE: (1) in the year debt is issued, FCFE rises by the amount of debt; (2) after debt issued, FCFE is reduced by FCInv purchases and after-tax interest expense.

Preferred stock and preferred dividends. Split out WACC into common stock, debt, and preferred stock. Preferred stock is *not* equity. FCFF includes preferred dividends because FCFF is for the entire firm:

$$FCFF = NI + NCE + I(1 - t) + PD - FCInv - WCInv$$

FCFE excludes preferred dividends, though NB = net new deb borrowing + net issuances of new preferred stock:

$$FCFE = NI + NCE - FCInv - WCInv + NB$$

If preferred dividends were paid in calculating net income, need to add them back for FCFF, though want to remove for FCFE.

Discretionary FCFF. Management wants to know how much 'discretionary' FCF there is.

$$FCFF = NI + NCE + I(1 - t) - Dividends - FCInv - WCInv$$

Approximate FCFF. Understates FCFF by after-tax interest expense, but it's a rough shortcut:

$$FCFF = CFO - capex$$

Multistage FCF models. Analogous to DDMs. Generally 2 or 3-stage models. E.g. simple 2-stage models:

$$FirmValue = \sum_1^n \frac{FCFF_t}{(1 + WACC)^t} + \frac{FCFF_{n+1}/(WACC - g)}{(1 + WACC)^n} - Debt$$

$$EquityValue = \sum_1^n \frac{FCFE_t}{(1 + t)^t} + \frac{FCFF_{n+1}/(r - g)}{(1 + r)^n}$$

Note: equity value can be obtained by FirmValue minus *market value* of debt. Might get slightly different answers to EquityValue calculated manually.

Operating versus non-operating assets. The above focus on operating assets. If there are non-operating assets then add (but must be *market* values, not book values).

$$\text{Value of firm} = \text{Value of operating assets} + \text{Value of non-operating assets}$$

Note: operating assets = FCInv - D + WCInv. Non-operating assets include noncurrent investments (market value, not book value), overfunded portion of pension plans.

Effects of changes in the financial statements. Assume below a 100 rise in everything and a tax rate of 40%.

- Depreciation. If increases by 100, then FCFF and FCFE rise by 40. EBIT falls by 100, but tax shield of 40 so NI falls by 60 + 100 NCE = 40. Getting the tax shield.
- Interest expense. If increases by 100, then FCFF doesn't change but FCFE declines by 60.
- Cash dividends, proceeds from issuing new common shares, common shares repurchase. Don't affect FCFF nor FCFE.

Note: real required return = real country return with adjustments, but **don't** subtract off the real long-term growth rate (that's the real growth rate)!

5.4 Market-based valuation: Price and EV Multiples

Can calculate P-E ratio here by (1) simple average of EPS and then using given P and divide by simple average EPS, (2) simple arithmetic average of ROE, and then apply to BV to get EPS, and take given P and divide by this EPS.

Method of comparables (comps, guideline assets or companies) - compare price or value to a current metric (which is not forecasted) such as earnings. Rely on law of one price - it's a comparison of price multiples or EV multiples (lower is more underpriced).

Method of forecasted fundamentals. Forecasting the metric, e.g. DCF which requires forecasting future values.

Justified price multiple or EV multiple - compare with estimated FV, e.g. median of industry or estimated intrinsic value.

P-E ratios. Numerator: usually use *common stock* price. Note disadvantage: can't use when EPS (earnings) are 0 or negative, or positive and tiny - though can use instead forward P-E or normal P-E or inverse price ratio (e.g. earnings yield).

- Trailing P-E: EPS usually last 12 months or 4 quarters. Generally use **diluted EPS** over normal EPS because different companies are more comparable this way. Adjust earnings to **exclude non-recurring items**, to get underlying or persistent or continuing or core earnings (requires analyst adjustment). Analyst should also adjust for cyclicalities and business cycles (due to 'Molodovsky effect' of high EPS at top of cycle and low EPS at bottom of cycle): use **normalized or normal EPS**.
- Current P-E: EPS last 6 months and forward forecast 6 months. Can also be defined as P-E using 'last-reported EPS'.
- Forward or leading or prospective P-E: EPS is next year. Next year can mean next 12 months, next 4 quarters, next FY, next calendar year, etc.
- Normalized P-E: because of business cycles, negative or 0 or tiny EPS, company size changes, use long-run expected average EPS. Normalized = mid-cycle EPS. Normalized or normal EPS: (1) historical average EPS, (2) average ROE multiplied by current book value per share (take whatever ROE is given and multiply by the common equity book value per share), (3) total assets multiplied by estimate of long-run ROA and divide by shares. (2) is better because it adjusts for size because based on *current* book value per share.
- Justified P-E. Trailing or forward. With $g > 0$, trailing P-E will be greater than forward P-E assuming E grows to support a growing D. Use an intrinsic P-E like from DCF or median P-E of group of similar companies.

- Predicted P-E. Regress P-E on other variables such as DPR, β , and earnings growth rate (EGR).
- NTM next twelve months P-E - can break up monthly, quarterly, annual of past and combine with part of future monthly, quarterly, annual earnings respectively.
- Inverse price ratio of P-E is **earnings yield**, E-P. Can address 0 or negative or small positive earnings.

If after comparing a specific company's P-E with peer or benchmark P-E and it's different, could be due to (1) different earnings growth, (2) different risk (operating or financial), or (3) other factors. If other factors, try looking at **PEG** ratio: P-E to expected EGR (P-E divided by percentage growth, e.g. 52 P-E divided by 35% = 1.51). Lower PEG ratio stocks are more attractive. PEG ratio disadvantages: (1) linear relationship assumption between P-E and EGR, but DDM shows it's this isn't true; (2) PEG doesn't factor in risk; (3) PEG doesn't factor in duration of EGR growth; (4) PEG doesn't capture growth of industry or economy as whole.

Average P-Es. Compare against own history average for relative valuation. If using market P-E average, these are often market cap-weighted; individual P-Es are then weighted by company market caps - this means average P-E is often influenced by large market-cap companies; so think about using median P-E instead.

Earnings yield, E-P. Inverse of P-E.

- Fed model: regress E-P on stocks and bond yields. $E/P \propto r_f$ where r is the bond yield. Intuition: when risk-free T-bonds offer higher yields than stocks, stocks become unattractive. Criticisms: (1) ignores equity risk premium, (2) assumes E-P and r relationship is linear but not true when r small, for instance.
- E-P is expected rate of return on no-growth stock. $P_0 = E_1/r_f + PVGO$, and with $PVGO = 0$, we get the Fed model.
- Yardeni model. $E/P = CEY = CBY - b \times LTEG$, where CBY is corporate bond yield, b is weight market gives to 5-year earnings projections, and LTEG is long-term earnings growth rate forecast for market index. If $b \uparrow \Rightarrow CEY \downarrow \Rightarrow P/E \uparrow$.

Caution when using P-E ratios.

- Business mix and leverage might change over time.
- Interest rates and economic fundamentals might be changing.
- Different companies have different abilities to pass-on their inflationary costs.

- Accounting standards might not be the same.
- Comparisons based on different markets might not be appropriate.

Pass-through ability. Start with Gordon growth model, but instead of g , use inflation = I . Assume company with no actual growth, i.e. $g = 0$.

$$P_0 = \frac{E_0(1 + I)}{r - I}$$

Use E (EPS) because assume all dividends passed to shareholders (e.g. unleveraged company), i.e. $b = 0$.

$$P_0 = \frac{D_0(1 + g)}{r - g}$$

$$\frac{D_0}{E_0} = 1 - b \Rightarrow D_0 = E_0(1 - b)$$

and with $g = I, b = 0$.

$$P_0 = \frac{E_0(1 + I)}{r - I}$$

Let λ ($0 \leq \lambda \leq 1$) represent percentage of inflation in costs the company can pass through to earnings.

$$P_0 = \frac{E_0(1 + \lambda I)}{r - \lambda I} = \frac{E_1}{r - I + I - \lambda I} = \frac{E_1}{(r - I) + (1 - \lambda)I}$$

Define $\rho = r - I$ to make above easier to calculate from facts given in questions:

$$P_0 = \frac{E_1}{\rho + (1 - \lambda)I}$$

$$\frac{P_0}{E_1} = \frac{1}{\rho + (1 - \lambda)I}$$

If $\lambda = 0$, then

$$P_0 = \frac{E_1}{r}$$

which is a perpetuity and the Fed Model. If a company has low cost inflation I and/or can pass through higher cost increases to customers, i.e. λ is small.

P-B value. B is book-value per share, take total equity and subtract off outstanding obligations (usually just preferred stock), and divide by number of outstanding **common shares**. In other words, common shareholders' equity is calculated as total shareholders' equity minus the value of preferred stock.

- Unlike earnings, B is generally positive.
- Book value per share suitable for valuing companies with liquid assets, such as finance, investment, insurance companies, as book values might approximate market values.

- Book value suitable if earnings are highly variable, 0 or negative.
- Book value suitable if company not going to be going-concern: then, earnings not relevant and book value much more important.
- Disadvantages: doesn't consider IP, human capital. Many assets and liabilities measured differently - not necessarily book value. Have to consider share repurchases - note: **if company repurchases shares at price above book value per share, then this reduces the book value of the company.**

Adjustments to book values. Sometimes use 'tangible' book values (removing intangible value - IP + goodwill). If one company using FIFO and another using LIFO, generally convert to FIFO. Add off-balance sheet items. Impairments, when FV doesn't match BV.

Justified P-B ratio.

$$\frac{P_0}{B_0} = \frac{ROE - g}{r - g}$$

This is intuitive; the P-B ratio should as a first intuition grow at ROE since ROE is based on book value equity. Proof. First define

$$ROE = \frac{E_1}{B_0} \rightarrow E_1 = B_0 \times ROE$$

and substitute into

$$V_0 = \frac{E_1(1 - b)}{r - g}$$

which comes from

$$\frac{D_0}{E_0} = 1 - b \Rightarrow D_0 = E_0(1 - b)$$

$$V_0 = P_0 = \frac{D_0(1 + g)}{r - g} = \frac{E_0(1 - b)(1 + g)}{r - g}$$

to get

$$V_0 = \frac{B_0 \times ROE \times (1 - b)}{r - g}$$

$$\frac{V_0}{B_0} = \frac{ROE \times (1 - b)}{r - g} = \frac{ROE - g}{r - g}$$

Residual income model.

$$\frac{P_0}{B_0} = 1 + \frac{\text{PV of expected future residual earnings}}{B_0}$$

which is intuitive because

$$P_0 = B_0 + \text{PV of expected future residual earnings}$$

Proof. Use the justified P-B ratio.

$$\frac{P_0}{B_0} = \frac{ROE - g}{r - g}$$

and since

$$ROE - g = ROE - g + r - r = (r - g) + (ROE - r)$$

then

$$\frac{P_0}{B_0} = \frac{ROE - g}{r - g} = 1 + \frac{ROE - r}{r - g} = 1 + \frac{B_0(ROE - r)}{B_0(r - g)} = 1 + \frac{B_0(ROE - r)/(r - g)}{B_0}$$

where the second value is the PV of expected future earnings over the opportunity cost.

Price-sales ratio. Sales = revenue per share, less subject to manipulation (but watch out for bill-and-hold manipulation). Sales positive even when earnings negative; can use sales even when negative, 0, or small positive. P-S ratio good for mature, cyclical, and zero-income companies. However, using sales is a mismatch because doesn't consider expenses.

Justified P-S ratio.

$$\frac{P_0}{S_1} = \frac{(E_1/S_1)(1 - b)}{r - g} = \frac{\text{Profit margin}(1 - b)}{r - g}$$

Proof. Begin with

$$P_0 = \frac{D_1}{r - g}$$

and note that

$$\frac{D_1}{E_1} = 1 - b \rightarrow D_1 = E_1(1 - b)$$

substituting,

$$P_0 = \frac{E_1(1 - b)}{r - g}$$

and then divide both sides by S_1 . Then with growth,

$$\frac{P_0}{S_0} = \frac{(E_0/S_0)(1 - b)(1 + g)}{r - g} = \frac{(\text{PM})(1 - b)(1 + g)}{r - g}$$

Note that profit margin (PM) also affects the justified P-S ratio indirectly through g:

$$g = b \times ROE = b \times PM_0 \times \frac{\text{Sales}}{\text{Total assets}} \times \frac{\text{Total assets}}{\text{Shareholder equity}}$$

So, higher PM boosts the increase in P-S directly and indirectly (in the numerator g and denominator g).

$$\frac{P}{E} \times PM = \frac{P}{S}$$

Price to cashflow. Be careful of 'cashflow' definition! Ratio of share price to any of multiple CF definitions. However, 'P-CF' ratio is more specific: share price to CF = E + NCE.

- EPS + per-share D, A, depletion (CF). Limitation: ignores ΔWC and non-cash revenue (e.g. deferred revenue recognition like subscription revenue before cash received, deferred tax assets), and not an FCF concept.
- CFO. Limitation: not FCF.
- FCFE. Limitation: often negative and variable.
- EBITDA. Limitation: ignores ΔWC and non-cash revenue, not an FCF concept.

P-cashflow. Cashflows are generally more stable than earnings. It can also sometimes address aggressive accounting (e.g. accelerated PPE depreciation can lead to low NI and then fast NI growth versus using linear depreciation but cashflows will be the same). Using FCFE instead of cashflows, but FCFE can be volatile because PPE is effectively uncapped and FCFE is often therefore negative; however, could use average FCFEs. EBITDA might alternatively be used for cashflows, though it tends to be better for debt-adjusted value comparisons, e.g. EV-EBITDA.

$$V_0 = \frac{(1+g)FCFE_0}{r-g}$$

Trailing P-cashflow. Current price divided by most recent 4Q cashflows.

Dividend yield, D-P. Sometimes P-D appears, but many stocks pay 0 dividends so not practical. Trailing DY = **current dividend rate** divided by current market price per share. Dividend rate when quarterly regular dividends = take most current dividend $\times 4$ (sometimes, take last 4 dividends); if semi-annual interim + final dividends (which differ in magnitude), then sum them to get dividend rate. Leading DY = **forecasted DR** divided by current market price.

$$\frac{D_0}{P_0} = \frac{r-g}{1+g}$$

If g is small, then value stock because value stocks have low prices and low growth. So value stocks have high DY.

Total return, %. TR.

$$TR = \frac{\Delta P + D}{P} = \frac{Pg + D}{P} = g + \frac{D}{P} = EGR + \frac{D}{P}$$

where $g = EGR$ from above, or intuitively,

$$P = E \times \frac{P}{E}$$

so,

$$\Delta P = \Delta E \times \frac{P}{E}$$

$$\frac{\Delta E}{E} = \frac{\Delta P}{P}$$

EV-EBITDA. EV is not very sensitive to financial leverage, which is advantageous compared to other measures above. Takes a control perspective. Better than P-CF when comparing companies with different capital structures. $EV = MV (\text{debt} + \text{equity} + \text{preferred equity}) - \text{cash} - \text{short-term investments}$. EBITDA is pre-interest (and therefore less sensitive to financial leverage), pre-dividend, and pre-tax. EBITDA often positive when EPS negative. Can use EV-EBITDA to value common equity, even though it has debt in it.

EBITDA variations. EBIT, EBITA, EBITDAR, FCFF.

Note: $\text{Equity} = EV - \text{debt value}$. EV should in simple cases $= DCF = DDM = RI$ firm value estimates.

EBITDA disadvantages.

- WC is growing. EBITDA doesn't subtract accruing charges like rising AR and inventories or shrinking AP - it's not 'cash' or 'short-term investments'. However, CFO does. So, EBITDA overestimates operational cashflows.
- FCFF is better valuation. **Only if depreciation = capex is EBITDA like FCFF.**

EV. $MV (\text{common equity} + \text{debt} + \text{preferred equity} + \text{minority interest} = \text{Total invested capital TIC}) - \text{non-earning interests (cash, cash equivalents, short-term investments)}$. But often just use BV for debt and minority interests. Note: current portion of long-term debt is counted as part of 'debt'.

Note:

$$EV = E + D - C = V_0 - C$$

so, $EV = DCF (= DDM = RI)$ valuation minus cash, or if there's nearly 0 excess cash, then they're equivalent.

EV-EBITDA. Should be positively related to (1) expected FCFF growth rate, (2) expected profitability as measured by ROIC (return on invested capital; operating profit after tax divided by invested capital), and negatively related to (3) WACC.

EV-S ratio. P-S ratio is not good because some company sales will be used to pay down debt, but P is just for equityholders. EV-S is better and addresses problem even if capital structures are different. EV-S is similar to profit margin.

Reconciliations of multiples - between different countries and accounting standards. GAAP used to require reconciliation, even for non-IFRS but IFRS-reporting entities don't have to anymore. Of multiples, P-CFO and P-FCFE are generally least affected by accounting differences, and P-B, P-E and multiples involving EBITDA are most affected. If P-EBITDA conflicts with EV-EBITDA, then use latter.

Momentum valuation indicators (a technical indicator, i.e. based on relative price movement). Relate values (e.g. multiples) to their own past values, or to an expected value (which represents a difference in time of information). Earnings surprise and relative-strength. Earnings surprise:

$$UE_t = EPS_t - E(EPS_t)$$

UE is earnings surprise. Scaled earnings surprise is

$$SEP_t = \frac{EPS_t - E(EPS_t)}{\sigma(E(EPS_t))}$$

Standardized unexpected earnings (SUE)

$$SUE_t = \frac{EPS_t - E(EPS_t)}{\sigma(EPS_t - E(EPS_t))}$$

Relative strength indicators (another technical indicator). Price momentum of stock's own history or compare price momentum to other stocks. But lots of opportunity for data snooping. Can be used to see if a stock is moving closer to or further from analyst's guess of intrinsic value.

Harmonic mean. Reduces impact of large outliers, but multiples effect of small outliers. Use simple

$$X_{simple} = \frac{n}{\sum_i^n \frac{1}{X_i}}$$

for weighted harmonic means, divide by n,

$$X_{weighted} = \frac{1}{\sum_i^n \frac{w_i}{X_i}}$$

Backtesting. Watch out for look-ahead bias, where backtesting involves quantities that could not have been known at that time based on other data in the model - i.e. time-inconsistency of data.

Stock selection and screens. Often just value-based (e.g. minimum dividend yield, large cap, etc.). But should also include growth-factors, momentum, risk, WACC. If use max P-E ratio, specify that $E > 0$. Have diversification of industries and companies in stock screens.

5.5 More on Gordon growth model

Gordon growth model.

- E, D, and P must grow at same rate, resulting in constant P-E.
- Capital appreciation cannot be greater than EGR unless P-E increases.

P-E ratio and EGR. If EGR rises, then P-E ratio **rises** paradoxically. This is because:

$$\frac{P_0}{E_1} = \frac{D_1}{r - g} = \frac{DPR \times E_1}{r - g}$$

So, P rises even faster than E.

5.6 Residual income valuation

$$Value = B_0 + RI$$

Residual income = NI minus common shareholder opportunity cost minus preferred shareholder dividends. Want to get to earnings for **common shareholders**. It considers all costs of capital (including debt, which is subtracted to get to NI). Sometimes called economic profit, abnormal earnings, economic value added (EVA), Edwards-Bell-Ohlson model, discounted abnormal earnings model, or others.

Clean surplus accounting. NI reflects all changes in book value of equity other than ownership transactions. Generally assume 'ownership transactions' cancel out, e.g. new share issuances offset share repurchases. Note this can be violated by e.g. items in OCI which bypass the IS but end up affecting shareholder book value equity.

Note: in some countries, interest expenses are not tax deductible - so make appropriate adjustments.

Calculating RI. Approaches:

- $RI = NI$ minus equity charge. Equity charge = equity capital (the BV of common shareholder equity) $\times$ equity cost of capital (r). This is just a comparison of ROE v r .
- $RI = NOPAT$ - capital charge. Capital charge = debt capital $\times$ debt capital cost (debt charge) + equity capital $\times$ equity capital cost (equity charge). $NOPAT$ = net operating profit minus taxes. If uncomplicated finances, gives same answer as first method.

- $RI = (ROIC - \text{effective capital charge}) \times \text{beginning capital}$. $ROIC = \text{return on invested capital} = \text{NOPAT divided by total (e.g. D+E) capital}$. $\text{Effective capital charge} = \text{capital charge divided by total capital}$.
- Commercial implementations of RI. Most common: EVA. Aims for approximation of economic profit: $EVA = \text{adjNOPAT} - (WACC \times TC)$, where NOPAT is adjusted.
- Market value added = MV of company - BV of capital.

If $RI < 0$, share price should be lower than BV. Take the RI and divide by number of common shares. This is the PV of the perpetuity which is added to the BV of equity. If negative, then lower than BV. This is the P_0 . From above,

$$r = \frac{E_0}{P_0}$$

so, the $EPS = rP_0$.

Uses GAAP. Some propriety adjustments concern R&D, matching investments with revenue, use only cash taxes, adjust for LIFO reserves, turn all operating leases into capital leases, adjust non-recurring items.

Math formulation of RI.

$$V_0 = B_0 + \sum_1^{\infty} \frac{RI_t}{(1+r)^t} = B_0 + \sum_i^{\infty} \frac{E_t - rB_{t-1}}{(1+r)^t}$$

To calculate RI using above, the first year: use current B and get r (from CAPM) and E forecast. For second year: calculate B at beginning of year (from last year's B + forecasted EPS - forecasted DPS) and repeat.

RI captures most of its value from BV at the beginning, in contrast to DDM and DCF which capture most value at terminal value.

Comparison with DDM. Results same, but RI recognizes value **earlier** because there's BV at the beginning, while DDM weights on dividends which come later (especially terminal value).

$$V_0 = \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \frac{D_3}{(1+r)^3} + \dots$$

Use the **clean surplus relation**

$$B_t = B_{t-1} + E_t - D_t$$

$$D_t = E_t - (B_t - B_{t-1}) = E_t + B_{t-1} - B_t$$

Substitute into the above,

$$V_0 = \frac{E_1 + B_0 - B_1}{(1+r)^1} + \frac{E_2 + B_1 - B_2}{(1+r)^2} + \frac{E_3 + B_2 - B_3}{(1+r)^3} + \dots$$

and re-arranging,

$$V_0 = \sum_i^{\infty} \frac{E_i}{(1+r)^i} + \sum_i^{\infty} \frac{B_{i-1} - B_i}{(1+r)^i}$$

Take the second term, which telescopes,

$$\frac{B_0}{1+r} + \left(\frac{-B_1}{(1+r)} + \frac{B_1}{(1+r)^2} \right) + \left(\frac{-B_2}{(1+r)^2} + \frac{B_2}{(1+r)^3} \right) + \dots$$

and assume it goes to infinity so the 'final' term is 0. Then

$$\begin{aligned} &= \frac{B_0}{1+r} + B_1 \left(\frac{1 - (1+r)}{(1+r)^2} \right) + \left(\frac{1 - (1+r)}{(1+r)^3} \right) + \dots \\ &= \frac{B_0}{1+r} + \left(\frac{-rB_1}{(1+r)^2} \right) + \left(\frac{-rB_2}{(1+r)^3} \right) + \dots \\ &= \frac{B_0}{1+r} + \frac{-rB_0}{1+r} + \frac{rB_0}{1+r} + \left(\frac{-rB_1}{(1+r)^2} \right) + \left(\frac{-rB_2}{(1+r)^3} \right) + \dots \\ &= \frac{(1+r)B_0}{1+r} + \frac{-rB_0}{1+r} + \left(\frac{-rB_1}{(1+r)^2} \right) + \left(\frac{-rB_2}{(1+r)^3} \right) + \dots \\ &= B_0 + \frac{-rB_0}{1+r} + \left(\frac{-rB_1}{(1+r)^2} \right) + \left(\frac{-rB_2}{(1+r)^3} \right) + \dots \end{aligned}$$

Now recombine back with the first term,

$$V_0 = \sum_i^{\infty} \frac{E_i}{(1+r)^i} + B_0 + \frac{-rB_0}{1+r} + \left(\frac{-rB_1}{(1+r)^2} \right) + \left(\frac{-rB_2}{(1+r)^3} \right) + \dots$$

to get

$$\begin{aligned} V_0 &= B_0 + \frac{E_1 - rB_0}{(1+r)^1} + \frac{E_2 - rB_1}{(1+r)^2} + \frac{E_3 - rB_2}{(1+r)^3} + \dots \\ V_0 &= B_0 + \sum \frac{RI_t}{(1+r)^t} = B_0 + \sum \frac{E_t - rB_{t-1}}{(1+r)^t} \end{aligned}$$

Excess earnings method. Related to RI, used in tax valuation.

$$V_0 = B_0 + \sum \frac{(ROE_t - r)B_{t-1}}{(1+r)^t}$$

Note: financial analysis ROE is usually based on average annual equity, whereas tax ROE is based on beginning of year equity.

RI growth. RI model assumes no growth. However, adding growth (constant ROE and constant $g = EGR$), we get P-B ratios with growth:

$$\frac{P_0}{B_0} = \frac{ROE - g}{r - g}$$

Re-arranging,

$$\frac{P_0}{B_0} = 1 + \frac{ROE - r}{r - g}$$

$$P_0 = B_0 + B_0 \frac{ROE - r}{r - g}$$

Since $RI = E - \text{cost of equity} = B_0 ROE - r B_0$,

$$P_0 = B_0 + \frac{B_0 ROE - r B_0}{r - g}$$

which is RI model with growth. So, RI model is most related to P-B ratio.

Based on the above equation, if $ROE > r \Rightarrow MV > BV$ and if $ROE < r \Rightarrow MV < BV$.

Tobin's q. RI is also related to q.

$$\text{Tobin's q} = \frac{\text{MV debt + equity}}{\text{Replacement cost of total assets}}$$

Tobin's q is higher if greater productivity. Lack of data is problem. Tobin expected average q = 1.

Multistage RI. If terminal value is a defined premium over BV, then

$$V_0 = B_0 + \sum_t^T \frac{E - r B_{t-1}}{(1+r)^t} + \frac{P_T - B_T}{(1+r)^T}$$

Remember to subtract off B_T to get value *above* the book value (but, **don't** subtract for DDM!). In the formula, the terminal value is the discounted surplus share value over the book value. Alternatively,

$$V_0 = B_0 + \sum_t^T \frac{(ROE_t - r) B_{t-1}}{(1+r)^t} + \frac{P_T - B_T}{(1+r)^T}$$

If RI declines to terminal r, then

$$V_0 = B_0 + \sum_t^{T-1} \frac{E_t - r B_{t-1}}{(1+r)^t} + \frac{E_T - r B_{T-1}}{(1+r-\omega)(1+r)^{T-1}}$$

where $0 < \omega < 1$ and is a **persistence factor** which indicates how fast RI will fade. Note: the 'terminal value' is at T, but discount from T-1. $\omega = 1$ means no fading (RI same level in perpetuity) whereas $\omega = 0$ means RI will be 0 after initial forecast horizon - this can be seen from the equation: the terminal value 'should' be divided by simply r for a perpetuity (i.e. $\omega = 1$) and by $1+r$ for fast declining values.

Companies with extreme accounting rates of return typically have low persistence factors. Companies with strong market leadership positions and low dividend payouts are likely to have high persistence factors.

Note: if no dividends are specified, may need to assume a level of dividends that leads to a steady-state, e.g. all NI turn into dividends where EBIT is constant. If some NI is added to shareholder equity, then shareholder equity increases, so at a constant equity cost of capital, the opportunity cost of equity blows up and since EBIT is constant (and ROE is constant, not declining), then RI falls to negative quickly, leading to 0 intrinsic value!

RI models versus DDM and DCF.

- RI models have most value at beginning; DDM and DCF most value at end. RI models better when forecasting terminal values is uncertain.
- RI models work when **clean surplus accounting** is obeyed - if it exists, aggressive accounting of higher BV but lower E; but this theory doesn't actually occur in practice. RI models use readily available accounting data, but must be careful about manipulation of that accounting data.

Accounting. On the second point above, RI model accounting-derived inputs may need to adjust accounting for (1) off-balance sheet items and (2) for OCI. Some aggressive accounting, e.g. capitalizing + amortizing instead of expensing, will lead to *both* NI *and* BV being higher than actual.

- Restructuring charges. The restructuring charge is a non-recurring item and not indicative of future earnings. In applying a residual income model, it is important to develop a forecast of future residual income based on recurring items. By upward adjusting the firm's net income, by adding back the R2 million restructuring charge to reflect the fact that the charge is non-recurring, future earnings will be more accurately forecasted.
- Off-balance sheet items. Must adjust BV. E.g. operating leases don't affect shareholder equity as off-BS assets offset off-BS liabilities, but affect E. Other examples: inventories (LIFO, FIFO, etc.), deferred tax assets and liabilities, reserves and allowances (incl. bad debt allowances), intangible assets.
- Intangible assets. If a company A buys company B, and there is excess paid over the FMV of its assets - extra could be goodwill, but assign to 10-year licenses previously not in company B's books (because company B developed the intangible asset and paid just a small fee for registration, it exists but is not recorded as an asset) which is then amortized over 10 years. Then, the consolidated statement would show low E and therefore ROE because of the amortization - RI consolidated will be less than RI (A + B) prior to transaction. If company A issued shares to buy company B, it won't matter to RI - the BV rises but the rest of the formula falls.

- Research and development. Another intangible asset. GAAP only allows expensing except feasible to develop software; IFRS allows capitalization and amortization. They will give different RIs in different years. Regardless, if expensed or capitalized, decline in RI but if productive R&D, then E will eventually rise and offset. So don't have to adjust RI accounting inputs for R&D.
- EVA (the commercial version). R&D costs are added back to earnings to calculate NOPAT because these costs are capitalized and amortized rather than expensed as in GAAP and IFRS. Therefore, decreasing R&D will serve to lower NOPAT and thus lower EVA.
- Non-recurring items. BV is fine. But adjust E and ROE to account for non-recurring items.
- Other aggressive accounting. Early recognition, cookie jar accounting, etc.
- OCI. If income in OCI, then not in NI but still factored into shareholder equity. So, OCI items 'bypass' the IS. Shareholder equity is accurately adjusted, so adjust E or ROE in the RI formulas.

Note: 'Pretax income' = EBT

5.7 Public v private company valuation

Private companies differ from public companies: limited disclosures (reduces valuation), usually concentrated ownership (often reduces valuation because main shareholder can work to disadvantage of others), early-phase, illiquid (reduces valuation), smaller, usually higher required return on equity (reduces valuation). Includes companies in liquidation.

Private company valuations - for ownership stake sales, compliance, litigation, financing, IPO, LBO or MBO, share-based compensation, impairment testing, taxes, performance measurement, etc.

Private company valuation. If using DCF-type model, then consider adjusting numerator (FCFF), denominator (cost of capital), and then post-calculation adjustments for control, liquidity, etc.

Private company buyers offer prices for private firms reflecting improvements the buyer will make after acquisition. They *don't* reflect higher discount rates that apply to private companies.

Financial statements. Often no history for private companies. However, can get **reviewed financial statements** which are accompanied by auditor letter and **compiled financial statements** which don't have such letters.

Earnings normalization. While there can be normalization of revenue for cyclicals, here it's usually about adjusting for better comparison with public companies, non-recurring items, goodwill, and **related-party transactions** (versus arms-length transactions). Related party transaction adjustments, e.g. private company CEO pays themselves too much or too little or simply gives themselves a dividend instead (in which case, need to *increase* the expenses to account for a regular CEO income expense). Excess employee benefits, use of company property, life insurance, loans to shareholders, etc. might be recorded as private company expenses but need to be adjusted *down*.

WACC and debt. May need to be adjusted, e.g. private company may have low *debt capacity* - reduced ability to borrow because small, little public information, hard to value, illiquid, etc. - so their current capital structure may be sub-optimal. Note: if debt cost of capital for acquiror is less than debt cost of capital for acquiree, this is a *transfer of value from acquiror to acquiree* and should be incorporated into private company valuation.

WACC and equity (CAPM). Can use:

- CAPM. Regular CAPM.
- Expanded CAPM. CAPM + small-cap premium + company-specific premium.
- Build-Up Approach. CAPM (but with $\beta = 1$) + small-cap premium + industry-risk premium + company-specific premium

Private company premia. Strategic buyers would pay a synergistic control premium. Financial buyers would pay a control premium, but they wouldn't pay a synergistic control premium (they might want control, to e.g. leave management in place and have control to change if they want).

Private company discounts. Discounts for lack of control (DLOC) and lack of marketability (DLOM):

$$DLOC = 1 - \frac{1}{1 + \text{Control premium}} = \frac{CP}{1 + CP}$$

DCF valuations are generally assumed to have control. Model DLOM as at-the-money put option using options pricing models, e.g. with volatility. Calculate DLOM by:

- Calculate put option on the private company. Since there are no public shares, use a comparable public company. Calculate premium on at-the-money put options. So the strike price of the put option is discounted forward. Use BSM to calculate the premium.
- For the put option calculation, assume a suitable time to expiry.

- Take the premium and divide by the current share price of the public company.

But note: these models assume expiry dates and don't assume the underlying private company valuation changes as a result of the put options (which normally they should), so they aren't 100% proxies for DLOM.

$$\text{Total discount} = 1 - (1 - DLOC)(1 - DLOM)$$

Private company valuation approaches. Income approach: capitalized cashflow method and excess earnings method. Market approach. Asset-based approach.

Income-based approaches. FCF approach:

$$IV_t = \sum \frac{FCFF_{t+1}}{(1 + WACC)^t} + \frac{TV_{t+n}}{(1 + WACC)^n}$$

Terminal value calculated by different ways.

Capitalized cashflow method (CCM). Capitalized cashflow (CCF) is

$$V_t = \frac{FCFF_{t+1}}{WACC - g}$$

This is equal to EV (enterprise value). Derivation. The **reinvestment rate** is

$$RIR = \frac{g}{WACC}$$

which is analogous to b as $WACC \approx ROE$ in a steady state:

$$g = WACC \times RIR \approx ROE \times b$$

since b is the retention ratio, or the amount of capital that needs to be kept to maintain the company, it should be deducted to get the remaining FCFF. So,

$$FCFF = NI + NCE + I(1-t) - \Delta WC - \Delta FC = EBIT(1-t) - (\Delta WC + \Delta FC)$$

$$FCFF = EBIT(1-t) - EBIT(1-t)RIR = EBIT(1-t)(1-RIR)$$

So,

$$V_t = \frac{EBIT_{t+1}(1-t)(1-RIR)}{WACC - g}$$

This assumes WACC is constant, which means capital structure probably remains constant. To get the firm intrinsic value, subtract off debt (*market value of debt*). Note we assume capital structure is constant:

$$V_0 - MV(D) = IV_t$$

$$IV_t = \frac{FCFE_{t+1}}{r_e - g}$$

where $r_e - g$ is the **capitalization rate**, and r_e is the equity cost of capital.

Excess earnings method (EEM). Best for valuing intangible assets, rather than entire companies. Based on residual income model. Subjective, requires guesses on WC (working capital) and FA (fixed assets).

$$V_0 = FMV(\text{tangible assets}) + RV_t$$

where tangible assets = WC + fixed assets. Further,

$$RV_t = \frac{RI_t(1 + g)}{r_{RI} - g}$$

where r_{RI} is the cost of capital of intangible assets. Value generated beyond assets is considered to be intangible assets instead of WC and fixed capital. RI = residual income, which is total minus tangible assets.

$$RI_t = NI - (WC \times r_{WC}) - (FA \times r_{FA})$$

where WC (working capital) and FA (fixed assets) are valued at FMV. r_{WC} is usually low, r_{FA} is usually higher, and intangibles are usually highest-risk so highest r_{RI} .

Market-based approaches. Compare private companies to benchmark public companies. Most popular approach, favorable to theoretical approaches above. 3 types: (1) guideline public company method GPCM compare to publicly traded company, (2) guideline transactions method GTM based on data from control of entire public or private companies, (3) prior transactions method considering actual transactions of stock in the private company.

GPCM. Public and private companies of comparison will have different degrees of leverage. Moreover, CAPM β assumes owners and investors are well-diversified, but private company owners are rarely well-diversified and often have most of their wealth tied up in the company. Adjust β for these differences. First, unlever beta:

$$\beta_u = \frac{\beta_l}{1 + (1 - t)\frac{D}{E}}$$

where D, E, t are for the public company. Derivation. From Modigliani-Miller,

$$r_e = r_u + (r_u - r_d)\frac{D}{E}(1 - t)$$

From CAPM,

$$r_u = r_f + \beta_u(r_m - r_f)$$

$$r_e = r_f + \beta_e(r_m - r_f)$$

$$r_d = r_f + \beta_d(r_m - r_f)$$

Since $e = l$ and we assume near risk-free debt, i.e. $\beta_d \approx 0$,

$$\beta_l = \beta_u + \beta_u \frac{D}{E}(1 - t) = \beta_u \left(1 + \frac{D}{E}(1 - t)\right)$$

$$\beta_u = \frac{\beta_l}{1 + (1 - t) \frac{D}{E}}$$

After unlevering beta, re-lever with private company D^*, E^*, t^* :

$$\beta_l^* = \beta_u \left(1 + (1 - t^*) \frac{D^*}{E^*}\right)$$

We unlever and re-lever because public companies tend to have lots of leverage, while private companies typically do not.

6 Financial statement analysis

6.1 Intercompany investments

Investees. If less than 20% 'financial asset', if 20%-50% then associate, if 50% then controlling; sometimes can be joint ventures if joint control. Percentages are direct + indirect.

Financial assets. IFRS 9. Debt:

- Initially valued at 'fair value'.
- If (1) financial assets are held to collect cash flows (business model test) and (2) cash flows are solely principal + interest on principal (cash flow characteristic), then measured at **amortized cost**. Amortized cost - subtract dividends/interest income, repayments, impairment. Effective interest example: Buy 11 M USD of bonds which pay 5% coupons per annum on par, where par is 10 M USD of bonds. Interest at time of purchase of bonds was 4%. So, bond buyer pays 440,00 USD in interest but gets 500,000 in coupons. So amortize the 11 M USD by 60,000 USD. This is the actual economic yield.
- If meet two criteria above but may be sold 'hold-to-collect and sell' (IFRS) or 'available for sale' (*not* 'held for sale' which is for discontinued operations) (GAAP), **fair value through OCI** (FVOCI). Otherwise, FVPL.
- IFRS: Can choose **fair value through profit or loss** (FVPL) to avoid 'accounting mismatch'. Accounting mismatch. Some assets and liabilities are measured at amortized cost while others at fair value. To avoid, have option.
- Under all 3 treatments, debt is recorded in the BS. Under amortized cost, no fair market value changes are recorded until it is sold. Under OCI,

fair market value changes are only recorded in OCI and recognized upon immediately in the BS but only in the IS upon sale. Under FVPL, fair market value changes are recorded in the debt regularly.

- Under all 3 treatments, impairments are treated the same - there is a change in the BS and the change appears in the IS.
- Under all 3 treatments, interest income, impairments, and fair value changes are in the IS, though in the first 2 the FMV changes only occur upon sale.
- Under all 3 treatments, CF is the same. IFRS: debt investment is CFI while interest is CFO or CFI.

Debt reclassification. Irrevocable unless business model (i.e. objective) for holding assets changes a lot and affects operations.

GAAP debt: similar to IFRS. Note in CF that debt investment is CFI and interest is always CFO.

IFRS 9 equities:

- If held for trading, FVPL.
- Others: irrevocably choose FVPL or FVOCI.
- If FVOCI, only dividend income is profit or loss.
- GAAP equity: only FVPL.
- CF statement and classifications similar to debt for both IFRS and GAAP.

IFRS 9 derivatives: FVPL.

IFRS 9 loan impairment: 'expected losses model'. If performing loan (low credit risk or even heightened credit risk), then impair expected 12-month losses. If NPL, then impair expected lifetime losses.

Associate (20%-50%), but also if less than 20% + significant influence (e.g. BoD, dependencies). **Equity method.** Start with % of the cost of the business and place on balance sheet. Then ADD investee's income % in IS and BS and SUBTRACT dividend % in BS from the initial cost paid for the specified shares. The result is the **Equity Investment** in the BS as e.g. 'non-current asset'. If the result goes to 0, stop; start again if gains exceed all losses even unrecorded losses at 0. The income goes on IS. One line entry in each of IS and BS.

As the acquired company's PPE depreciates, this is not reflected in the acquirer's financial statements. Instead, the acquiree's depreciation is reflected in its net income and dividends which are in the acquirer's IS and BS respectively.

Joint ventures: (1) contract between 2 or more venturers, (2) contract establishes joint control. Equity method. Sometimes, **proportionate method** - show assets, liabilities, income, expenses of JV to be combined or proportioned. Should get same final answer as equity method, though ratio analysis will be different. No longer used by IFRS and GAAP, though there are legacy cases.

A joint venture's shareholder equity is just split as an asset and added to carrying values of the parent company's financial statements.

Bargain purchases. If price paid is less than FV net assets. IFRS and GAAP require a profit or loss be recorded at time of purchase. Subsequent changes are also profit or loss.

Excess purchase price - accounting. Excess purchase price = actual purchase price is in excess of *book value* of shares because investee's assets and liabilities reflect historical costs rather than fair value. PPE accounting: IFRS allows either historical or FV (minus accumulated depreciation) while GAAP only allows historical (minus accumulated depreciation). Take actual purchase price minus portion of investee's *net book assets* (i.e. after subtracting liabilities), subtract % of PPE and land (usually BV=FV for current assets or any liabilities) FV-BV and allot the residual to goodwill. The excess FV-BV PPE and land, as well as goodwill, is accounted by the investor separately, e.g. PPE above book value is amortized in its own right, and goodwill is tested regularly for impairment. Example:

	Book Value		Fair Value
Current assets	€10,000		€10,000
Plant and equipment	190,000		220,000
Land	120,000		140,000
	€320,000		€370,000
Liabilities	100,000		100,000
Net assets	€220,000		€270,000

Purchase price	€100,000
30% of book value of Brown ($30\% \times €220,000$)	66,000
Excess purchase price	€34,000
Attributable to net assets	
Plant and equipment ($30\% \times €30,000$)	€9,000
Land ($30\% \times €20,000$)	6,000
Goodwill (residual)	19,000
	€34,000

In the above, can alternatively take the purchase price minus 30% of the fair value net assets to get the goodwill. The 'above-bookvalue' PPE depreciation is done by the investor company as a side calculation, and then the depreciation is recorded in the IS and BS as part of the one line entries in each.

Excess purchase price = actual price - net asset **book value**. **Goodwill** = actual price - net asset **fair value**.

Fair value option. Instead of hassling with excess purchase prices and having to adjust the one-line BS entry for income and dividends, can simply make an initial and irrevocable choice (GAAP) to account the purchase at fair value. Under IFRS, can do this if acquiree shares 'held-for-sale' and only VCs, mutual funds, and other institutional investors can do this. Dividends, interest, unrealized gains and losses (i.e. $FV_{old} - FV_{new}$) are simply profit and losses in the IS.

Goodwill and impairment. Goodwill not separately reported and not specifically tested for impairment. If objective evidence of impairment, test entire equity method investment. IFRS: Test for impairment as usual: compare carrying amount to higher recoverable amount: higher of (a) VIU and (b) FV less selling costs. Impairment loss on IS, carrying value of investment on BS is reduced directly or reduced using an allowance account. Can reverse impairment under IFRS if underlying numbers changed. GAAP: If FV below carrying value *and* decline is permanent, recognize loss in IS and reduce investment on BS to FV.

Transactions with associates, e.g. goods and services. If investor sells to associate (downstream sale) or associate sells to investor (upstream sale), investor excludes the part of the sale until the sale is *used* or *sold onward to third parties* - until then, it's like an internal asset transfer. With-hold the % of **profit** from the sale, not the sale revenue.

Business combinations. IFRS doesn't have different types of business combinations; GAAP separates into merger (1 company remains), acquisition (re-

main separate companies), consolidation (new 1 company created), special purpose/variable interest entities. Use **acquisition method**.

Acquisition method. IFRS and GAAP. FV of acquisition assets and liabilities (including present and known contingent *obligations* and indemnification assets and liabilities) is basis. Fees for acquisition are expensed. IFRS allows 'partial' goodwill, but GAAP only allows whole goodwill as entity is one-whole.

Recognition and Measurement of Goodwill

Acquirer contributes \$800,000 for an 80% interest in Acquiree. The identifiable net assets have a fair value of \$900,000. The fair value of the entire entity is determined to be \$1 million.

	IFRS Partial Goodwill
Fair value of consideration	\$800,000
80% of Fair value of identifiable net assets	720,000
Goodwill recognized	<u>\$80,000</u>
	IFRS and US GAAP Full Goodwill
Fair value of entity	\$1,000,000
Fair value of identifiable assets	900,000
Goodwill recognized	<u>\$100,000</u>

Excess purchase price = actual price - net asset **book value**. **Goodwill** = actual price - net asset **fair value**.

Bargain purchases. If price paid is less than FV net assets. IFRS and GAAP require a profit or loss be recorded at time of purchase. Subsequent changes are also profit or loss.

Example 7 (100% acquisition). Acquisition method post-combination balance sheet. Note shares are issued for the purchase and no other purpose.

- Par common stock is 'capital stock' in shareholder equity; extra from extra market value is 'additional paid-in capital'. Acquiree puts shares back into company and removes capital stock + additional paid-in capital.
- Take acquirer's BV + 100% (regardless of 51% or 100% control) acquiree's FV (if double-negative debt, this is positive) except goodwill.
- Goodwill = full or partial goodwill methods.

- Note that current assets and liabilities can be included but usually BV=FV.

If a company A acquires 100% of company B and company B's shareholder equity = net assets = 100, but company A only paid 10. Then the extra would be 'gain on purchase' in shareholder equity. If company A paid 200, then 100 goes to goodwill, licenses, etc.

Note: under acquisition method, whether full or partial goodwill methods, even though total shareholder equity differs between them, the shareholder equity for the acquirer is **the same**.

Note: if balance sheets are consolidated versus using the equity method, then note that the assets of the acquirer will change because the 'investment' line entry will be removed if consolidated while it remains if equity:

	NinMount	Boswell
Cash	50	20
Receivables—net	70	45
Inventory	130	75
Total current assets	250	140
Property, plant, & equipment—net	1,570	930
Investment in Boswell	320	—
Total assets	2,140	1,070
Current liabilities	110	90
Long-term debt	600	400
Total liabilities	710	490
Common stock	850	535
Retained earnings	580	45
Total equity	1,430	580
Total liabilities and equity	2,140	1,070

After acquisition, depreciation and amortization of combined PPE. Combined PPE = BV of acquiror + FV of acquiree.

Consolidation process. Intercompany transactions eliminated. Mergers and consolidations: consolidated financial statements (FSs). Acquisitions: if less than 100%, then parent and subsidiary each prepare FS but parent also prepares consolidated.

Non-controlling interests - minority stake in an acquisition. In the consolidated BS shareholder's equity section, list a separate line 'noncontrolling interest' equity. IFRS: (1) full goodwill method - % of FV of entire subsidiary; or (2) partial goodwill method - NCI's % of acquiree's FV net identifiable assets. (1)

and (2) are whole NCIs, not just for goodwill. GAAP only allows (1).

Example 8 (less than 100% acquisition). BS will have acquirer BV + acquiree FV regardless of whether less than 100% acquisition.

PPE. FV of consolidated PPE is allocated to *both* controlling and non-controlling entities.

Consolidated income statement. NCI's income (after-tax) is deducted from IS and intercompany transfers are eliminated. This is NCI's % share of the *subsidiary* income. **This means the controlling entity gets the proportion of NI that it has control over.** Under the full goodwill *and* partial goodwill method, deduct the acquiree % of consolidated after-tax income. Note, however, that ROA and ROE will be slightly different as assets and equity will be slightly different for full and partial goodwill methods.

Goodwill impairment. Not amortized. Test for impairment at least annually. Once written down, can't be restored under IFRS and GAAP.

Impairment tests:

- IFRS. Goodwill assigned to 'cash-generating units'. Impairment test: if **recoverable amount** of unit is less than carrying value, impairment applied to goodwill of unit first, and then if more impairment then to other units pro-rata.
- GAAP. Goodwill assigned to 'reporting units'. Impairment test: if **fair value** of unit is less than carrying value, then impairment loss (= FV of unit minus FV of unit's assets + liabilities) is applied to goodwill. If more impairment, doesn't spread to other units.

Financial statement presentation (consolidated report). Financial assets (less than 20%) are found in BS as 'Liquid investments' and 'Other investments'. Associates are found in BS as 'Investments in associates and joint ventures'.

VIEs. SPEs can be legitimate to segregate activities and reduce risk - they can be LLC, corporation, trust, partnership, unincorporated. Created by legal arrangements. In the past, sponsors able to avoid consolidating SPEs because they often didn't have 'control' - so sponsors looked like they had improved asset turnover, lower leverage and higher profitability. Now, IFRS and GAAP revised to include SPEs and VIEs. For GAAP, VIE if (1) entity needs financial support for total equity at risk, or (2) equity investors can't make decisions, have obligation to absorb losses, or have right to receive returns. Primary beneficiary - often sponsor - must consolidate SPEs and VIEs. Sponsor's consolidated BS looks same whether directly borrow money or consolidate SPE.

Special topics.

- Contingent assets and liabilities. IFRS: record contingent liabilities, don't recognize contingent assets. GAAP: Contingent assets and liabilities recorded at FV at time of acquisition; non-contractual contingent assets and liabilities recorded if 'more likely than not' they are an asset or liability.
- Contingent consideration. IFRS/GAAP: measured at FV.
- In-process R and D: FV initially. Amortize if marketable product results; impair otherwise.
- Restructuring costs - expenses.

6.2 Employee compensation: Post-employment and share-based

Salaries and wages. Recognize compensation costs at FV when employee provides services, usually when compensation vests (employee becomes legally entitled). Steps: (1) grant, (2) vesting, (3) settlement. Most companies aggregate compensation in accordance with function, e.g. R&D, SGA, etc.

	Grant 1 January	Vesting 1 January – 14 January	Settlement 14 January
Income statement	No impact	General and administrative expense +3,162*	No impact
Balance sheet	No impact	Accrued compensation +3,162	Accrued compensation (3,162)
Statement of cash flows	No impact	No impact	Cash flows from operations (3,162)

	Grant 1 January	Vesting 1 January – 14 January	Settlement 14 January	Sale 3 April
Income statement	No impact	No impact	No impact	Cost of sales +3,931
Balance sheet	No impact	Inventories +3,931 Accrued compensation +3,931	Accrued compensation (3,931)	Inventories (3,931)
Statement of cash flows	No impact	No impact	Cash flows from operations (3,931)	No impact

Shares. Valued at FV at *time of grant*, and then on IS record annualized (simply divide by number of years to vesting as expense, e.g. 'R&D expense' if person is in research) - note that the expense reduces net income, which reduces retained earnings. In BS, each year to vesting, record the annualized amount as

'share-based compensation reserve' in *equity* (even on the last year of vesting) (offset by the lower retained earnings) and then transfer on the last year to common stock + paid-in capital. No CF entries. Note: changes in FV after grant don't make a difference.

- Share-based compensation vesting may require (1) service conditions, (2) performance conditions (of company), (3) market conditions.
- Restricted stock - common stock usually has voting rights and dividends, but not tradeable. If performance conditions, restricted stock is called 'performance shares'.
- RSUs (restricted stock units) - rights to receive shares upon vesting. No voting rights, no dividends, not tradeable. Usually adjusted downward to account for no dividends.

Stock options. Non-tradeable call options. Usually issued at the money. $FV = \text{intrinsic value (over strike price)} + \text{time-value}$. Estimate FV using a model (IFRS and GAAP allow choice). Accounting:

- IS. Record FV (of options, not shares) at grant-time in IS like stock-based compensation using FV divided by number of years to vest. If exercised, nothing happens to IS.
- BS. As with stock-based compensation, record 'increase in shareholder equity reserve' each year until vesting (of the options, not shares). If exercised, transfer (FV of options at grant x number of options) from equity reserve to paid-in capital + (strike price x number of options).
- CF. Nothing until options exercised. If exercised, cash *financing* inflow of (strike price x number of options).

The shareholder reserve is like cash outflows at the FV of the call option. It then returns + employee invests in company.

Share-based (shares, RSUs, options) compensation tax implications if the **settlement-day price** doesn't equal the grant-date price.

- IFRS. If settlement share price or share option value higher than grant, gain in shareholder equity in BS from DTA. If lower, then loss from DTL. No significant tax implications on IS.
- GAAP. If settlement share price higher than grant, *excess tax benefits* because tax expenses are higher (and so effective tax rate is lower). If lower, then higher taxes (so effective tax rate is higher). These are recorded under **income tax expenses** - not any other expenses!
- Deferred tax assets and liabilities held in BS, so they affect shareholder equity. Changes in them held in OCI.

- In IFRS and GAAP, if the share, RSU or share option price rises in value from grant to vesting, then company pays less tax because the higher value is tax-deductible. That will result in lower CFO payments and some of the lower tax benefits will be transferred to CFF inflows.

Note: If there is a tax windfall, then this will *overstate* $FCF = \text{after-tax EBIT} + D/A \pm \text{increase in WC} - \text{capex}$. Since the EBIT doesn't factor in tax windfalls because EBIT is earnings *before* taxes, etc. but gets the 'benefit' of lower effective tax rates.

To model and forecast share-based compensation expenses, use percentage of revenues.

Shares outstanding for diluted EPS. Affected by share-based compensation and used for diluted EPS calculations. Use treasury stock method:

- Begin with basic outstanding shares = weighted average for reporting period, *doesn't* include unsettled share-based awards.
- Add 'shares' equivalent from options. No. options outstanding (*all* outstanding options, unvested and vested but not exercised) - assumed repurchases. Assumed repurchases = $(\text{assumed proceeds} + \text{average unrecognized share-based compensation expense}) / (\text{average share price over reporting period})$. Assumed proceeds from cash exercise are = no. options outstanding $\times$ grant-date **strike-price**. Average unrecognized share-based compensation expense = $(\text{start} + \text{end}) / 2$, where start and end = will be given as information (e.g. 'the company reported unrecognized share-based compensation expenses of XXX related to options in 20XX') = no. of outstanding options $\times$ grant-date **fair values** of those options (i.e., the future expenses in the IS 'saved' by assuming everything vests today).
- Add 'shares' equivalent from RSUs. No. unvested RSUs outstanding (e.g. $(1 - 1/(\text{years to vest})) \times \text{RSUs}$) - assumed repurchases. Assumed repurchases = $(\text{assumed proceeds from cash exercise (for RSUs} = 0) + \text{average unrecognized share-based compensation expense}) / (\text{average share price})$. Average unrecognized share-based compensation expense = $(\text{start} + \text{end}) / 2$, where the start and end values should be given (e.g. 'the company reported unrecognized share-based compensation expenses of XXX related to RSUs in 20XX') = no. of unvested RSUs $\times$ FV at grant of RSUs.
- Note: don't dilute if negative earnings - diluted EPS will then rise above basic EPS!
- **In-the-money options are dilutive, out-of-the-money options are anti-dilutive.** RSUs are dilutive except if average stock price below RSU at grant date. This is measuring 'how many net shares are OUTSTANDING' with the addition of unvested assets minus assets to be re-purchased back into the treasury.

- Note: must disclose share-based payment arrangements, how FV determined, effect of share-based payments on net income.

To model and forecast shares outstanding, need to consider (1) grants of share-based awards, (2) forfeitures, and (3) settlements. For dilutive securities, just assume a percentage of shares are dilutive. Often use historical grants net of forfeiture rates, and assume percentage of outstanding awards settles each period. Use this framework:

- Basic shares outstanding at *beginning of period*
- + RSUs vested + options exercised
- + share issuances
- - share repurchases

Intuition of treasury stock method: if the share price is higher after grant, then stock options and RSUs are now 'worth more' (and the treasury stock buybacks are less effective), which is a net transfer from shareholders to employees, which dilutes shares. **Valuation effect from share-based compensation.**

- Share-based compensation is strictly speaking a non-cash expense. However, many companies repurchase equivalent number of shares when they award share-based compensation, making share-based compensation effectively a cash expense. Even if companies don't do this, since share-based compensation is a transfer of value from the company to employees, it should be considered in FCF and DCF valuations - which are cash-based.
- DCF models don't account for share-based compensation. For the current EPS, simply dilute (though here we assume all treasury stock method repurchases occur - so may want to 'extra' dilute). For future FCFs, just subtract share-based compensation (here we could adjust FCF by diluting shares, but this is too complicated).
- Share-based compensation = non-cash, and since $FCF = NI + \text{non-cash expenses} - \text{increase in WC} - \text{investments}$, it gets automatically added back to FCF, **but** we don't want this - we want share-based compensation excluded from FCF as mentioned above.

Pensions. DC or DB, more moving to DC so some DB plans are 'closed' or 'frozen'. OPEB = other post-employment benefits like health.

DC plan accounting like wages:

- IS. Like wages - expense when vested.
- BS. When vested, 'accrued compensation', then upon settlement accrued compensation is subtracted.
- CF. Operation cash outflow upon settlement.

DB (and OPEB) plan accounting:

- Income statement. Employer can make contributions whenever, as long as plan is well-funded - but these are **not** counted as IS expenses; instead, a more complex formula is recognized under IFRS and GAAP. IFRS: (1) *current* service cost (actuary calculates: projected unit credit method) is an operating expense, (2) *past service cost* if plan is amended is an operating expense; (3) net interest income - beginning plan net assets x discount rate; this is a financing expense; (4) net interest expense - beginning plan obligation x discount rate; this is a financing expense; (5) remeasurement (actual returns net of interest changes): actual plan returns - net interest expenses + net interest income - in OCI (+ stockholder equity); (6) remeasurement: actuarial gains and losses - in OCI (+ stockholder equity).
- Balance sheet. Net pension asset or net pension liability. Funded status = FV of plan assets - pension obligation. Plan assets protected in case of bankruptcy. Pension obligation depends on discount rate (IG corporate bonds, but can be government bonds) - if discount rate falls then pension obligations can rise quickly. Note: can't offset one plan by another; can't net them off.
- Balance sheet. Remeasurements in shareholder equity.
- Balance sheet. Net pension asset = beginning of year FV plan assets - beginning of year net obligation + net interest income - net interest expense + remeasurement - service costs - net interest expense = beginning of year FV plan assets - beginning of year net obligation + (beginning of year FV plan assets x actual % increase) - service costs - (beginning of year net obligation x discount rate).
- Cashflows. Operating cash outflows whenever employer makes plan contributions. However, DB plan is a separate entity and it makes plan benefit payments - these are both reductions in cash and benefit obligations which are exactly offsetting.

GAAP v IFRS DB pension accounting. Same for BS and CF but different for IS (and OCI).

- Current service costs. Same as IFRS.
- Past service costs. Recorded first in OCI. In subsequent periods, amortize over service lives of employees.
- Interest cost. Like IFRS, pension obligation at beginning of year x discount rate.
- Expected return on plan assets. Different from IFRS. Unlike IFRS, multiply beginning of year plan FV assets x *expected return*.

- Remeasurements (both types listed above for IFRS) in P&L or OCI ('corridor approach' of 10% of plan assets FV or plan obligation at beginning of year, and amortize excess to smooth earnings over expected average remaining working lives of employees in plan).

So, IFRS IS interest cost = (obligation at beginning of year x discount rate) - (assets at beginning of year x discount rate). GAAP IS interest cost = (obligation at beginning of year x discount rate) - (assets at beginning of year x expected return rate).

Disclosure requirements. DC is minimal (just discuss expense in notes). For DB and OPEB, must under IFRS (1) explain characteristics and risks of plans (e.g. sensitivity analysis and assumptions), (2) explain amounts, (3) how plans affect company cashflows.

Financial modeling and valuations. DC: part of operating expense forecasting. DB (and OPEB): model service cost, net interest expense/income, remeasurements, plan contributions unless 'well-funded', i.e. less than 5% of equity market cap (in which case, might ignore in DCF calculations). DB (and OPEB): if underfunded plan, debt in EV calculations; if overfunded plan, ignore because can't withdraw DB plan assets anyways. DB (and OPEB): future service costs are like share-based compensation; deduct from FCF and EV. Underfunded DB plans: include the difference in funding assets and obligations in FCF as debt. For FCF forecasting, don't add/subtract net interest income/expense of plans for valuation - if properly funded, then it'll be net-0, and if underfunded it is interest paid so included as debt in FCFF (though not in FCFE).

6.3 Multinational operations

International transactions lead to FX exposure as transaction date differs from payment date. Differences in FX from transaction to payment are in BS: cash *payment amount* and inventory is in *transaction amount* (i.e. cash and inventory often differ), with the difference between the two being in retained earnings; in the IS: FX gain or loss. If the transaction straddles a financial reporting date, then assume *unrealized* FX gain or loss - one of the few times accounting rules allow unrealized gain or loss.

Where to place FX gains and losses. IFRS and GAAP don't specify, but usually put under 'operating expenses or income' or 'non-operating expenses or income'. Note that while gross profit and net profit don't change, depending on the choice of 'operating' or 'non-operating' expenses, the operating profit will change. Foreign currency gains can be recorded as 'income' and losses as 'expenses'.

Balance sheet FX translation (of assets and liabilities in subsidiary to parent). General theoretical approaches:

1. Current rate method (close to IFRS/GAAP). All assets and liabilities translated at current FX.
2. Monetary/non-monetary method. Only monetary assets and liabilities translated at current FX while others at historical FX. Cash, AR, AP = current; inventory, fixed assets, intangibles, non-monetary liabilities and stock = historical.
3. Temporal method (a variation of the monetary/non-monetary method) (close to IFRS/GAAP). All monetary assets and liabilities translated at current FX but also non-monetary assets and liabilities measured at current values are translated at current FX. **Accounts receivables and accounts payable are monetary assets so are current, inventory at historical, accumulated depreciation at historical.** Historical = beginning of year. Non-monetary assets valued at current values: inventory at net realizable value, PPE under revaluation model, and investment properties.

When translated from subsidiary to parent, the assets won't balance with liabilities, so enter FX translation adjustment in parent's OCI (current rate) and IS (temporal method) and consolidated stockholder equity (an unrealized gain/loss). Under method (1) above, unrealized becomes realized if parent sells subsidiary; under method (2), unrealized becomes realized if subsidiary uses its own cash to pay liabilities and parent sends enough cash to the subsidiary to pay the rest of liabilities (why?).

Net balance sheet exposures for 'exposed' assets and liabilities. If assets which are converted for accounting purposes *at current FX* is greater than liabilities *at current FX*, then **net asset balance sheet exposure** and otherwise, **net liability balance sheet exposure**.

Balance Sheet Exposure	Foreign Currency (FC)	
	Strengthens	Weakens
Net asset	Positive translation adjustment	Negative translation adjustment
Net liability	Negative translation adjustment	Positive translation adjustment

Presentation currency (in financial statements) and **functional currency** (in which the company works - can be more than one). Factors to consider in determining subsidiary's functional currency: (1) currency that mainly influences sales prices, (2) currency of country, and others (see Exhibit 3, page 135).

Subsidiary is to turn everything into its functional currency: (1) identify its functional currency, (2) translate foreign currency balances into its functional currency (e.g. translate ARs in foreign currency into functional currency using current FX), (3) translate functional currency into parent's presentation currency.

Parent presentation currency $\neq$ subsidiary's functional currency. Essentially the **current rate** method (IFRS/GAAP):

1. IS. Revenues and expenses translated at FX when transactions took place. An annual average is usually used.
2. IS. Dividends FX translated at date issued.
3. BS. All assets and liabilities translated at current FX at balance sheet date.
4. BS. Shareholder equity translated at historical FX.

Translation adjustment in OCI and consolidated **stockholder equity**. **Translation adjustment**: if parent currency strengthens and foreign currency weakens, then **negative FX translation** in parents' consolidated statement. If/when subsidiary is sold, then the sum of these unrealized FX translation adjustments throughout the years is realized and recorded in net income. Results in *net asset balance sheet exposure* because shareholder equity is usually positive; net liability balance sheet exposure if shareholder equity is negative.

Parent presentation currency = subsidiary functional currency. Though subsidiary initially records in local currency, then convert all accounts as-if in functional currency (GAAP calls this 'remeasurement'; IFRS calls it 'reporting foreign currency transactions in the functional currency'). Essentially temporal method (IFRS/GAAP):

1. IS. Revenues and expenses translated at FX when transactions took place. Inventories and COGS depend on FIFO/LIFO/WA method. Often use annual average FX (different annual averages, e.g. one for whole-year, one for inventories, etc.).
2. IS. Depreciation, amortization translated at beginning of year FX.
3. IS. Dividends FX translated at date issued.
4. BS. Monetary assets and liabilities translated at current FX.
5. BS. Non-monetary assets and liabilities measured at historical cost translate at historical FX. E.g. inventory measured at cost under the lower of cost or market rule; PPE, intangible assets.
6. BS. Non-monetary assets and liabilities measured at current value (e.g. marketable securities and inventory measured at market value under the lower of cost or market rule) translate at FX when transaction took place.
7. BS. Non-monetary assets and liabilities NOT measured at current value (e.g. deferred revenue) measured at historical cost.
8. BS. Shareholder equity translated at historical FX.

Above is essentially temporal method. Translation adjustment in subsidiary and parent **net income**. Could generate either net asset or liability balance sheet exposure. If subsidiary is sold, the sum of the unrealized net income FX translation adjustments will be realized. Only cash and AR are monetary assets, while most liabilities are monetary - so, usually net liability balance sheet exposure.

Temporal method notes. (1) Inventory (FIFO/LIFO) under temporal method has different historical FX than depreciation (beginning of year). (2) COGS (on IS) and inventory (on BS) have different FX's under temporal method.

Inventories. Under the current rate method, COGS are valued at the time they are sold. This occurs at the same time as revenues. So, use average for the year for both in the IS. Under the temporal method, COGS are valued historically and revenues at the time they are sold - so, there's a mismatch. Under FIFO, COGS will be valued at an earlier FX rate than revenue. Under LIFO, COGS will be valued at a later FX rate.

Retained earnings. Accumulation over company lifetime. RE = accumulation of NI minus sum of dividends. Tricky because shareholder equity is always translated at historical FX. Same under current and temporal method:

Net income in FC		[Translated according to the method used to translate the income statement]	=	Net income in PC
- Dividends in FC	×	Exchange rate when dividends declared	=	- Dividends in PC
R/E in FC				R/E in PC

Retained earnings in parent currency at the end of the first year become the beginning retained earnings in parent currency for the second year, and the translated retained earnings in the second year (and subsequent years) are then calculated in the following manner:

Beginning R/E in FC		[From last year's translation]	→	Beginning R/E in PC
+ Net income in FC		[Translated according to method used to translate the income statement]	=	+ Net income in PC
- Dividends in FC	×	Exchange rate when dividends declared	=	- Dividends in PC
Ending R/E in FC				Ending R/E in PC

See pages 145-146. Summary.

- Subsidiary uses functional currency, i.e. current rate method. Translate IS using current FX, average FX, or if available the transaction-time FX,

to get net income. Subtract dividends translated at FX of time of dividend issue, to get translated RE. *Copy translated RE to BS*, translate rest of BS using end-of-year current FX (except shareholder equity, which is at historical FX). The residual = translation adjustment.

- Subsidiary uses parent presentation currency, i.e. temporal method. COGS and PPE are translated at transaction-date FX (historical FX) and depreciation of COGS and PPE at beginning of year FX; otherwise same as current rate method. Translate BS *first*, with the residual = translated RE. *Copy translated RE to IS*, translate rest of IS, then residual = translation adjustment.
- Note: In either case above - whether the subsidiary is using the functional currency or parent presentation currency, **ignore the subsidiary's retained earnings calculation!**

Current rate method views subsidiary operations as autonomous (e.g. independent operating entity), so the income statement drives retained earnings - the parent just incorporates the BS FX translation as an independent input for consolidated statements (it feels like a 'minor' or 'independent' input into the parent business). Temporal method views subsidiary operations as integrated (e.g. sales outlet), so the balance sheet drives retained earnings - the parent incorporates the 'action' in the IS into the consolidated statements (it feels more 'recent' and 'up-to-date' and detailed).

It is possible for current rate method and temporal method to result in opposing-sign translation adjustments. For instance, suppose the foreign currency strengthens, and under current rate method translation adjustment might be positive because assets bigger than liabilities, and temporal method might translation adjustment might be negative because non-monetary assets might be less than non-monetary liabilities.

Note: under temporal method, usually net liability balance sheet exposure because PPE is such a big portion of assets, and these use historical FX under the temporal method.

	Temporal Method, Net Monetary Liabil- ity Exposure	Temporal Method, Net Monetary Asset Exposure	Current Rate Method
Foreign currency strengthens relative to parent's presen- tation currency	↑ Revenues ↑ Assets ↑ Liabilities ↓ Net income ↓ Shareholders' equity Translation loss	↑ Revenues ↑ Assets ↑ Liabilities ↑ Net income ↑ Shareholders' equity Translation gain	↑ Revenues ↑ Assets ↑ Liabilities ↑ Net income ↑ Shareholders' equity Positive translation adjustment
Foreign currency weakens relative to parent's presenta- tion currency	↓ Revenues ↓ Assets ↓ Liabilities ↑ Net income ↑ Shareholders' equity Translation gain	↓ Revenues ↓ Assets ↓ Liabilities ↓ Net income ↓ Shareholders' equity Translation loss	↓ Revenues ↓ Assets ↓ Liabilities ↓ Net income ↓ Shareholders' equity Negative translation adjustment

High inflation. Generally, 100% in 3 years, which is about 26% per year, though trend also matters. If naively use current FX rates, then PPE disappears (e.g. PPE carrying value is same but with depreciating currency); it doesn't consider how the PPE value is also rising relatively in the local economy.

- GAAP (temporal method): continue to use functional currency but use temporal method - PPE presentation currency value remains the same regardless of inflation.
- IFRS (restate for inflation + current rate method): stop using functional currency, restate the asset or liability value by multiplying with inflation rate, then convert using current FX - this is the current rate method - PPE presentation currency value fluctuates depending on relation between inflation rate and current FX rate but PPE value remains around the same.

High inflation - IFRS in more detail. Restate BS and IS by 'scaling up' using general price index (GPI) PPE and capital stock from beginning of year to end of year. Scale up revenues and interest expenses in IS by GPI from average to end of year. Don't scale cash and debt. Then add **purchasing power gain/loss** to net income of scaled-up IS to get final net income. Then translate everything in IS and BS at current FX.

Purchasing power gain/loss (cash + debt):

1. $-(\text{beginning of year cash balance}) \times (\text{end-of-year GPI} - \text{beginning-of-year GPI}) / (\text{beginning of year GPI})$
2. $-(\text{increase in cash balance during year}) \times (\text{end-year GPI} - \text{average GPI}) / (\text{average GPI})$

3. $+(\text{note payable}) \times (\text{start-year GPI} - \text{end-year GPI}) / (\text{start-year GPI})$. Add these to get net purchasing power gain/loss

These factors are monetary - where restatement factors = 1 (i.e. no inflation scale-up).

High inflation note: if the GPI and FX movements are exactly inverse (i.e. if GPI rises 100% then FX halves) then GAAP and IFRS will give exact same answer.

	FC	Restatement Factor	Infla- tion-Ad- justed FC	Exch. Rate	US\$
Cash	1,750	200/200	1,750	0.50	875
Land	9,000	200/100	18,000	0.50	9,000
Total	10,750		19,750		9,875
Note payable	5,000	200/200	5,000	0.50	2,500
Capital stock	5,000	200/100	10,000	0.50	5,000
Retained earnings	750		4,750	0.50	2,375
Total	10,750		19,750		9,875
Revenues	1,000	200/125	1,600	0.50	800
Interest expense	(250)	200/125	(400)	0.50	(200)
Subtotal	750		1,200		600
Purchasing power gain/loss			3,550	0.50	1,775
Net income			4,750		2,375

Some companies will have some subsidiaries using the current rate method and other subsidiaries using the temporal method. IFRS and GAAP require disclosure of the effect of FX changes on net income and shareholder equity but don't have to split them out. Net income includes both FX *transactions* (unrealized FX gains and losses) and FX *translations* (from subsidiaries) from the temporal method in 'other operating income/expenses'. FX transactions in the current method are in OCI. Shareholder equity includes OCI where OCI is broken down e.g. in MD&A.

IFRS requires disclosure of 'the amount of exchange differences recognized in profit and loss' in net income.

Multinational operations and company effective tax rate. Compare company's home tax rate with its effective tax rate; if higher, could be the company is getting more profit from higher-tax jurisdictions. Lower taxes in consolidated

income statement could mean more business in jurisdictions with lower tax rates.

Net sales versus organic sales growth - strip out FX changes. Net sales growth = organic sales growth + effects of acquisitions, divestitures, and FX. FX translation loss reduces net sales growth.

6.4 Analysis of financial institutions

Mainly banks and insurance companies, though there are others - credit unions, cooperatives, building societies, etc.

Global organizations. Help stabilize and spread risk - e.g. reinsurers. Minimize regulatory arbitrage. Basel Committee on Banking Supervision, supported by the Bank for International Settlements. Basel III framework: 'to improve the banking sector's ability to absorb shocks arising from financial and economic stress, whatever the source, improve risk management and governance, and strengthen banks' transparency and disclosures'. Basel III: minimum capital requirements, minimum liquidity, and stable funding - e.g. through longer tenor of bonds, higher-quality bonds, improve risk assessment process, etc.

CAMELS approach. Developed to examine banks. Rate 1-5 for each component; 1 is best. Weightings of each of components differs depending on bank and even the examiner.

- Capital adequacy. Can be calculated by weighting assets by their 'risk-proportion' (e.g. 0% for cash - i.e. no capital needed to fund cash, 100% for corporate bonds, 150% for NPLs), and add off-balance sheet exposures, to get total risk-weighted assets (RWAs). Basel III and other standards specify minimum 'Common Equity Tier 1 Capital' (common stock, retained earnings, OCI, DTA, etc.), the highest-quality capital because it can absorb losses quickly and permanently, minimum 4.5% of RWAs (a capital ratio). **CET-1 capital is the most important ratio.** Total Tier-1 (subordinated to deposits and debt, no fixed maturity, no dividends/interest, e.g. preferred stock) at least 6%. Tier-2 (loan and lease losses) within Total Capital (T1 + T2) at least 8% of risk-weighted assets.
- Asset quality. Mainly about credit risk, e.g. counterparty risk. Liquid assets are better, with 'held for sale' and 'held to maturity' assets being riskier; loans underwritten by bank most riskiest. Loans typically largest portion of bank assets, and need to check loan qualities - e.g. divide by credit quality and divide by total assets to get percentages. Look at e.g. loans portfolio - what is the fair value of the loans versus book (amortized cost) value? See below about loans.
- Management capabilities. Compliance with laws and regulation. Quality of managers from proxy statement. Good corporate and risk governance - e.g. more independent directors, separate CEO and Chairman, many

meetings, auditor statements on effectiveness of internal controls. For large companies, difficult to eliminate all related-party transactions. Appropriately make profit.

- Earnings sufficiency. High quality earnings: high and trending up. Can use statistics, but need judgment in the end. Use fair-value hierarchy: tier 1, 2, 3 depending on inputs and observability. Sustainable earnings are best earnings. If adjustments in provisions for credit losses on the IS is what is increasing income, then this isn't sustainable. Trading income tends to be unsustainable. Overseas income is subject to currency risks; macroeconomic risks are everywhere - these should be considered.
- Liquidity position. About timing. Even if have adequate capital, sudden illiquidity can be negative. Basel III: (a) liquidity coverage ratio (LCR) of highly liquid assets divided by expected cash outflows - mostly about 1-month liquidity - target minimum 100%, and (b) net stable funding ratio (NSFR) of available stable funding divided by require stable funding - mostly about 1-year or more liquidity - target minimum 100%. For (b), use 'available stable funding' ASF factors, with regulatory capital (e.g. CET1) and capital with 1+ year maturity as best (100%), and non-maturity deposits like bank deposits being partially stable (retail more stable than corporate accounts), and short-term funding like repos the worst (0%). Beyond (a) and (b), watch out for concentration of duration and contractual maturity mismatch.
- Sensitivity to market risk. Banks usually have more assets than liabilities so if interest rate rises, better and worse if it falls. But *repricing frequency* also matters: if the structure of assets and liabilities is such that there's more repricing on assets, then bank interest income rises faster and falls faster with correlative changes in interest rates. VaR is good for intra-company comparisons, but VaR not useful for inter-company because different calculation assumptions. Note VaR covariance adjustments are for not-perfect correlations. VaR good for short-term shocks, not long-term issues.

Note: CAMELS doesn't address bank's competitive environment, FX risk, or off balance-sheet obligations.

Total Regulatory Capital = Common Equity Tier 1 (CET1) + Additional Tier 1 (AT1) + Tier 2 Capital. Each of these is divided by risk-weighted assets (RWAs). Each, including total regulatory capital, should be rising.

Notes on asset quality:

- **Allowance for loan losses** is a contra-account on the BS. A **provision for loan losses** is an IS expense, and it is the change to the allowance for loan losses. Total loans minus allowances for loan losses = expected

value of loans. Allowances are discretionary; if they're too small, then net income and assets overestimated.

- Allowances - want to have more, not less - this helps anticipate and prepare for losses in the future.
- **Net charge-offs** = charge-offs minus recoveries. More objective than allowances - are for NPLs. Net charge-offs reduce the allowances for loan losses.
- Three **ratios** are well-used: (1) allowances to NPLs, (2) **allowances to net charge-offs**, (3) provisions to net charge-offs.
- Non-accrual loans - loans which experienced some nonpayments but are basically fine.

Liquidity. Example of net stable funding ratio calculation:

<i>(In billions of dollars)</i>	31 Dec. 2016	30 Sep. 2016	31 Dec. 2015
Available stable funding:			
Total deposits	\$929.4	\$940.3	\$907.9
Long-term debt	206.2	209.1	201.3
Common equity	205.9	212.3	205.1
Total available stable funding	\$1,341.5	\$1,361.6	\$1,314.3
Required stable funding:			
Total investments	\$353.3	\$354.9	\$343.0
Total loans, net	612.3	626.0	605.0
Goodwill	21.7	22.5	22.3
Intangible assets (other than MSRs)	5.1	5.4	3.7
Mortgage servicing rights (MSRs)	1.6	1.3	1.8
Other assets	128.0	116.5	133.7
Total required stable funding	\$1,122.0	\$1,126.6	\$1,109.5
Net Stable Funding Ratio	120%	121%	118%

Non-CAMEL factors:

- Government support. Is the bank too big to fail.
- Government ownership. Can increase likelihood of bailout.
- Mission of bank. Might be for ag or community.
- Corporate culture. Is it more risk-taking?
- Competitive environment of bank.

- Off balance-sheet items. Can make visibility opaque. VIEs. Benefit plans (sort-of off balance sheet) - falling interest rates can lead to plans blowing up. Size of AUM for asset managers.
- FX exposure.
- Legal and regulatory.

Derivatives accounting

- Must state notional amount of contract. This better indicates scale of exposure.
- Measure MTM and put them into assets and/or liabilities. Changes in A and L lead to changes in PNL on IS.
- Hedge (cash flow or net investment in foreign subsidiary) is OCI and trickled into net income over course of investment life.
- Free-standing derivative or hedge of fair value: changes in PNL which means volatile earnings.

Property and casualty insurance companies. More 'lumpy' and volatile claims. Earn money on premiums and interest on 'float'. Some countries like US require insurance companies to use accounting standards other than GAAP to focus on solvency. Insurers should diversify their risk and coverage and not overconcentrate insurance and investments. P&C insurer industry is cyclical: soft and hard phases with fluctuating levels of competition. Insurers report 'reserves' = loss reserves = incurred but not paid (reserved for losses but unpaid). If pay high proportion of claims and claim adjustments, it's short-term insurer like P&C. Downward revisions to prior year estimates of losses are conservative and have a big effect on income. Net premiums written are those to be paid, net premiums earned are those earned over the insurance period.

- *Loss and loss adjustment expense ratio* = $(\text{Loss expense} + \text{Loss adjustment expense}) / \text{Net premiums earned}$. This ratio indicates the degree of success an underwriter has achieved in estimating the risks insured. The lower the ratio, the greater the success.
- *Underwriting expense ratio* = $\text{Underwriting expense} / \text{Net premiums written}$. This ratio measures the efficiency of money spent in obtaining new premiums. A lower ratio indicates higher success.
- *Combined ratio* = *Loss and loss adjustment expense ratio* + *Underwriting expense ratio*. This ratio indicates the overall efficiency of an underwriting operation. A combined ratio of less than 100 is considered efficient.
- *Dividends to policyholders (shareholders) ratio* = $\text{Dividends to policyholders (shareholders)} / \text{Net premiums earned}$. This ratio is a measure of liquidity, in that it relates the cash outflow of dividends to the premiums earned in the same period.
- *Combined ratio after dividends* = *Combined ratio* + *Dividends to policyholders (shareholders) ratio*. This ratio is a stricter measure of efficiency than the ordinary combined ratio, in that it takes into account the cash satisfaction of policyholders or shareholders after consideration of the total underwriting efforts. Dividends are discretionary cash outlays, and factoring them into the combined ratio presents a fuller description of total cash requirements.³³

Life and health insurance companies. Claims take awhile to come through, in contrast to P&C insurers and more stable because they correspond closely to actuarial death, etc. calculations. Lower capital requirements, can take greater risks. More exposure to interest rate risk because long-term investments. Also provide investment products and hybrid life insurance (payment upon death) and investment products. Insurance company-specific metrics:

- Income. Total benefits paid as a % of net premiums written + deposits. Commissions and expenses incurred as % of net premiums written and deposit
- Investment performance. Investment income divided by invested assets (cash + investments). Investment income + realized gains/losses with/without unrealized capital gains/losses.
- Compare assets to obligations.
- Note: current ratio not applicable to L&H companies because BS often don't distinguish 'current' and 'non-current'.

Note mismatches between assets (subject to interest rate risk because reported as market values) and liabilities (not subject to rate risk because reported as

historical cost).

Note: banks usually borrow over short term (e.g. retail deposits) and lend at longer term. So if yield curve inverts, bad for bank.

6.5 Evaluating quality of financial reports

Earnings quality spectrum. Only high-quality reports permit accurate assessment. Earnings management = earnings smoothing. Conservative = reduce earnings. Aggressive = inflate earnings.

Potential problems. Compare revenue growth with asset growth. Check 'long-term trade receivables' and 'unbilled revenue accounts'.

Reclassification. Removing AR from BS and reclassify as something else, making DSO, receivables turnover better. Stuffing non-recurring revenue into 'operating income' and 'operating' cash inflows.

M&A. Acquirer makes acquisition to use acquiree's assets and cashflows to hide acquirer's problems. Acquiror pumps up its stock in all-stock acquisitions, acquiree pumps up value before being bought. Acquirer finds companies that reduce comparability of their own financial statements by targeting companies having less public information and less similar operations. Reduce fair value of acquired assets in consolidated statements, which will (a) reduce future amortization and therefore expenses, and (b) pump up goodwill which is only tested for impairment, and when impaired is a one-time expense which can be easily discussed with investors.

Accounting judgment. R&D, items in OCI (e.g. FX translations, available-for-sale security price changes), sales backlogs, etc.

Financial statement evaluation. Many things to look out for. Here are some: declining receivables turnover (some revenues recorded prematurely or allowance for doubtful accounts is insufficient), declining inventory turnover (inventory turnover = COGS over average inventory), net income greater than cash inflow (suggests aggressive accrual accounting to shift current expenses to later periods). Companies in multiple geographies or product segments might 'shift' revenues or income to the 'hot' geographies or products to get more attention and lower capital costs.

Beneish quantitative model of earnings manipulation.

$$\begin{aligned} \text{M-score} = & -4.84 + 0.920(DSR) + 0.528(GMI) + 0.404(AQI) + 0.892(SGI) \\ & + 0.115(DEPI) - 0.172(SGAI) + 4.679(Accruals) - 0.327(LEVI) \end{aligned}$$

M-score is N-distributed random variable with mean 0 and std 1. Beneish thinks earnings manipulation of 3.8% (M-score -1.78) or higher is a red flag. The 3.8%

is dependent on relative Type I error (incorrectly classifying manipulator as non-manipulator) and Type II error (incorrectly classifying non-manipulator as manipulator).

Notes: (1) Non-manipulator curve average is less than 0 (i.e. to the left of the manipulator null hypothesis curve). (2) There are many other potential factors other than Beneish factors: auditor change, book-to-market values, etc. (3) Predictive power of Beneish model declining due to strategic behavior.

High earnings quality = good reporting, high and sustainable earnings, and earnings in excess of company's capital cost. Recurring earnings used (no or little invalid 'classification shifting' of earnings from non-recurring to recurring, and no or little shifting of expenses from recurring to non-recurring).

Earnings persistence. Higher b_1 indicates more persistent earnings:

$$\text{Earnings}_{t+1} = a + b_1 \text{Earnings}_t + e_t$$

Earnings = cashflow + accruals. If cash collection occurs at different time than revenues recorded = accrual, e.g. accounts receivable.

$$\text{Earnings}_{t+1} = a + b_1 \text{Cashflow}_t + b_2 \text{Accruals}_t + e_t$$

Look for cashflow persistence. Accruals = non-discretionary (normal) + discretionary (outliers - possible manipulation). Lots of discretionary accruals = low quality earnings.

$$\text{Total accruals}_t = c + \sum d_i \text{Normal accrual factors}_t + e_t$$

where normal accrual factors = growth in credit sales (leads to more ARs), growth in depreciable assets (more capital investment, because it's not recorded as an expense and is instead capitalized, leads initially to unaffected net income and cashflow, even though future depreciation per year is accrued - but once depreciation begins, it leads to reduced accruals because depreciation lowers the difference between net income and cashflow). **The residual is the abnormal accrual.** SEC uses something like this (Accounting Quality model). Another way to look at accruals is to take total accruals and divide (scale) by net operating income. Higher (i.e. more negative) depreciation expense = lower accrual.

$$\text{Accruals} = \frac{\text{Net income} - \text{Cashflow}}{\text{Total assets}}$$

Positive income, negative operating cash inflows. This could happen if (a) accruals are large (e.g. early revenue recognition before cash received, fast increase in AR), (b) large investments, others. However, this isn't sustainable. World-Com leased network capacity from others - these should have been classed as expenses and therefore reduced net income and operating cashflows, but instead they were capitalized - so assets rose and amortized in investing cash outflows.

So, net income and operating cashflows looked healthy.

Mean reversion. Can't extrapolate high earnings forever. If accruals are high relative to cash, then hastens mean reversion. Meeting or narrowly beating earnings expectations - can be good, but can also be sign of manipulation.

Revenue. Most common area of misrepresentation because of its discretionary nature. Pay attention to: one-time disposals treated as recurring revenue, channel-stuffing, bill-and-hold sales. Even a small proportion of bill-and-hold in revenue can change net income a lot. DSO can rise fast and AR turnover can decline fast. Compare AR to revenue.

Multiple-deliverables, like software licenses + support obligations. Misallocation of revenue can occur. Support services should be billed when delivered, not at the beginning - check for irregular licence versus support revenue trends.

Expenses. Another major area of misrepresentation. WorldCom 'line cost' capitalization is an example. Look out for non-current assets which change suddenly. If revenues and net income steady but assets are building fast, then possible improper cost capitalization. Sudden increase or decrease in turnover ratios. Compute depreciation expense compared to asset base. CAPEX to PPE ratio - is it rising quickly?

Check relationships with company insiders. Propping - company insider might transfer resources to inflate revenue temporarily and mislead investors, and then later extracts outsized profits from company. Company might also transfer or dump old inventories to company insider for low prices but inflate prices on its income statement.

Bankruptcy prediction models. Quantify likelihood of default. **Altman model** unifies ratio analysis - addresses contradictions in ratios, Z-score of 3 is low probability of bankruptcy, below 1.81 is high probability:

$$\text{Z-score} = 1.2 (\text{Net working capital/Total assets}) + 1.4 (\text{Retained earnings/Total assets}) + 3.3 (\text{EBIT/Total assets}) + 0.6 (\text{Market value of equity/Book value of liabilities}) + 1.0 (\text{Sales/Total assets})$$

Net WC/TA = short-term liquidity risk. RE/TA = RE accumulates over time. EBIT/TA = profitability. MV equity/BV liabilities = leverage. Sales/TA = activity ratio.

Side note on WC. Changes in WC can be from changes in trade and other receivables, unbilled revenue, trade and other payables, unearned revenue, other liabilities, retirement benefit obligations, etc.

Altman model weaknesses. Single point in time. Doesn't incorporate market-

based data, e.g. CDS data. Assumes company is solvent, when in fact it might not be. Many improvements afterwards using accounting and market-based data.

Cashflow statement. Less subject to manipulation, but still opportunities for misrepresentation.

Reporting quality: complete, unbiased, clear.

Cashflow quality. Operating cashflow (OCF) is what everyone looks at. OCF can be manipulated by timing changes in AR/AP and therefore DSO, DPO; also by classification changes in investing/financing and operating cashflows. Examples to watch out for:

- Using the indirect method from net income to OCF by adding non-cash reductions - improper additions such as FX gains/losses are red flags because that is double-counting; it's already part of net income.
- Big changes (esp. negative) in receivables and inventories as AR rises because revenue is recognized prematurely and inventory rises because of customer returns. Note: a big rise in revenue which outpaces relatively AR is fine.
- Classification changes. Especially for IFRS which permits interest and dividends to be moved between operating and investing cashflows. Changes y/y are a potential red flag. Also, changes in classification from non-trading securities (investing CF) to trading (OCF).

Balance sheet quality: optimal leverage, adequate liquidity, efficient asset allocation. Use ratio analysis. Watch out for:

- Off-balance sheet items. Use 'constructive capitalization' to estimate off balance-sheet items and reconstruct and then adjust BS.
- Unconsolidated statements, even though ownership in subsidiaries approaches 50%. Under equity method, profitability ratios can be overstated in unconsolidated statements.
- Market value is less than book value goodwill or even book value equity.
- Understated valuation allowance for DTAs.
- Securities which are tier-3 in fair value hierarchy.
- Pension liabilities and their assumptions (e.g. interest rate assumptions).

Risk information.

- Audit opinions. But limited usefulness - often too late. However, a discretionary change in auditor is a potential red flag.

- Notes to financial statements, MD&A.
- Capital raising. Other financial report filings, management changes, M&A. Financial press.

IFRS management commentary: (1) nature of business, (2) objectives and strategies, (3) resources, risks, and relationships, (4) results and prospects, (5) performance measures and indicators. GAAP MD&A: (1) liquidity, (2) capital resources, (3) results of operations, (4) off-BS arrangements, (5) contractual arrangements. Only significant risks should be identified - not a sea of generic risks.

Other required disclosures. Specific to an event, e.g. capital raising, non-timely filing of financial reports (red flag), etc.

Financial press can be a source of risks.

6.6 Integration of financial statement analysis techniques

Financial statement analysis:

1. Define purpose and context. Note: if company shows prospect of future asset writedowns or new legal liabilities, then they may issue dilutive securities to 'repair the balance sheet' - want to avoid this.
2. Collect input data.
3. Process data. Sometimes want to remove effect of associates from equity method. Note: impairments are recurring expenses - they are in the ordinary course of business.
4. Analyze/interpret data. Sometimes use DuPont analysis (there, can sometimes assume financial leverage between parent and associates are similar). Check also capital structure - e.g. bond debt (fixed payments) are more onerous than benefit plan obligations (flexible). Check also asset composition; use ratio of total Capex % to total asset %. Note accrual performance measures like EBIT (because subtract non-cash expenses like D/A and include AR/AP accruals) often differ from cashflows. Can reconstruct cashflows per unit by taking operating profit + NCE.
5. Develop conclusions + communicate. Want high cash and low leverage, OCF greater than operating profit, want to show parent standalone and together with any subsidiaries.
6. Follow up.

Accruals. Balance sheet (NOA = net operating assets):

$$\text{BS accruals} = \frac{NOA_t - NOA_{t-1}}{\text{Average NOA}}$$

NOA = Operating assets - Operating liabilities. Operating assets = total assets - (cash + short-term investments). Operating liabilities = total liabilities - long-term debt - current debt liabilities. Operating assets include AR, inventories; operating liabilities include AP. This is not a good measure of accruals - as using cash to buy PPE can raise BS-accrual ratio. Instead perhaps better use cashflow accruals:

$$\text{CF accruals} = \frac{NI_t - (CFO_t + CFI_t)}{\text{Average NOA}}$$

Operating assets and liabilities = part of core business, used to generate revenue, and excludes financing (how the core business is funded). Operating assets: AR, inventory, PPE, intangible assets. Operating liabilities: AP, accrued expenses - unpaid wages/rent/etc, deferred revenue (payments received in advance for goods/services yet to be delivered).

Don't include financing cashflows in the above formulas because accruals are about operations, and NOA doesn't include financing.

Cashflow is about operational success. Total assets are about management resource allocations over time.

Calculate company's consolidated earnings (minus non-controlling interests) to total market capitalization. Calculate company's 'standalone' earnings to 'standalone' market cap: consolidated earnings minus 100% of subsidiary earnings minus non-controlling interests (if want to calculate for controlling interests only of the parent company), to total market cap minus % of subsidiary market cap.

7 Corporate Issuers

7.1 Shareholder equity

Shareholder equity (on BS): contributed capital + RE.

Contributed capital = common stock + APIC (additional paid-in capital). Common stock = par value. APIC is market price minus par value. APIC = contributed surplus. If company issues shares at price below par value, then still count as common stock or common stock at par value + negative APIC.

RE = RE from previous year + (NI - dividends). It can be negative.

7.2 Analysis of dividends and share repurchases

This chapter focuses on common share (not preferred share) dividends. Payout (distribution) policy - dividends, share repurchases, etc.

- Regular cash dividends. Signal confidence.

- Extra or special or irregular dividend. Might do this in good years.
- Liquidating dividends. A return of capital - e.g. sell part or entire business - beyond its retained earnings - and returns it to shareholders.
- Stock dividends (bonus issue of shares or scrip dividend). Generally not taxable. Helps increase share liquidity. Companies in some cases might pay the same dividend rate on the new shares as the old shares - in which case, the dividends for shareholders will rise. Shareholder equity remains the same: retained earnings reduced by value of stock dividends, but the contributed capital increases by the exact same amount.
- Note: cash dividend affects capital structure, while stock dividends do not.
- Stock splits. Similar to stock dividends. 2-for-1 split is same as 100% stock dividend, though doesn't have same accounting treatment as stock dividend; for stock splits, reduce the par value in contributed capital, and increase the number of shares; shareholder equity remains the same. Stock splits have declined in frequency as ETFs have de-linked transactions costs to shares traded. There's no more optimal 'marketable price range'.
- Reverse stock splits. Generally to increase share price above 5 USD (penny stock) or to get listed (for standardized trading lot sizes).

Dividend policy theory. Some think dividends don't matter, others think they matter (because of taxes, asymmetric information, agency costs, transaction costs, risk).

- Perfect capital market assumptions - no taxes, no transaction costs, no information asymmetry - dividend policy has no impact on shareholder wealth. In other words, dividend decision is independent of company's investment and financing decisions. If company has no NPV options, then excess cash - this can be distributed as cash dividends, or by share repurchases, which have the same effect and keep the share price the same. Shareholder with lots of shares can also sell shares to get cash if company doesn't issue dividends, or shareholder can re-invest dividend cash if company issues dividends but shareholder doesn't want them (**homemade dividends**).
- Risk. Cash dividend better than potential value in company.
- Agency costs. Don't want managers to have too much cash.
- Taxes. Dividends typically taxed at higher rates than capital gains.
- Laws and regulations. Sometimes require companies pay dividends from positive earnings.

- Signalling. Difficult or costly to mimic. Brings attention to undervalued companies, which raises their share price; overvalued companies won't do this because it brings unwanted scrutiny.
- Agency costs. Dividends to reduce overinvestment in negative-NPV projects (Jensen's FCF hypothesis).
- Equity market to monitor utility managers and regulators. Pay dividends on old shares and issue new shares - new shares will determine if regulators have set the right price.

Dividend irrelevance. Company has E equity C cash to distribute, N shares. Dividends:

$$P' = \frac{E - C}{N} + \frac{C}{N} = \frac{E}{N} = P$$

Share repurchases:

$$P' = \frac{E - C}{N - (C/P)} = \frac{PN - C}{N - (C/P)} = \frac{P(N - (C/P))}{N - (C/P)} = P$$

Smaller, growing companies like tech companies will not pay dividends as they reinvest and have high-NPV projects. They are also in riskier industries so want to hoard cash.

Bondholders. Don't want dividends sometimes - especially if using borrowed cash or selling assets (almost same thing). So use debt covenants to enforce conditions (e.g. can't use new borrowed money for dividends, must have min earnings).

Factors affecting dividend policy.

- Investment opportunities.
- Expected volatility of future earnings.
- Financial flexibility. Repurchase shares in open market might be better than dividends - these don't create future expectations for more repurchases unlike dividends.
- Taxes. Three systems: (1) Double-taxation: corporate tax + dividend tax, so total tax rate = $1 - (1 - t_c)(1 - t_d)$; (2) Dividend imputation tax system: earnings first taxed at corporate tax rate, then if dividends then shareholder gets tax credit (franking credit)- in the end, shareholder pays only shareholder tax rate; (3) Split-rate tax system: earnings distributed as dividends taxed at corporate level at lower rate + shareholder tax rate. Shareholders may prefer dividends if capital gains taxes high, but they can just hold onto shares and some put into tax-exempt retirement accounts. Further, institutional shareholders (e.g. endowments) might be tax-exempt.

- Flotation costs. Generally new shares more expensive than equivalent debt. Flotation costs higher for small companies per share than larger companies.
- Contractual and legal restrictions. Brazil mandates dividends. Canada + US require dividends (unless they're liquidating) to first meet min capital requirements (impairment of capital rule). Bondholders require covenants in bond indentures.

Payout policies: stable and constant dividend policies.

Stable dividend policy - regular dividends paid and don't reflect short-term earnings volatility (most common). Usually follows a target payout ratio - dividends to earnings.

Lintner's 'target payout adjustment model': (1) companies have a target payout ratio based on LT sustainability, (2) managers more concerned about changes in dividends rather than levels, (3) companies reduce dividends only in extreme cases. Expected increase in dividends = (expected earnings x target payout ratio - previous dividend) x adjustment. Adjustment = 1 divided by number of years of adjustment.

Constant dividend policy - same % DPR. So, dividend levels fluctuate with earnings.

More mature markets (e.g. developed countries) have more dividends but this has been in the decline - though rising since GFC. Share repurchases have been rising. If corporate governance rises (as it has been rising in most places), dividends decline (because of their signalling value).

Share repurchases. Often subject to laws and regulations, e.g. need shareholder approval (EU but not US), limit to repurchases, measures to protect creditors. Note: a dividend authorization commits company to issue dividends, but repurchase authorization doesn't.

Share repurchase execution methods: (1) open market (most common, most flexible; can reduce cost - buy opportunistically), (2) buy back fixed number of shares at fixed price (fixed price tender offer; can be done quickly), (3) Dutch auction (shareholders submit bid prices and number of shares; can reduce cost), (4) direct negotiation (e.g. with major shareholder, or buy back to thwart takeover by greenmail to the detriment of other shareholders).

Share repurchases affect IS, EPS. Smaller number of shares, increase EPS.

- If share repurchase financed by high-cost borrowing, then NI is lower and EPS could be lower than before repurchase.

- If share repurchase financed by debt, EPS rises if earnings yields is higher than after-tax debt cost (i.e. $E/P > r \Rightarrow E > Pr \Rightarrow EPS > \text{Debt cost per share}$); RHS is debt cost per share and LHS is for EPS.
- If share repurchase financed by cash, EPS rises if cash would NOT earn cost of capital if retained by company.
- Even if EPS rises (= capital gains g rises), the dividend yield might fall (less cash) from share repurchase.
- If MV of share is greater than book value per share (BVPS) before share repurchase, BVPS will fall (because intuitively buying more 'expensive' shares); if MV is less than BVPS then BVPS will rise. However, **shareholder wealth changes in response to changes in market price, not BVPS.**

Share repurchases = dividends, from shareholder wealth perspective under simple assumptions. For share repurchases, take market value (= total no. of shares x market price) minus share repurchase value, and then divide by total remaining shares after repurchases. This should equal the market price. **If pay more than market price in share repurchase, this depresses the share price after repurchase.**

Share repurchase advantages for company:

- Potential tax advantages over cash dividends.
- If low price, then good investment - and managers have some better inside information than rest of market.
- Support share price and therefore shareholders, though this can be abused to thwart takeovers.
- Signal financial health.
- Managerial flexibility.
- Increase financial leverage (i.e. less E versus D). Note: using 100 debt for share repurchases (debt ratio = $(D+100) / ((E-100)+(D+100))$) increases leverage slightly more than using 100 cash (debt ratio = $D / ((E-100)+D)$).
- Use repurchased shares as reserve for future acquisitions.
- Use for share-based compensation, preventing dilution to existing shareholders.
- Boost financial metrics, e.g. by reducing number of outstanding shares, increase EPS.

Share repurchase disadvantage: signal of no new good investment projects. Check the investing cashflows from CF statement for this - if very negative, then might assume lots of projects.

When a company repurchases shares and stores them in its treasury, there is a cash outflow (decline in total assets in the BS) and a 'contra' shareholder equity (i.e. negative) account of 'treasury shares' is recorded in the BS, so the fundamental accounting equation is maintained in the BS.

Treasury stock. Even if repurchased, now have no voting rights and don't receive dividends (and don't have rights to the residual of the company upon bankruptcy, aren't counted as outstanding shares anymore). Treasury shares are stripped of these rights because they create conflicts of interest, e.g. company can vote what it wants to do itself, exacerbating agency problems. If resold, they regain voting and dividend eligibility rights, and other regular share rights.

Company can choose to keep treasury shares or retire them permanently (canceled shares). Advantages of retiring:

- Reduce shareholder equity, thereby boosting ROE and possibly other financial metrics.
- Shareholders get nervous if company holds too many treasury shares, as it indicates possible dilution in the future.

If the company retires treasury shares permanently, the additional paid-in capital (APIC) might be adjusted downward as the contra shareholder equity 'treasury shares' is removed from the BS.

Some countries restrict the proportion of treasury shares that a company can hold, or require companies to disclose intentions of why they're buying back stock:

- Buying back too many shares can artificially inflate stock prices.
- Company might reissue treasury stock too strategically and unfairly to the detriment of existing shareholders.
- Reduces share liquidity.
- Share repurchases can be subject to insider trading.
- Might increase leverage too much.

Dividend safety. Inverse of DPR is **dividend coverage ratio** which is NI divided by D. High DPR = small % fall in NI = big DPR cut. However, using FCFE is better than NI for considering dividend safety. And include share repurchases to cash dividends:

$$\text{FCFE coverage ratio} = \frac{\text{FCFE}}{\text{Dividends} + \text{Repurchases}}$$

If this is 1, company returning all cash to shareholders. Less than 1 is unsustainable. Risk factors:

- High financial leverage
- Macroeconomic conditions

Stock dividend. Generally not taxed, in contrast to cash dividends. The value of the stock dividend = no. of stocks x market price, and this is subtracted from RE and added to contributed capital. If there's not enough RE, then RE goes negative.

7.3 ESG considerations in investment analysis

Concentrated versus dispersed ownership. Concentrated more common - tend to have less stringent requirements for independent directors. But dispersed ownership more common in UK and US - more stringent requirements for independent directors. Concentrated ownership = less principal-agent problems.

Controlling shareholders. Can be minority controlling shareholders through horizontal or vertical ownership or using dual-class shares.

Principal and agent problems.

- Dispersed ownership + dispersed voting power = strong managers + weak shareholders $\Rightarrow$ principal-agent problems with managers taking advantage.
- Concentrated ownership + concentrated voting power = strong shareholders + weak managers $\Rightarrow$ principal-principal problem as controlling shareholder can allocate resources to themselves against minority shareholders.
- Dispersed ownership + concentrated voting power. Again, principal-principal problems.
- Concentrated ownership + dispersed voting power. Principal-agent problems. When there are **voting caps** (e.g. to restrict foreign investors from taking control).

Types of shareholders.

- Banks. If both equityholder and creditor, potential conflict of interest. So corporate governance has to balance interests.
- Families. Some individuals from same family serve on multiple boards - interlocking directorates. Lack of minority interest consideration.
- SOEs. Mixed-ownership models if public can own shares.
- Institutional investors. Lots of expertise, can improve corporate governance.

- Groups, e.g. Samsung. Cross-holding arrangements. But potential for related transactions at expense of minority shareholders.
- PE, foreign investors, managers and board directors.

Board structures. One-tier or two-tiered (e.g. Germany). When evaluating corporate governance, look at board: structure, independence, committees, skills and experience, tenure, diversity of composition. Evaluate remuneration (too high?). Look at shareholder voting structure - too much concentrated power?

ESG factors and analysis. Look at materiality of any ESG disclosures, time over which they're material. Use proprietary methods (e.g. developed in-house), ESG provider data and information (but provider rankings and assessments can be subjective), non-profit frameworks.

ESG integration. Equities - ESG used to identify opportunities and risks (e.g. stranded assets). Fixed-income - ESG to mitigate risks.

Green bonds. Issuer decides whether green, might use 'Green Bond Principles' from International Capital Market Association. Same as regular bonds, same credit ratings *ceteris paribus*. Only purpose is different, and some have a premium. But greenwashing can mean people overpaying for green bonds. Green bonds aren't very liquid.

7.4 Cost of capital: advanced topics

WACC. Cost of debt = after-tax, lower risk than equity. Cost of equity = required equity return from CAPM. Don't forget preferred equity. Note: in CAPM, equity risk premium (ERP) is higher for common equity than preferred equity because of relative risk. To analyze WACC, look at top-down (external) factors and bottom-up (internal) factors.

Top-down factors: capital availability, market conditions, legal and regulatory, taxes. Note: **increasing tax rate** leads to **lower debt costs**, assuming interest costs are tax-deductible.

Bottom-up factors: revenue, earnings, cash flow, assets, liquidity, leverage, ESG, security features, others. Sales: volatility, customer concentration. Leverage: more operating (fixed costs to total costs) and financial, more risk. Assets: tangible assets are better than intangible assets for reducing cost of capital, more fungible and liquid assets are better. Security features: callability increases bond interest, putability and convertibility reduces bond interest rates, cumulative preferred shares reduce equity capital costs, etc.

Note: **Straight debt** = debt without embedded options. If a company only has straight debt, debt cost approximated by YTM on longest existing maturity debt for that company. Alternatively, could use short-term bonds if traded

frequently.

Estimating debt cost. Type of debt (traded? what is the debt mix composition?), credit rating, fundamental ratios (e.g. interest coverage (IC) ratios, leverage) or models.

Debt cost: bank debt. Can be fully amortizing, partially amortizing, or non-amortizing. Amortizing loans have lowest cost - lower default risk as principal is being repaid. Non-amortizing loans, where entire principal repaid at maturity, is like bullet bond and riskier. Check company's newest debt financing rates.

Debt cost: leases. Financial (capital) leases - amortized loans (capitalized + amortized). Operating leases - just expenses. Leases are cheaper than loans because securitized. Rate implicit in lease (RIIL) calculated from first term or third term in:

$$PV(\text{lease payments}) + PV(\text{residual value}) = FV(\text{lease}) + \text{Lessor costs}$$

Decision to lease or not: compare company's incremental borrowing rate (IBR), the debt cost on collateralized loan over same time period, on the open market with the implicit (calculated) lease cost.

Debt cost: International considerations. Country risk rating (CRRs). Add this onto the debt cost.

Note: In some countries, there are limits on tax-deductibility for interest. Once interest expense reaches limit, no more tax shield capacity.

ERP - equity risk premium. Compensation for holding risky equity securities over r_f . Hard to estimate. Use historical approach or forward-looking approach.

- Historical approach. Use if good historical reliable data. Use broad-based equity index as proxy - but be careful of survivorship bias. If construct equity index using only 'survivors', then survivorship bias, so the ERP (equity risk premiums) = equity **returns** will be **biased higher**. Arithmetic mean is best for one-period averages; e.g. using CAPM and multifactor models. For r_f , use short-term government bond (advantage: know r_f and no reinvestment costs; disadvantage: duration doesn't match infinite-life equity) or YTM of long-term government bond (but has coupon reinvestment risk).
- Forward-looking approach. Forward-looking approach. Can use survey to figure out ERP (but caution on recency bias and confirmation bias. DDM (solve for r of Gordon growth model; guess DY and g to get r_e and the subtract r_f).

Q. The equity risk premium (ERP) demanded by investors will *most likely*:

- A. decrease during recessions.
- B. move inversely with relative rates of inflation.
- C. be lower in developed economies than in developing economies.

Solution

- A. Incorrect because a company's cost of capital is highly influenced by market conditions such as interest rates, inflation rates, and the macroeconomic environment. The ERP demanded by investors tends to increase during recessions and decrease during expansionary times, not vice versa.
- B. Incorrect because macroeconomic and country-specific economic factors, such as inflation rates. When interest rates are relatively low and credit spreads are tight, the costs of debt and equity capital are lower. Higher relative rates of inflation, represented in a higher risk-free rate, increase the cost of capital for companies. Thus, as relative rates of inflation increase, the equity risk premium demanded by investors would also most likely increase.
- C. **Correct** because developed economies typically have more established, liquid capital markets with greater capital availability, more stable currencies, better property protection, and a greater strength in the rule of law than those of developing economies. Consequently, the perceived risk associated with investing in companies in more mature capital markets is lower than for companies in less mature economies. Lower perceived risk translates into lower credit spreads, ERPs, and costs of capital for companies in more mature, or developed, economies.

Can have multiple stages. Then, r_e is from the LHS term of:

$$V = PV_0 + PV_1 + PV_2$$

for a 3-stage growth model.

ERP from macroeconomics. **Grinold-Kroner**:

$$ERP = DY + \text{Expected capital gain} - r_f$$

$$ERP = DY + \text{Expected repricing} + \text{Earnings growth per share} - r_f$$

where Earnings growth per share = Real economic growth + inflation - % change in shares = $i + g - \Delta S$. So,

$$ERP = DY + \Delta(P/E) + i + g - \Delta S - r_f$$

which is valid for small ΔS . If $\Delta S > 0$ then earnings dilution from more shares. If ΔS gets large, then $ERP = DY + \Delta(P/E) + \frac{i+g}{1+\Delta S} - r_f$. Note: Grinold-Kroner not appropriate for developing country where stock market not sufficiently large portion of economy because the model assumes share prices grow with real economic growth.

Cost of equity for public companies. DDM, BYPRP, risk-based models.

- DDM. $r_e = \frac{D_1}{P_0} + g$. Or IRR.
- BYPRP (bond yield + risk premium). $r_e = r_d + RP$, use historical mean difference in returns between equity market index and corporate bond index.
- Risk-based models. r_e = Compensation for time value of money r_f + Compensation for risk (factors). (1) CAPM using regression series of $r_{it} = r_{ft} + \beta(r_{mt} - r_{ft})$. (2) Unlever and lever existing β . (3) Fama-French model

Cost of equity for private companies. (1) Size premium, (2) industry risk premium, (3) specific-company risk premium. Use

- Expanded CAPM. $r_e = r_f + \beta_{peer}(ERP) + SP + SCRP + CRP$
- Build-up approach. $r_e = r_f + ERP_{public} + SP + IP + SCRP + CRP$

International companies.

- Country risk premium (CRP). $ERP_{developing} = ERP_{developed} + (\lambda \times CRP)$, where lambda is the company exposure to the country (e.g. proportion of company sales). CRP can be approximated using sovereign bond yield spread (SYB). Can adjust $CRP = SYB \times \frac{\sigma_{equity}}{\sigma_{bond}}$
- Extended CAPM. (1) Global CAPM - use one global market index variable (but probably insignificant beta, so add a domestic equity term), (2) International CAPM - $r_e = r_f + \beta_G(r_{gm} - r_f) + \beta_C(r_c - r_f)$ where gm is global market index and C is for FX index; β_G indicates how closely related to global economy, (3) Country spread and risk rating models.

Aspects that would increase cost of capital:

- High FC - high operating leverage
- Low proportion of liquid assets
- Lots of intangible assets, less cash
- Volatile revenues (opposite: subscriptions which are stable and recurring)
- Geographic concentration
- Key person risk

7.5 Corporate restructuring

Transactions: investments (e.g. acquisitions, JV, equity investment), divestments (sale, spinoff), restructuring, LBOs (which combine elements of the previous 3). Equity prices correlate with all transaction volumes. So, strong economy deals are overpriced; weak economy deals give better shareholder returns.

Investments. Equity investments, JV, acquisitions. Cost synergies through scale economies, e.g. sharing GSAs. Revenue synergies through scope economies. Look for undervaluations

Note on investments: even if the acquisition of a company leads to increased costs which are higher than the expected synergies - i.e. a net increase in operating costs - if the target is high-margin, the combined revenues and income might be higher than the acquirer prior to the acquisition.

Divestments. Sales and spinoffs. Look for and get rid of overvalued targets. Issuer trading at a valuation lower than sum of its parts = conglomerate discount - an example of diseconomies of scale or scope. Example: after Alcon was spinoff from Novartis, its P-E ratio rose.

Restructuring. Opportunistic improvement (e.g. franchising) and forced improvement. Forced improvement:

- Cost restructuring, e.g. outsourcing and offshoring.
- Balance sheet restructuring, e.g. franchising, shifting asset composition, changing capital structure, sale leasebacks. **Sale leasebacks** often more expensive than depreciation + amortization, though can be cheaper if lessor has low-cost financing. Sale leasebacks used to get quick liquidity. Another BS restructuring: **dividend recapitalization**, which is debt-financing share repurchases - returns value to shareholders but increases financial leverage.
- Reorganization. In or outside of bankruptcy proceedings, to reach new agreement with creditors.

LBOs. Investment + restructuring + divestment. Usually, target company has less leverage before LBO and then more after the PE sale - PE use leverage to buyout and then transfer leverage to target company, securing the debt using the target company, and then keeping leverage high to reduce share dilution.

Evaluating restructurings.

- Size. A 'large' acquisition is 10% or more of acquirer's EV prior to transaction. 95% of acquisitions are under 1 bn and over 80% of targets are private companies.
- Stock price change upon transaction announcement. But initial changes often don't mean anything in the long-run according to research.
- Timing. Small simple transactions = several months. Larger = more than one year.

Preliminary valuation. Relative valuation usually to see if r_e rises or falls or find target value.

Define peer group companies and get data and use comparable company analysis (but flawed - what if peers themselves are overvalued; and doesn't have takeover premium), but also comparable transaction analysis (which include takeover premiums unlike comparable company analysis). EVs often used - not sensitive to capital structure.

Example of comparable transaction analysis. Find EPS, CF per share, BVPS, Sales per share of similar transactions and target, then assemble P-E, P-CF, P-BV, P-S for similar transactions to see if averages fluctuate too much - and if they don't, multiply by target values to get prices and average these prices.

Takeover premium PRM usually 20%-40%. Reflects **control premium**.

$$PRM = \frac{DP - SP}{SP}$$

DP = deal price, SP = unaffected stock price. Have to exclude pre-announcement increase from rumors or speculation.

Modeling and valuation of restructurings. Use pro form financial statements.

Revenue	1. Combine acquirer and target revenues. 2. Add revenue synergies or subtract the cost of incompatible activities (dis-synergies).
Operating expenses	1. Combine acquirer and target operating expenses. 2. Subtract cost synergies or add the cost of incompatible activities (dis-synergies).
Depreciation and amortization	1. Combine acquirer and target depreciation and amortization. 2. Add amortization of acquired intangible assets.
Other expense or income	1. Combine acquirer and target other expense or income.
Interest expense	1. Start with current acquirer interest expense. 2. Add increased interest from new debt issuance and revised interest rate.
Income taxes	1. EBT-weighted average of tax rates of acquirer and target; estimate usually provided by issuers
Shares outstanding	1. Start with current acquirer shares outstanding 2. Add shares from any share issuance

Notes:

- 'Total consideration' in a transaction = EV. This is because the acquirer pays for the target shares + ASSUMES target debt and then gets cash

(so, minus the cash). If the debt is transferred from the target to acquirer, the target owners 'receive' the value of the debt (they don't have to pay it anymore). So, the EV in a transaction = acquirer cash payment + acquirer shares issued and given to target owners + debt assumed - target cash.

- Materiality of deal: size (10% of EV prior to transaction announcement) or fit (change of strategy or focus).
- Conglomerate discount - when total company EV is less than sum of division or segment EVs.
- If considering selling a unit, consider it versus spinoff depending on which holds more value. Also consider reduced cost synergies.
- Cap rate = Operating income divided by property value.

7.6 Synthetic assets

Synthetic bonds. Made from a real fixed-income base, but then properties adjusted using options, swaps, CDOs, etc. Advantages are cost, liquidity, flexibility.

Synthetic credit rating. Custom rating system based on research, can be tied to actual rating systems used by e.g. Moody's.

8 Portfolio management

8.1 Economics and investment markets

PV model. Intertemporal rate of substitution $m_{t,s}$, the ratio of MU tomorrow / MU today. One-period real riskfree rate is inversely related to the intertemporal rate of substitution (if good times now and bad times in future, intertemporal rate is high $m_{t,1}$, so bond price now $P_{t,1}$ is high, so that one-period real risk free rate is low). That is, the higher the return the investor can earn, the more important current consumption becomes relative to future consumption.

- If good economy now, spend more now, so MU today is lower, so $m_{t,s}$ and therefore the price of an asset $P(m_{t,s})$ is higher, which means relative to \$ 1 bond, the bond price is lower today.
- If uncertainty tomorrow increases, MU tomorrow declines.
- If wealthy, then MU is low. So any risk leads only to a small decrease in MU. So wealthy people more ready to take risks and therefore need a smaller risk premium. So they buy more but eventually have the same willingness to buy as poor people as financial market equilibrium is reached.

- If person has 'decreasing absolute risk version', asset prices decline and returns rise. If more wealthy like this, then spend more today so MU today is low, so willing to buy more investments until MU tomorrow is low enough - until the market has one price. However, initially, if there are many wealthy people, then the $m_{t,s}$ is high, so asset prices are low, and poor people also benefit.
- If future economic growth is high, then low MU because know that will be secure tomorrow. So, increase borrowing now.
- If future uncertainty rises, MU tomorrow is lower, so $m_{t,s}$ is lower, P is lower, so 'level of relative utility' is low, will take low compensation per \$ 1 of bond (i.e. high bond price), so there will be more lower bond prices to choose from, so they will purchase more bonds.
- During a recession, the yield curve will probably steepen. During a bad economic time, buy more current bonds.

Future price uncertainty creates a discount for risk. The RHS below in the inequality is smaller; the covariance is the discount for risk. For a two-period model at the end of the first period:

$$P_{t,s} = E_t(P_{t+1,s-1}m_{t,1}) = \frac{E_t(P_{t+1,s-1})}{1 + i_{t,1}} + Cov_t(P_{t+1,s-1}, m_{t,1})$$

where

$$E_t(P_{t+1,s-1})E_t(m_{t,1}) > E_t(P_{t+1,s-1}m_{t,1})$$

because by analogy

$$E(xy) = E(x)E(y) + cov(x, y)$$

and if the covariance is negative. So,

$$E_t(P_{t+1,s-1})E_t(m_{t,1}) = E_t(P_{t+1,s-1}m_{t,1}) + Cov_t(P_{t+1,s-1}, m_{t,1})$$

where P is the actual modelled market price, and $E_t(P_{t+1,s-1}m_{t,1})$ is the risk-neutral price. For risk-average investors, Cov for most risky assets is negative because of risk aversion: the inverse future price - intertemporal rate relationship, e.g. when expected future price of investment is high, marginal utility of future consumption is low for that future.

Interest rates. If GDP is rising, real interest rate rises too - because if GDP is rising there will be more to buy later. If GDP is volatile, real interest rate is higher to account for a risk premium.

Notes: (1) If an asset such as equity is negatively correlated with GDP (ie rises when economy does badly) then Cov is positive and therefore its required rate of return might be lower than the real interest rate. (2) If GDP is rising, then cashflow numerator in PV is rising but so is real interest rate, so unclear

if PV is positive or not.

Upward-sloping yield curve. Returns to short-dated bonds are MORE negatively correlated with bad times than are returns to long-dated bonds. If a bond is a good hedge against bad times, then it has high price and low return - if the yield curve is upward sloping, investors value short-dated bonds more than long-dated bonds; the explanation might be that investors value short-dated bonds more because of recession-protection more than long-dated bonds where the future is far-off.

Taylor rule for interest rates

$$pr_t = st_t + \pi_t + 0.5(\pi_t - \pi^*) + 0.5(\ln(Y_t) - \ln(Y_t^*))$$

where pr_t = policy rate, st_t = short-term interest rate that balances long-term savings and borrowing in the economy.

Breakeven inflation rate = difference in rates between nominal and real bond rates = inflation + uncertainty over inflation. Riskfree bonds and risk premiums. Correlation between government bond returns and economic growth is generally negative - government bonds pay off in bad times most so investors pay for being able to hedge against bad economies.

Risk premiums. Subtract yield on an index-linked government bond from yield on conventional government bond (nominal) and subtract inflation expectations (e.g. from survey). This is the bond risk premium, the extra yield investors require for above the real required return and expected inflation. This is about uncertainty, e.g. in inflation. Bond risk premiums fall in bad economic times.

Short-dated default-free bonds are the best hedge against bad consumption outcomes - compared to long-dated default-free bonds or equity.

Equities and equity risk premium. Equities are correlated to economic performance so need extra premium.

P-E ratios have been increasing. Why? (1) increase in expected earnings, (2) falling real interest rates and lower GDP growth volatility, (3) decrease in inflation expectations, (4) falling uncertainty about future inflation, (5) lower equity risk premia. They also fluctuate a lot depending on the business cycle.

Robert Shiller's CAPE (real cyclically-adjusted P-E ratio). Real equity price (to remove inflation), E is 10-year MA of real earnings - deflate the volatility. Won't have short-term P-E fluctuations.

Commercial real estate. Like a bond because it pays rent but also is like equity as it has a resale value. Credit quality depends on tenants. There is also a

illiquidity premium. Finally, because commercial real estate capital values are correlated with economic performance, there is a further risk premium.

8.2 Analysis of active portfolio management

First choose a benchmark portfolio **measure**, e.g. market cap, to compare performance of active portfolio management. Can use float-adjusted market cap (share price x outstanding shares, where shares of insiders + restricted shares are excluded), but then active management is a zero-sum game (depending on whether the float-adjusted index has all assets or only some assets of a certain sector, industry, etc.) - but it's still fine to use and actually a good benchmark for individual portfolios or a subset of market portfolios. Because the market portfolio represents the average performance across all investors that own securities before costs, active investors as a group cannot outperform the market.

Then choose **benchmark portfolio**, e.g. Fidelity's Magellan fund. It may or may not have all the securities from which the actively-managed portfolio will choose from. However, assume that the universe of securities from which the benchmark and actively-managed portfolio is N.

Benchmark portfolio:

$$R_B = \sum_1^N w_{B,i} R_i$$

Actively-managed portfolio:

$$R_P = \sum_1^N w_{P,i} R_i$$

Active management value added, or **active return**:

$$R_A = R_P - R_B = \sum_{i=1}^N \Delta w_i R_i$$

$$\Delta w_i = w_{P,i} - w_{B,i}$$

Since $\sum \Delta w_i = 0$, $R_B \sum_1^N \Delta w_i = \sum_1^N \Delta w_i R_B = 0$. So, above can be rewritten as

$$R_A = \sum_{i=1}^N \Delta w_i R_i = \sum_{i=1}^N \Delta w_i R_i - \sum_{i=1}^N \Delta w_i R_B = \sum_{i=1}^N \Delta w_i (R_i - R_B) = \sum_{i=1}^N \Delta w_i R_{A,i}$$

where R_i is the return on an individual security, R_B is the return on the entire benchmark portfolio, and $R_{A,i}$ is the return on an individual security over or under the entire benchmark portfolio. The above more informatively shows where the total return is composed of returns from being overweight ($\Delta w_i > 0$

and underweight ($\Delta w_i < 0$ from stocks that have returns above the benchmark total return ($R_i - R_B > 0$) and below ($R_i - R_B < 0$).

Risk-adjusted value added = alpha based on portfolio beta:

$$\alpha_P = R_P - \beta_P R_B$$

Note that alpha is **not** necessarily the active return; if this is true, $\beta = 1$.

The above is just about changing asset weights. In the more complex situation, the asset allocation weights differ from a composite benchmark **and** each asset class is actively managed by individual security selection (so, introduce $R_{P,j}$). **Total value added** is then:

$$R_A = \sum_{j=1}^M w_{P,j} R_{P,j} - \sum_{j=1}^M w_{B,j} R_{B,j}$$

Rewriting this to alternatively decompose value added ('performance attribution').

$$\begin{aligned} R_A &= \sum_{j=1}^M w_{P,j} R_{P,j} - \sum_{j=1}^M w_{B,j} R_{B,j} + \left(\sum_{j=1}^M w_{P,j} R_{B,j} - \sum_{j=1}^M w_{P,j} R_{B,j} \right) \\ R_A &= \left(\sum_{j=1}^M w_{P,j} R_{P,j} - \sum_{j=1}^M w_{P,j} R_{B,j} \right) + \left(\sum_{j=1}^M w_{P,j} R_{B,j} - \sum_{j=1}^M w_{B,j} R_{B,j} \right) \\ R_A &= \left(\sum_{j=1}^M w_{P,j} R_{B,j} - \sum_{j=1}^M w_{B,j} R_{B,j} \right) + \left(\sum_{j=1}^M w_{P,j} R_{P,j} - \sum_{j=1}^M w_{P,j} R_{B,j} \right) \\ R_A &= \sum_{j=1}^M \Delta w_j R_{B,j} + \sum_{j=1}^M w_{P,j} R_{A,j} \end{aligned}$$

The first term = value added from **asset allocation** (different weights Δw) from benchmark or policy portfolio). Second term = value added from **security selection** = Fund return (using altered asset allocation weights) - benchmark/policy return (using **UNaltered** asset allocation weights).

Example. Fidelity had a return of -5.6% and its benchmark S&P500 had a return of -4.5%; PIMCO had a return of -0.3% while its benchmark Bloomberg Barclays US Aggregate had a 0% return. Policy weights 0.6 and 0.4 for Fidelity and PIMCO respectively. Portfolio weights are 0.68 and 0.32. We use fund return minus benchmark return to generate the value added for each fund.

Fund	Fund Return (%)	Benchmark Return (%)	Value Added (%)
Fidelity Magellan	-5.6	-4.5	-1.1
PIMCO Total Return	-0.3	0.0	-0.3
Portfolio Return	-3.9	-2.7	-1.2

$$R_A = ((0.68 - 0.6)(-4.5) + (0.32 - 0.4)(0)) + 0.68(-1.1) + 0.32(-0.3) = -1.2$$

Absolute and relative risk-return tradeoffs. Sharpe ratio is absolute risk-return tradeoff. Information ratio is relative risk-return tradeoff. IR is like a relative Sharpe ratio.

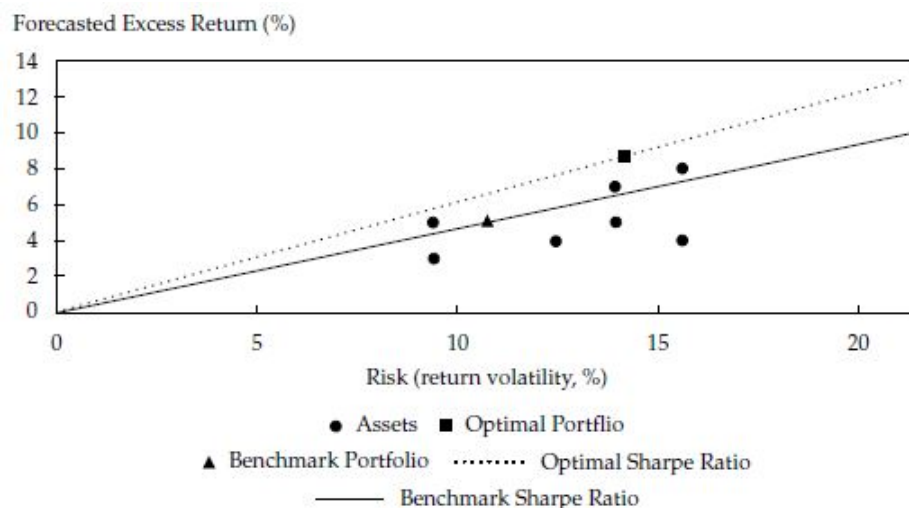
Sharpe ratio. Numerator is $E(R)$ if ex-ante (forecast/expected) or average R if ex-post:

$$SR = \frac{R_p - R_f}{\sigma_p}$$

Annualize the numbers: if monthly average values for R and σ , then multiply R by 12 and σ by $\sqrt{12}$ (variance increases by 12, so standard deviation is $\sqrt{12}$). These assumptions work if returns are independent and continuously compounded or logarithmic. Can use compounded or arithmetic returns - there's a small difference but shouldn't matter for purposes here. SRs historically are 0.2-0.6, and bond portfolio SRs have been the highest due to secular declines in interest rates.

Leverage and lending (i.e. adding cash or leverage in a portfolio) don't affect Sharpe ratio - though they affect where along the Sharpe ratio slope you are, depending on the level of risk (which can be adjusted by leveraging or lending). If w_p in portfolio and $1 - w_p$ in risk-free cash. Then, the combined portfolio c is:

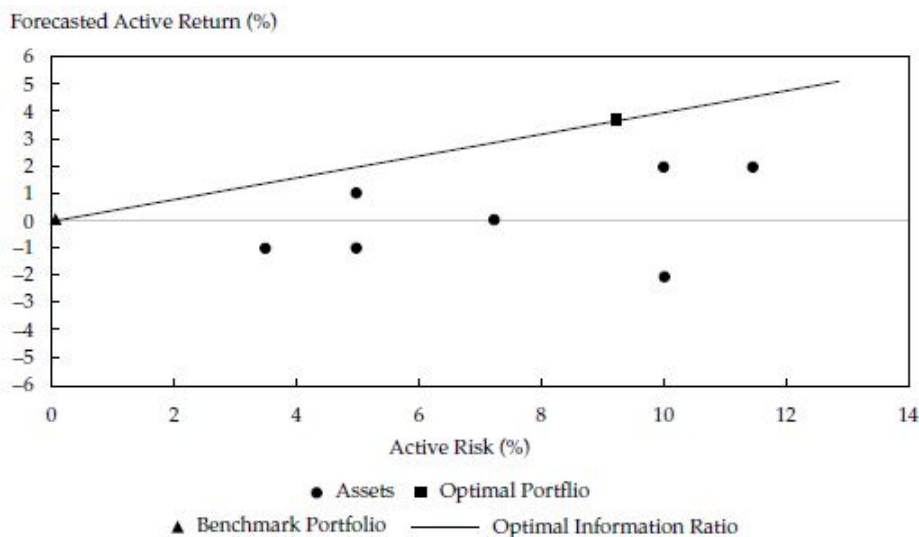
$$\begin{aligned}
 R_c &= w_p R_p + (1 - w_p) R_f \\
 \sigma_c^2 &= w_p^2 \sigma_p^2 + (1 - w_p)^2 \sigma_f^2 + 2w_p(1 - w_p)Cov(R_p, R_f) = w_p^2 \sigma_p^2 \\
 \sigma_c &= w_p \sigma_p \\
 SR_c &= \frac{R_c - R_f}{\sigma_c} = \frac{w_p R_p - w_p R_f}{w_p \sigma_p} = SR_p
 \end{aligned}$$



Information ratio. *Measures consistency of active return.* It is also a measure of PM skill (even more than information coefficient). It is like 'extra' ratio above the benchmark.

$$IR = \frac{R_p - R_b}{\sigma_{R_p - R_b}} = \frac{R_A}{\sigma_A}$$

Note: σ_A is called the 'tracking error' (not tracking risk, which is different).



In the above, as will be shown below, IR of optimal portfolio is higher than IR of any individual asset. The optimal risk is the one from

$$\sigma_A = \frac{IR}{SR_B} \sigma_B$$

where A is active portfolio, B is benchmark portfolio.

Unlike the Sharpe ratio, the information ratio is affected by the addition of cash or the use of leverage. For example, if the investor adds cash to a portfolio of risky assets, the information ratio for the combined portfolio will generally shrink. However, the information ratio of an unconstrained portfolio is unaffected by the aggressiveness of active weights. Specifically, if the active security weights, Δw_i , defined as deviations from the benchmark portfolio weights, are all multiplied by some constant, c , the information ratio of an actively managed portfolio will remain unchanged.

$$R_c = \sum c\Delta w_i R_i = c \sum \Delta w_i R_i = cR_A$$

$$IR_{\text{scaled by } c} = \frac{cR_A}{c\sigma_A} = IR$$

The Sharpe ratio of the actively-managed portfolio can be calculated from the benchmark portfolio Sharpe ratio + actively-managed portfolio information ratio IR:

$$SR_P^2 = SR_B^2 + IR^2$$

which means SR and IR are orthogonal as in Pythagoras. This means active portfolio with highest IR = highest Sharpe ratio (though above not useful if IR is negative). According to mean-variance theory (see below), expected IR is best criterion for assessing active performance among managed funds with **same benchmark**.

Above equation comes from

$$\frac{R_P - R_F}{\sigma_P^2} = \frac{R_B - R_F}{\sigma_B^2}$$

Squaring both sides,

$$\frac{SR_P^2}{\sigma_P^2} = \frac{SR_B^2}{\sigma_B^2}$$

$$SR_P^2 = SR_B^2 \frac{\sigma_P^2}{\sigma_B^2} = SR_B^2 \left(1 + \frac{\sigma_P^2 - \sigma_B^2}{\sigma_B^2} \right) = SR_B^2 \left(1 + \frac{\sigma_A^2}{\sigma_B^2} \right)$$

and since

$$IR = SR_B \frac{\sigma_A}{\sigma_B}$$

$$SR_P^2 = SR_B^2 + IR^2$$

This also means the SR of the actively-managed portfolio is higher than the benchmark portfolio. The equation also implies that the higher the IR, the higher the SR of the actively-managed portfolio.

Using the mean-variance optimality condition

$$\frac{E(R_A)}{\sigma_A^2} = \frac{E(R_B - R_F)}{\sigma_B^2}$$

which comes from maximizing the portfolio SR with respect to w:

$$SR_P = \frac{R_P - R_F}{\sigma_P}$$

which is squared to make it easier for differentiation:

$$SR_P^2 = \frac{(R_P - R_F)^2}{\sigma_P^2}$$

Expand the numerator and denominator:

$$SR_P^2 = \frac{(wR_A + R_B - R_F)^2}{(w^2\sigma_A^2 + \sigma_B^2)}$$

Differentiate, using the chain and quotient rules, and optimize:

$$\frac{d(SR_P^2)}{dw} = \frac{2(wR_A + R_B - R_F)R_A(w^2\sigma_A^2 + \sigma_B^2)^2}{(w^2\sigma_A^2 + \sigma_B^2)^2} - \frac{(wR_A + R_B - R_F)^2 2w\sigma_A^2}{(w^2\sigma_A^2 + \sigma_B^2)^2} = 0$$

The denominator is not relevant:

$$2(wR_A + R_B - R_F)R_A(w^2\sigma_A^2 + \sigma_B^2)^2 - (wR_A + R_B - R_F)^2 2w\sigma_A^2 = 0$$

Reduce:

$$2(wR_A + R_B - R_F)R_A(w^2\sigma_A^2 + \sigma_B^2)^2 = (wR_A + R_B - R_F)^2 2w\sigma_A^2$$

$$R_A(w^2\sigma_A^2 + \sigma_B^2)^2 = (wR_A + R_B - R_F)w\sigma_A^2$$

and with w=1 which is what we assume $R_P = R_A + R_B$:

$$\frac{R_A}{\sigma_A^2} = \frac{(wR_A + R_B - R_F)}{\sigma_A^2}$$

which is

$$\frac{E(R_A)}{\sigma_A^2} = \frac{E(R_B - R_F)}{\sigma_B^2}$$

and continuing,

$$\frac{IR}{\sigma_A} = \frac{SR}{\sigma_B}$$

rearranging,

$$\sigma_A = \frac{IR}{SR_B} \sigma_B$$

which is the **optiaml amount of active risk**. Note this should actually be

$$\sigma_A = \frac{IR}{SR_B} \sigma_B \beta$$

but assume here $\beta = 1$. Using a FIXED IR, calculate the OPTIMAL ACTIVE RISK (aggressiveness of weights) of the actively-managed portfolio. From this, calculate the total risk of the actively-managed portfolio - see below for derivation and notes about this:

$$\sigma_P^2 = \sigma_B^2 + \sigma_A^2$$

and then calculate the optimal active risk by

$$\sigma_A IR = R_A$$

which is then added to the benchmark portfolio above the riskfree rate = $R_B + R_A = R_P$. Then calculate the OPTIMAL PORTFOLIO SHARPE RATIO:

$$SR_P = \frac{R_P}{\sigma_P}$$

and confirm it's the same as

$$SR_P^2 = SR_B^2 + IR^2$$

The total risk of the active-managed portfolio is

$$\sigma_P^2 = \sigma_B^2 + \sigma_A^2$$

comes from

$$R_A = R_P - R_B$$

$$R_P = R_A + R_B$$

$$\sigma_P^2 = \sigma_A^2 + \sigma_B^2 + 2Cov(R_A, R_B)$$

since R_A (the actively-managed portfolio EXCESS return) is not correlated (probably independent) of R_B ,

$$\sigma_P^2 = \sigma_A^2 + \sigma_B^2$$

We could also derive directly from

$$R_A = R_P - R_B$$

$$\sigma_A^2 = \sigma_P^2 + \sigma_B^2 - 2Cov(R_P, R_B)$$

but since R_P and R_B aren't independent nor uncorrelated,

$$\sigma_A^2 = \sigma_P^2 + \sigma_B^2 - 2Cov(R_A + R_B, R_B)$$

$$\sigma_A^2 = \sigma_P^2 + \sigma_B^2 - 2Cov(R_A, R_B) + 2Cov(R_B, R_B)$$

$$\begin{aligned}\sigma_A^2 &= \sigma_P^2 + \sigma_B^2 - 0 - 2\sigma_B^2 \\ \sigma_A^2 &= \sigma_P^2 + \sigma_B^2 - 2\sigma_B^2 \\ \sigma_A^2 + \sigma_B^2 &= \sigma_P^2\end{aligned}$$

Sometimes, we want to keep the covariance term when calculating the active risk, e.g. if the correlation coefficient of two active returns is CORR,

$$\sigma_P = \sqrt{\sigma_A^2 + \sigma_B^2 + 2(\sigma_A)(\sigma_B)CORR(r_A, r_B)}$$

Optimal IR. The optimal portfolio IR is higher than the IR of any individual asset of the portfolio (this is intuitive) and is mathematically:

$$IR_P^2 = IR_1^2 + IR_2^2 + \dots$$

which means portfolio IR and asset IRs are orthogonal as in Pythagoras. It is proved below.

Basic fundamental law of active portfolio management (Grinold). Framework for conceptualizing portfolio active weights and returns. The active return of each security is:

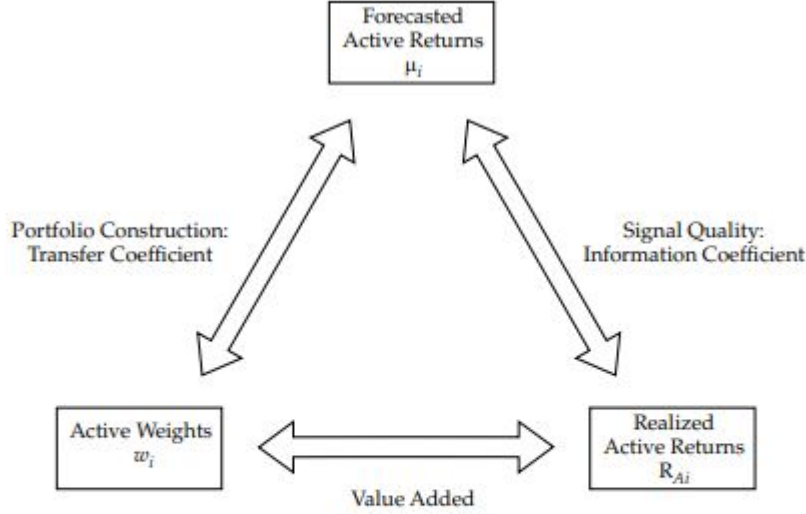
$$R_{Ai} = R_i - R_B$$

or equivalently

$$R_{Ai} = R_i - \sum_{j=1}^K \beta_{j,i} R_j$$

which is a multi-factor model.

The basic fundamental law is based on the **correlation triangle**:



To understand the role of the correlation coefficient, consider the following algebraic expansion of Equation 3 that uses COV, STD (σ), and COR (ρ) to designate the covariance, standard deviation, and correlation coefficient functions, respectively:

$$\begin{aligned}
 R_A &= \sum_{i=1}^N \Delta w_i R_{Ai} \\
 &= \text{COV}(\Delta w_i, R_{Ai}) N \\
 &= \rho(\Delta w_i, R_{Ai}) \sigma(\Delta w_i) \sigma(R_{Ai}) N
 \end{aligned}$$

There is little hope of adding value if the investor's forecasts of active returns do not correspond at least loosely to the realized active returns. Signal quality is measured by the correlation between the forecasted active returns, μ_i , at the top of the triangle, and the realized active returns, R_{Ai} , at the right corner, commonly called the information coefficient (IC). Investors with higher IC, or ability to forecast returns, will add more value over time but only to the extent that those forecasts are exploited in the construction of the managed portfolio. The correlation between any set of active weights Δw_i , in the left corner, and forecasted active returns, μ_i , at the top of the triangle—measures the degree to which the investor's forecasts are translated into active weights, called the transfer coefficient (TC). Note the portfolio manager's **forecasted** active return is

$$\mu_i = E(R_{Ai})$$

The total active return is

$$R_A = \sum_{i=1}^N \Delta w_i R_{Ai}$$

which is the bottom part of the correlation triangle.

$$\Delta w_i^* = \frac{\mu_i \sigma_A}{\sigma_i^2 \sqrt{\sum_{i=1}^N \frac{\mu_i^2}{\sigma_i^2}}}$$

is the left side of the correlation triangle. A larger forecast return μ_i leads to higher weights; given a higher active portfolio risk σ_A the active weight must be higher; higher individual security i volatilities lead to lower weights as don't need to so much active weights with so much individual volatility, and the square root term is a divisor - more returns means less weight per security.

The Δw_i^* is derived by maximizing IR (which in turn will maximize the portfolio Sharpe ratio):

$$IR = \frac{\mu_A}{\sigma_A} = \frac{\sum \Delta w_i \mu_i}{\sqrt{\sum \Delta w_i^2 \sigma_i^2}}$$

But this is a Rayleigh quotient, so the optimal value is

$$\Delta w_i \propto \frac{\mu_i}{\sigma_i^2}$$

$$\Delta w_i = k \frac{\mu_i}{\sigma_i^2}$$

To determine k,

$$\begin{aligned} \sum \Delta w_i^2 \sigma_i^2 &= \sigma_A^2 \\ \sum \left(k \frac{\mu_i}{\sigma_i^2} \right)^2 \sigma_i^2 &= k^2 \sum \frac{\mu_i^2}{\sigma_i^2} = \sigma_A^2 \\ k &= \frac{\sigma_A}{\sqrt{\sum \left(\frac{\mu_i^2}{\sigma_i^2} \right)}} \end{aligned}$$

Substituting back,

$$\Delta w_i^* = k \frac{\mu_i}{\sigma_i^2} = \frac{\mu_i \sigma_A}{\sigma_i^2 \sqrt{\sum \left(\frac{\mu_i^2}{\sigma_i^2} \right)}}$$

which is the result. This result can be used to prove:

$$IR_P^2 = \sum IR_i^2$$

by taking Δw_i^* and plugging it into

$$\mu_A = \sum \Delta w_i^* \mu_i = \sum \left(\frac{\mu_i \sigma_A}{\sigma_i^2 \sqrt{\sum \left(\frac{\mu_i^2}{\sigma_i^2} \right)}} \right) \mu_i = \frac{\sigma_A}{\sqrt{\sum \left(\frac{\mu_i^2}{\sigma_i^2} \right)}} \sum \left(\frac{\mu_i^2}{\sigma_i^2} \right) = \sigma_A \sqrt{\sum IR_i^2}$$

$$\begin{aligned} IR_P &= \frac{\mu_A}{\sigma_A} = \sigma_A \sqrt{\sum IR_i^2} \frac{1}{\sigma_A} = \sqrt{\sum IR_i^2} \\ IR_P^2 &= \sum IR_i^2 \end{aligned}$$

IC is the expected information coefficient - how much information there is between the expected forecast and the actual forecast. Ex-ante IC is cross-sectional correlation between the forecasted active returns μ and the realized active returns R . Ex-post IC is about how good a portfolio manager can forecast. It is a correlation, so varies between -1 (perfectly bad PM) and +1 (perfectly good PM). This can be seen from the graphic above:

$$IC = \rho \left(\frac{R_{Ai}}{\sigma_i}, \frac{\mu_i}{\sigma_i} \right)$$

where returns are divided by σ above and elsewhere for risk-adjusted returns. Assume:

$$\mu_i = E(R_A) = \rho \sigma_i S_i = IC \sigma_i S_i$$

which makes μ_i more like a standard deviation of the portfolio returns. Note σ_i is like a scaling because it is the denominator in the IC formula. S is the direction 'score' - which way the security will go (up/down), and is like a standard deviation of the forecasted return of security i ; it is needed because the correlation formula of IC is generally positive (+ with +, - with -; and we need the - with - positive correlations to be negative). S is either +1 for up or -1 for down. $\sum S = 0, \sigma_S = 1$. Scores affect IC's - if the scores are *accurate* and *result in large active returns*, then it will raise ICs. Substituting μ_i above into the active weights,

$$\Delta w_i^* = \frac{\mu_i \sigma_A}{\sigma_i^2 \sqrt{IC^2 N}} = \frac{\mu_i \sigma_A}{\sigma_i^2 IC \sqrt{BR}}$$

where BR is the breadth, which is less than N if there are positive correlations, equal to N if no correlations between asset active returns, and more than N if there are other factors being considered by the portfolio manager (e.g. macro factors for each asset). If there is more than one period (e.g. 12 months instead of 1 year), and the decisions across periods are uncorrelated, then the original BR is multiplied by 12. BR can be much higher than N if there is arbitrage, as the decision is not just about whether an asset will move higher and another asset will move lower; it's also about the *relative* movement between the two. Breadth can be calculated assuming all cross-correlations other than the diagonal in the correlation matrix are ρ :

$$BR = \frac{N}{1 + (N - 1)\rho}$$

but returning to the proof...

$$\sigma_A^2 = \sum (\Delta w_i^*)^2 \sigma_i^2$$

where if the square root terms aren't correlated, then there are N terms and therefore the 'breadth' (BR) is N . Substituting in μ_i and Δw above into the R_A equation, we get, for unconstrained portfolios:

$$E(R_A) = \sum_{i=1}^N \Delta w_i \mu_i$$

and with optimal weights:

$$E(R_A) = \sum_{i=1}^N \frac{\mu_i \sigma_A}{\sigma_i^2 IC \sqrt{BR}} \mu_i = \sum_{i=1}^N \frac{\mu_i^2 \sigma_A}{\sigma_i^2 IC \sqrt{BR}} = \sum_{i=1}^N \frac{IC^2 S_i^2 \sigma_A}{IC \sqrt{BR}} = \frac{IC \times BR \sigma_A}{\sqrt{BR}} = IC \sqrt{BR} \sigma_A$$

$$E(R_A)^* = IC \sqrt{BR} \sigma_A$$

which is the **Basic Fundamental Law**. The above can be reduced to

$$\frac{E(R_A)^*}{\sigma_A} = IR^* = IC \sqrt{BR}$$

So the IR of the active manager depends on the number of opportunities for uncorrelated investment decisions BR and the skill of forecasting, IC.

Defintions

- Total return = benchmark return + active return
- Total weights = benchmark weights + active weights
- Active risk = square of (active weights)(active return volatility) and sum, and square root.

From the graphic above, the transfer coefficient (for **non-optimal weights**) is

$$TC = \rho \left(\frac{\mu_i}{\sigma_i}, \Delta w_i \sigma_i \right)$$

and using

$$\Delta w_i^* = \frac{\mu_i \sigma_A}{\sigma_i^2 \sqrt{\sum_{i=1}^N \frac{\mu_i^2}{\sigma_i^2}}}$$

$$\mu_i \propto \Delta w_i^* \sigma_i^2$$

so,

$$TC = \rho(\Delta w_i^* \sigma_i, \Delta w_i \sigma_i)$$

since TC is a correlation,

$$|TC| \leq 1$$

It makes sense that TC is the correlation between the forecasted active return μ_i and the active weight Δw_i . Moreover, the active weight is scaled by the active risk volatility to represent the aggressiveness of the active weight.

TC is used because it captures constraints on the portfolio, e.g. no shorting. If Δw_i is very negative, it could mean shorting the stock, but this might not be allowed by some institutions. TC measures the efficiency with which a manager translates their insights into their portfolio bets, i.e. they look at the optimal active weights (i.e, the unconstrained portfolio weights Δw_i^* and then constrain

it and get constrained weights Δw_i . TC = 'tracking risk' (not tracking error, which is something else). TC = 0 means no correlation between active return forecasts and active weights, so no expectation of value added by active management, and PM has no skill so just invest in the benchmark portfolio. TC = 1 means perfect correlation and expected value fully in active weights (TC = 1 is also **unconstrained** portfolio). TC negative means weights need to be rebalanced.

The TC is basically a modification of the IR^* to constrain it when there are 'no shorts allowed', etc. So, adding TC to the Basic Fundamental Law, we get the Full Fundamental Law (constrained portfolios):

$$E(R_A) = (TC)(IC)\sqrt{BR}\sigma_A$$

and

$$\frac{E(R_A)}{\sigma_A} = IR = (TC)(IC)\sqrt{BR}$$

Definition(s):

- Portfolio 'risk constraint' = σ_A . The optimal weights might lead to a certain σ_A^* , but perhaps the portfolio manager wants a different risk constraint.

TC added to active risk:

$$\sigma_A = \frac{(TC)(IR^*)}{SR_B}\sigma_B$$

rearranging,

$$SR_B \frac{\sigma_A}{\sigma_B} = (TC)(IR^*)$$

and inserting into

$$SR_P^2 = SR_B^2 + IR^2$$

$$SR_P^2 = SR_B^2 + [(TC)(IR^*)]^2$$

so that if TC = 0, better to just invest in the benchmark portfolio.

Ex-post (realized) performance measurement. Now we know IC_R the realized IC, so

$$E(R_A|IC_R) = (TC)(IC_R)\sqrt{BR}\sigma_A$$

$$R_A = E(R_A|IC_R) + Noise$$

where Noise = noise from constraining the optimal portfolio. An academic study finding that the above equation's realized variation (= tracking error = performance variation = realized variation) can be decomposed into:

$$\sigma_A^2(E(R_A|IC_R)) \propto TC^2$$

$$\sigma_A^2(noise) \propto (1 - TC^2)$$

Portfolio active variance (performance variance) can be divided into (1) realized information proportional to TC^2 and (2) noise proportional to $(1 - TC)^2$.

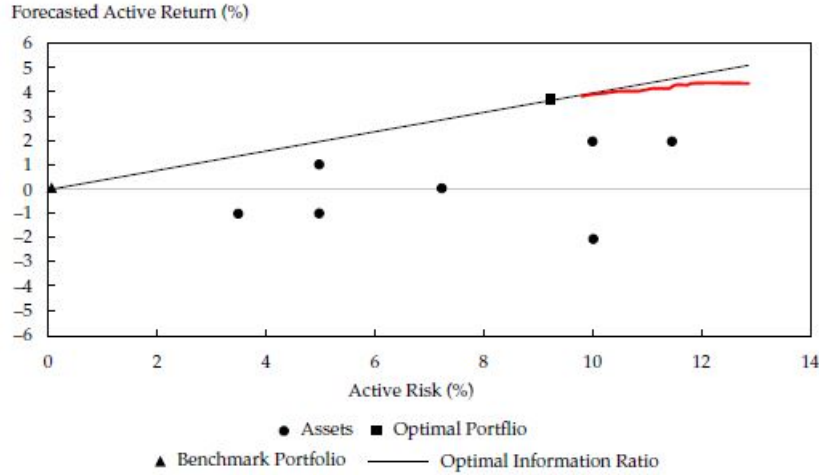
Constrained portfolios. Optimization of active weights can be constrained, e.g. due to sum of active weights = 0 (i.e. cannot have leverage or lending) or active risk must be less than X%, or cannot have negative (i.e. shorted positions) in the portfolio (= active weight + benchmark weight). If optimization is constrained, $TC < 1$; otherwise, $TC = 1$. Since

$$IR = (TC)(IC)\sqrt{BR}$$

If TC falls, then IR also falls. Since

$$\frac{E(R_A)}{\sigma_A} = IR$$

Increasing the allowed active risk doesn't lead to a proportional increase in $E(R_A)$. This means increasing the allowed active risk of a constrained portfolio leads to the **IR curve curving downwards** because of diminishing returns as the PM encounters more and more problems with the constraints; so the active return rises but slower compared to the increase in allowed active risk. However, increasing the allowed active risk of an unconstrained portfolio doesn't do that, since $TC = 1$, so IR is stable, so $E(R_A)$ increases proportionally and the IR curve is straight.



Uncertainty over the IC. Incorporate uncertainty about IC:

$$\sigma_A = \sigma_{IC}(\sqrt{N}\sigma_{RM}) \approx \sigma_{IC}\sqrt{BR}$$

so, solving for $\sqrt{BR}$ and inserting into the fundamental law and assuming $TC = 1$:

$$E(R_A) = (TC)(IC)\sqrt{BR}\sigma_A = \frac{IC}{\sigma_{IC}}\sigma_A$$

The higher the uncertainty about forecasting ability σ_{IC} , the smaller the likely expected value added.

Time-series IC. E.g. if 55% of the time correct and 45% of the time incorrect, then $IC = 0.55 - 0.45 = 0.1$. If a time series of T **predicted** (+1 or -1) scores, S_p , and a time series of T **realized** dichotomous scores (+1 or -1) S_r , and both have mean = 0, then

$$Cov(S_p, S_r) = \frac{1}{T} \sum S_{p,t} S_{r,t}$$

Application example. Start with historical active return volatility. Then add scores manually, ensuring they sum to 0 and have a cross-sectional $\sigma = 1$. Assume an ex-ante IC and use $\mu_i = IC\sigma_i S_i$ to generate active returns. Note that ex-ante IC will have to be adjusted for consistency with the forecast scores. Use Excel solver or other app to optimize the active weights Δw_i^* using the above equation. Then confirm $E(R_A) = (TC)(IC)\sqrt{Br}\sigma_A$ is the same as $\sum \Delta w_i^* \mu_i$. Now the ex-post IC will be different because of the scores.

Application example - with portfolio constraint. E.g. if no shorting or σ_A is set to a limit. Taking latter example,

$$\frac{\sigma_{constraint}}{\sigma_A} = x$$

Take x and apply to returns, e.g. x is added to one of the benchmark asset weights and x is subtracted from the other, and then vice versa for the opposite case.

8.3 ETFs: Mechanics and applications

ETF issuers transact with authorized participants (APs) in primary market who then deal into the secondary market. APs have incentive by arbitrage to keep the ETF in a tight price band.

ETFs are more fair than mutual fund. If mutual fund buys a security, everyone in the MF pays the transaction cost. However, those holding ETFs will not incur any transaction cost if others are buying or selling. Moreover, if APs create or redeem from ETF issuer, AP bears cost and pass it on to its buyers so that the other ETF holders and ETF issuer do not bear the cost.

Expense ratios. Usually lower than mutual funds.

Index tracking and tracking errors. Daily standard deviation to ETF index. Fees and expenses, accounting, regulations, taxes, sampling (ETF PM only holds a sample of the index), etc. lead to tracking errors. Usually underperforms index because expense ratio is subtracted. Note: sampling (also: optimization) leads to high tracking errors.

Tracking errors. Rolling average of ETF versus its benchmark. Tracking error is just an annualization of daily standard deviations - one number, doesn't say what happens over time and when. Tracking difference does - basically tracking error over a long time. More informative than just expense ratio

Although additions and deletions of securities from the underlying benchmark index may occur and result in tracking error, such index changes generally occur infrequently (often quarterly). In addition, ETF portfolio managers may work with APs for index rebalance trades to ensure market-on-close pricing to minimize this source of tracking error. Therefore, the resulting tracking error caused by index changes will not likely be as large as the tracking error caused by representative sampling or by fees and expenses incurred by the ETF.

Bid-ask spreads.

Q. The bid-ask spread for very liquid, high-volume ETFs will be *least* influenced by the:

Practice Problem

- A. market maker's desired profit spread.
- B. creation/redemption fees and other direct costs.
- C. likelihood of receiving an offsetting ETF order in a short time frame.

Solution

B is correct. ETF bid-ask spreads are generally less than or equal to the combination of the following:

- ± Creation/redemption fees and other direct costs, such as brokerage and exchange fees
- + Bid-ask spread of the underlying securities held by the ETF
- + Compensation for the risk of hedging or carrying positions by liquidity providers (market makers) for the remainder of the trading day
- + Market maker's desired profit spread
- - Discount related to the likelihood of receiving an offsetting ETF order in a short time frame

For very liquid and high-volume ETFs, buyers and sellers are active throughout the trading day. Therefore, because most of these ETF trades are matched extremely quickly and never involve the creation/redemption process, the first three factors listed do not contribute heavily to their bid-ask spreads. So, creation/redemption fees and other direct costs are not likely to have much influence on these ETFs' bid-ask spreads.

Dividends. ETFs generally distribute dividends - they send the dividends to brokerages which keep the ETFs for investors. Mutual funds in contrast may or may not distribute; they can reinvest (but this is still generally taxable).

Capital gains. ETF funds sell securities it has, realize capital gains, which is then sent as cash distributions to ETF holders. However, ETF funds send

much fewer capital gains to investors than mutual funds because mutual funds actively buy and sell securities a lot for the fund, whereas ETFs buy and sell to rebalance the fund, tracking index composition changes, or for creation and redemptions.

Return-of-capital (ROC) distributions. An ETF can distribute more than it earns in income, dividends, etc. by eating into the investor capital. This might be done to distribute funds steadily to ETF holders. But this can't be indefinitely sustained, because it affects NAV. ROC distributions **are generally not taxable**.

Taxes. Capital gains: ETFs are more efficient than mutual funds because of (1) tax fairness - as with transaction fees, only parties to the transaction incur capital gains, whereas in MFs, the capital gain is passed onto the fund and therefore all fundholders; (2) at ETF redemption, PM can sell alternative, low-capital gain shares, reducing cost.

ETF trading costs. Bid-ask spreads - can be higher or smaller depending on many factors (including taxes, regulations, creations/redemption trading, etc.); if can hedge ETF's benchmarked index (e.g. if the benchmark has a corresponding future like S&P 500 index and S&P 500 index futures) then bid-ask spread is narrower. Net asset values (NAVs) at end of day, intraday-NAVs (iNAVs) intraday. NAVs are based on 'creation baskets'. (i)NAV premiums and discounts:

$$\frac{\text{ETF price} - (\text{i})\text{NAV per share}}{(\text{i})\text{NAV per share}}$$

Note there can be timing differences within the ETF portfolio assets for valuation, and ETFs themselves might not be traded so might go stale, so mispricings can occur.

Total ETF costs. **Trading costs and management costs** (taken from ETF portfolio holdings). ETF PMs might lend securities of portfolio, reducing fees. If hold ETF for short time, trading costs are large; if hold for long time, management fees become more important.

Round-trip holding cost. If trading commission is 10 on 20000 trade (0.05% each way), with 0.15% bid-ask spread on purchase and sale. Round-trip cost = $(0.05\% \times 2) + (0.5 \times \text{bid-ask spread} \times 2) = 0.25\%$.

ETPs (exchange-traded products), e.g. ETNs (exchange-traded notes). ETNs not truly funds - don't hold underlying securities; they are basically unsecured debt, and have large counterparty risk (credit risk). Synthetic ETFs replicate index or fund performance using swaps, derivatives, etc. and are subject to counterparty risk as well. ETN and ETF issuers, and their counterparty risk, can be monitored by CDS pricing - if it goes above 5%, then possible danger.

Leveraged and inverse funds. Leveraged funds: e.g. a 3x-ETF has 300% exposure to an index relative to NAV - but only a daily basis.

	Index Level	One-Day Index Return (%)	3× ETF NAV (£)	Notional Swap Exposure (£)	3× Swap Exposure (£)	Swap Exposure Adjustment (£)
Day 1	100	—	100	300	300	0
Day 2	105	5%	115	315	345	30

	Index Level	One-Day Index Return (%)	Index Period Return (%)	−2× ETF NAV	One-Day ETF Return (%)	2× ETF Holding Period Return (%)
Day 1	100	—		100	—	—
Day 2	110	10%	10%	80	−20%	−20%
Day 3	99	−10%	−1%	96	20%	−4%

ETF strategies. Use for efficient portfolio management: (1) use excess cash balances quickly and minimize cash drag, (2) rapid portfolio rebalancing and more frequently, (3) fill temporary gaps in portfolio, (4) when transition PMs or strategies - a placeholder.

Fund closure risk. Fund-closure risk is the risk that an ETF may shut down. The reasons that lead to an ETF closing down often have to do with changes in regulations, increased competition, and corporate activity (merger and acquisition activity within the ETF industry).

Expectations risk. Expectation-related risk is the risk that some ETF investors may not fully understand how more complex ETFs will perform because of a lack of understanding of sophisticated assets classes and strategies.

Other ETF advantages: (1) use for fixed-income which is more illiquid, (2) exposure to other assets, including commodities.

Disadvantages of ETFs for large asset owners: (1) can negotiate lower fees for SMAs, (2) SMA can be more customized, (3) many regulators require large ETF holdings be disclosed, adding to regulatory burdens.

Active versus factor investing. Active = discretionary. Factor (smart beta) ETFs - use fundamental factors, often quantitative and openly-disclosed models; can be higher or lower risk than market, and use to manage portfolio risks. Smart beta (factor) strategies aren't tactical; since they're fundamental, they're long-term investment strategies.

Factor (smart beta) strategy ETFs are usually benchmarked to an index created with predefined rules for screening and/or weighting stock holdings and are considered longer-term, buy-and-hold investment options rather than tactical trading instruments. The strategy index rules are structured around return drivers or factors, such as value, dividend yield, earnings or dividend growth, quality, stock volatility, or momentum. Investors using factor-based investing seek outperformance versus a benchmark or portfolio risk modification.

8.4 Multifactor models

CAPM doesn't do well empirically. Arbitrage pricing theory (APT) - alternative to CAPM. If investor can form well-diversified portfolio and eliminate asset-specific risk; and there are no arbitrage opportunities:

$$E(R_p) = r_f + \lambda_j \beta_{p,j} + ..$$

where λ is the reward for bearing the risk (factor risk premium or factor price) - return **surprises** so have an expected value = 0 (not just returns), β is the sensitivity of the portfolio to the factor j .

CAPM says, based on mean-variance optimization and max Sharpe ratio, that all investors hold only a combination of the market portfolio and riskfree asset (two-fund separation theorem). This market portfolio and the riskfree asset are then plotted on the SML; the market portfolio has $\beta = 1$ and r_m , while the riskfree asset is $\beta = 0$ and r_f ; the line connecting them has slope $r_m - r_f$. So though all investors hold only the market portfolio and/or riskfree asset, if they were to hold an individual asset, given the specific asset return or specific asset beta, we can calculate the other.

CAPM assumes that each investor will hold a unique portfolio with its own particular array of betas, as opposed to the identical "market portfolio". In some ways, the CAPM can be considered a "special case" of the APT in that the securities market line represents a single-factor model of the asset price, where beta is exposed to changes in value of the market.

CAPM estimates cost of equity using only correlations, returns, variances. It is based on mean-variance optimization. CAPM has strict assumptions: (1) markets are efficient, (2) investors are rational and risk-averse, (3) all investors have same expectations about returns. APT is based on no-arbitrage. APT assumes only (2), and from this, that investors form well-diversified portfolios, eliminate asset-specific risk, and no arbitrage opportunities (between portfolios). APT doesn't require market efficiency - otherwise, smart beta strategies won't work. Extensions of CAPM such as Fama-French and Carhart, can be seen as APTs.

The APT assumption of requiring only no-arbitrage instead of CAPM's market equilibrium is looser. Even if there is no arbitrage, there might not be market

equilibrium because people are not holding mean-variance optimized portfolios.

Checking arbitrage between multifactor models based on $E(R) = 0.05 + \beta \times 0.05$ - short $0.5A + 0.5C$ and buy D. To find arbitrage opportunity, equalize factor sensitivity and make sure $\sum a \times r_f = r_f$

Portfolio	Expected Return	Factor Sensitivity
A	0.0750	0.50
B	0.1500	2.00
C	0.0700	0.40
D	0.0800	0.45
0.5A + 0.5C	0.0725	0.45

Carhart 4-factor model.

$$E(R_p) = r_f + \beta_1 RMRF + \beta_2 SMB + \beta_3 HML + \beta_4 WML$$

where RMRF (return minus risk free) = return on a value-weighted equity index in excess of a 1-month T-bill rate; SMB = small minus big; average return on 3 small-cap portfolios minus average return on 3 large-cap portfolios. HML = high minus low, average return on 2 high book-to-market (value) portfolios minus average return on two low book-to-market (growth) portfolios. WML = winners minus losers; return on a portfolio of past year's winners minus return on losers.

Multifactor models: (1) macroeconomic, (2) fundamental, and (3) statistical (e.g. Fama-French, Carhart). Can have either of these for fixed-income, equity, etc. models. Statistical factor models reproduce historical return covariances ('factor analysis models') and historical return variances ('principal component models'). (3) is simple in theory but hard to interpret the factors.

Macroeconomic models. Factors are *surprise* returns, intercept is expected return. Define factor series first and then regress for coefficients (factor sensitivities / factor betas / factor loadings), error represents asset-specific risk. Portfolio returns usually have negative sensitivities for inflation - if they have positive ones, they are 'worth' more for hedging so have lower required returns.

$$R_i = E(R_i) + b_1 F_{inflation} + b_2 F_{GDP} + \epsilon_i$$

and ϵ_i is the firm-specific *surprise*. Can combine multiple macroeconomic models by simply adding them.

Fundamental models. Factor sensitivities are standardized beta:

$$b_{i,k} = \frac{\text{Value of attribute k for asset i} - \text{Average value of attribute k}}{\sigma_{\text{Value of k}}}$$

This standardization leads to each factor sensitivity intuitively being a 'pure factor' with standard deviation = 1.

Unlike macroeconomic models, in fundamental models define the factor sensitivities first (standardized betas first) and then regress for factors, e.g. define beta as P-E ratio and then regress for the factor (i.e. contribution of return from P-E ratio) for that security. Note that for each time period, the factor betas are updated and there is a new regression to find the factor returns (e.g. value premium of 3%, momentum premium of 2%). To forecast, need to forecast both the factor betas and factor returns.

$$R_i = a_i + b_1 F_{\text{value}} + b_2 F_{\text{capsize}} + \epsilon_i$$

where ϵ_i is the return not specified by the model.

Intercepts. In a macroeconomic model, it is the expected return if no surprises. In a fundamental model, if the sensitivities were not standardized, because sensitivities and 0 asset specific risk mean the intercept is the riskfree rate. However, since they are standardized, the intercept is just there to adjust the regression line to fit the asset specific returns $E(\epsilon_i) = 0$. This is true even if the standardized sensitivities are 0 and asset specific returns are 0, because the meaning of the sensitivities has changed with standardization. So, the intercept is now purely statistical.

Return attribution. Active return is based on factor tilts + security selection:

$$R_p - R_b = \sum (\text{Portfolio sensitivity} - \text{Benchmark sensitivity})_k \times (\text{Factor return})_k + \text{Security selection}$$

Do this for each of k factors. Factor sensitivities are b_k .

Factor	Factor Sensitivity			Factor Return (4)	Contribution to Active Return	
	Portfolio (1)	Benchmark (2)	Difference (3) = (1) - (2)		Absolute (3) × (4)	Proportion of Total Active
RMRF	0.95	1.00	-0.05	5.52%	-0.2760%	-13.3%
SMB	-1.05	-1.00	-0.05	-3.35%	0.1675%	8.1%
HML	0.40	0.00	0.40	5.10%	2.0400%	98.4%
WML	0.05	0.03	0.02	9.63%	0.1926%	9.3%
A. Return from Factor Tilts =					2.1241%	102.4%
B. Security Selection =					-0.0500%	-2.4%
C. Active Return (A + B) =					2.0741%	100.0%

Risk attribution. Active risk = tracking error = tracking risk.

$$TE = s(R_p - R_b)$$

Assuming returns are serially uncorrelated, if 12 months, then multiply by $\sqrt{12}$.

Active risk squared = $s^2(R_p - R_B)$ = Active factor risk + Active specific risk

$$\text{Active factor risk} = \sum \sum b_i b_j \text{Cov}(F_i, F_j)$$

where

$$b_i = \sum w_i b_{ji}$$

$$\text{Active specific risk} = \sum (w_i)^2 (\sigma_{\epsilon_i})^2$$

Portfolio	Active Factor			Active Specific	Active Risk Squared
	Industry	Style Factor	Total Factor		
A	12.25	17.15	29.40	19.60	49
B	1.25	13.75	15.00	10.00	25
C	1.25	17.50	18.75	6.25	25
D	0.03	0.47	0.50	0.50	1

Note in the example above portfolio D is a passive portfolio.

Factor models in portfolio construction. **Factor portfolios** = only have exposure to one factor, and sensitivity to that factor is 1 and 0 for all other factors. This is good for hedging. Two common types of multifactor portfolios.

- Benchmark portfolios. Pure factors can be created by long top 20% of each time period and short bottom 20% ranked for that pure factor (long-short portfolios), so that there's a rolling window frame (can also use ranking stocks by Pearson or Spearman and using correlation). Often have higher returns.
- Risk parity portfolios. Take the factor series from the benchmark portfolio and turn into 5-year variance-covariance matrix. Taking into considerations correlations between factors, create a multifactor model where risk contributions are weighted equally from each of the factors - so the weights will all be different. Often have higher Sharpe ratios than benchmark portfolios because the volatility is much lower.

If we form BM and RP portfolios long-only, i.e. only buy these portfolios and don't short them, then the factor weights must add up to 100%.

Note: For one-factor models, assume the intercept is r_f and that the expected return is $E(R)$. However, for two-factor and higher-factor models, assume the intercept is the *expected return*!

8.5 Measuring and managing risk

VaR is the minimum loss expected a certain percentage over a timeperiod given assumed market conditions. Expressed as currency units or portfolio value percentage. Its complement is the probability to lose **no more** than the amount in that time, not 'lose less' because fund is usually profitable.

VaRs 'ignore' right-tail events, though they do actually 'indirectly' account for them, e.g. if there's a big right-tail event, the return distribution can shift to the right.

Three methods to estimate VaR:

- Parametric (analytical, variance-covariance) method. Typically uses normal distribution for simplicity. E.g. for 5% VaR, $(E(R_p) - Z_{normal}\sigma_p)(-1)(\text{Portfolio value})$ where $Z = 1.65$ for 5%, $Z = 2.33$ for 1%, and $Z = 1$ for 16%. All sample instances weighted equally. Not good if options included (any unexercised options, specifically).
- Historical (simulation) method. Rank gains and losses in lookback period, take e.g. 5th percentile for 5% VaR. All sample instances weighted equally as default, but can be weighted differently. Can absorb options.
- Monte Carlo. Specify e.g. expected returns, standard deviation, correlation between assets, and distribution. Then it approximates the parametric method. Avoids complexity of math, just needs computational power. Then take percentiles to get VaRs. Can handle options (specify option parameters). Can use with any distribution, including fat-tailed distributions for large losses, as well as customized parameters.

VaR advantages: simple, easily communicated, facilitates capital and risk allocation decisions. Disadvantages: somewhat subjective (calculation methods), underestimates extreme events, doesn't consider liquidity/illiquidity, sensitive to correlation risk (e.g. during downturns everything is correlated), vulnerable to trends and repeating nearly-reaching VaR, misunderstanding/oversimplifies risk, doesn't consider right-tail events.

VaR extensions.

- Conditional VaR. Answers how much expect to lose if VaR is exceeded. An average of losses below VaR. Especially suitable for expected tail-loss or expected shortfall.
- Incremental VaR. If change assets, what's the difference or change in VaR.
- Marginal VaR. If change assets by \$1 how much does VaR change. VaR actually = sum of mVaR. Similar but not same as iVaR.
- Relative VaR. Ex-ante tracking error. VaR with constraint of over and above benchmark. Like shorting benchmark.

Sensitivity analysis measures.

- Equity. CAPM: $E(R_i) = R_F + \beta_i(E(R_m) - R_F)$
- Fixed-income. Duration. $\frac{\Delta B}{B} = -D \frac{\Delta y}{1+y} + \frac{1}{2} C \frac{\Delta y^2}{(1+y)^2}$.
- Futures, forwards, swaps can be evaluated using underlying as they are linear. However, for options, use Greeks.

Greeks.

$$Delta = \frac{\Delta \text{option value}}{\Delta S}$$

$$Gamma = \frac{Delta^2}{\Delta S}$$

Note *Gamma* gets large when options approach expiration usually, unless they're deeply in/out of money.

$$Vega = \frac{\Delta \text{Option value}}{\Delta \sigma}$$

Call option price.

$$c + \Delta c = c + Delta(\Delta S) + \frac{1}{2} Gamma(\Delta S)^2 + Vega(\Delta \sigma)$$

Overall portfolio Greeks are sums of the Greeks of the components.

Scenario risk. For bond portfolio scenarios (including historical scenarios), use yields - prices are indirect (blurred by inflation and other factors) and even matching durations (which implicitly carry yield data) don't carry absolute yield data. Stress tests - most useful for leveraged positions (unleveraged positions: unlikely to become insolvent) and for exploring *known* risks to a *particular* exposure, historical scenarios, hypothetical scenarios (including reverse stress-testing by identifying biggest and most vulnerable risks and designing scenarios exposing them) for testing *unknown* risks. Often have increased correlation between assets in scenarios. Note **Greeks only work for small changes**, so not usually appropriate for scenarios which often have big changes.

Sensitivity and granularity. E.g. if put all bonds in one 'group' for analysis, then longs and shorts might be assumed to, for each unit of risk, hedge each other perfectly when they can't (e.g. because of non-parallel interest rate changes for long and short bond positions of different tenors). Granularity v efficiency.

Constraints in market risk management. If too tight, no profit. If too loose, chance of big losses. If impose at very low levels of organization, can't take advantage of offsetting risks. Some organizations therefore might use mVaR, which always sums to total risk.

- Risk budgeting. Usually focuses on 1 standard deviation. Divide risk into business units. VaR or ex-ante tracking error usually. Pension funds might start with assets minus liabilities = surplus at risk.
- Position limits. VaR is problematic if portfolios are unusually concentrated. So spread using position limits on e.g. asset classes, geographies, etc. Unlike VaR however, don't take into consideration duration, volatility, correlation, etc.
- Scenario limits.
- Stop-loss limits. Especially useful when portfolio losses are 'trending', i.e. don't break VaR limit but racking up losses slowly.
- Hedge. Drawdown control or portfolio insurance. Using protective options.

Capital allocation. Assign more risk where more reward in organization. Economic capital - amount of capital firm needs to hold to survive severe losses. Use economic capital to figure out slack.

Who uses what risk measures.

- Highly leveraged. Shorter period, high confidence VaR is often used.
- Fair-value accounting (marked-to-market accounting). VaR and economic capital.
- No leverage. Longer duration, ex-ante tracking error. If VaR, focus on more common scenarios.
- Fixed-income. Focus on instantaneous yield and interest rate changes.
- Book value accounting. Asset-liability gaps.
- Banks. Liquidity and regulatory concerns.
- Traditional asset managers. Most concerned about ex-ante tracking error to benchmark and underperformance.
- Hedge funds. Highly-leveraged, so high-confidence VaR with short periods. Maximum drawdown (worst historical case) considered for high-risk transactions.

Pension (defined benefit) funds. Interest rate and curve risk, surplus at risk (an application of VaR, assets into VaR as long as pension liabilities are short fixed-income positions) - estimating how much assets might underperform liabilities. If surplus at risk exceeds limit, then change asset allocation to make assets better match liabilities - 'liability-driven investing'. If there is excess, make a separate portfolio.

Insurers. Property and casualty - use reinsurance and geographic dispersion. But also note sensitivities, economic capital and VaR, and scenario analysis. For life insurance and annuities, more like DB pensions: sensitivities, asset and liability matching, and scenario analysis.

8.6 Backtesting and simulation

Backtesting: (1) design strategy, (2) simulation, (3) analysis of backtesting output.

Backtesting problems. (1) Survivorship bias, e.g. in stock indices - use point-in-time data to address. (2) Look-ahead bias. Survivorship bias is actually a type of look-ahead bias. Look-ahead bias from: (a) reporting lags, (b) data revisions, (c) data vendors add new companies not there until later. Add lags, don't use revised data, etc. to address. (3) Data snooping, p-hacking. A type of model selection bias. Use higher t-statistics and cross-validation to address.

Notes: (1) If PMs don't hedge currencies, then backtest using single-currency-denominated returns (after conversion). (2) Problem with using P-E ratios for 'cheapness' of stock is that it makes no sense when earnings are negative; so, use earnings yield (the inverse). (3) If cover short lookback period, may not capture whole business cycles. (4) Maximum drawdown - biggest peak-to-trough fall. (5) Positive kurtosis - much higher tip than ND but fatter tails. (6) Risk-parity portfolios created by using rolling window for both returns and building covariance-variance matrices for aligning each constituent factor portfolio with equal risk contribution to the overall risk-parity portfolio (rebalanced each month or other time period) - compare to the benchmark portfolio which usually has equal weights (compare their returns, volatility, Sharpe ratios, VaR). (7) Bootstrap in historical simulations because the number of simulations is much bigger than historical dataset usually.

Regime changes and structural breaks. RP and BM portfolios. During recessions, have different returns (lower) and higher kurtosis (and therefore more concentration of returns around center but fatter tails).

Simulations: (1) historical - sample randomly from different historical periods without regard to chronology and (2) Monte Carlo - define probability distributions for each input variable (e.g. factor returns) by calibration and apply. For Monte Carlo simulations and risk-parity portfolios, need to calculate the multivariate distribution (e.g. multivariate normal) for the RP portfolio (for the historical RP portfolio, calculate variance-covariance matrix). Note that a multivariate normal distribution can lead to original normal distributions with lots of skew and kurtosis to lead to faulty results.

Sensitivity analysis. This includes changing the probability distribution in Monte Carlo analysis - e.g. multivariate normal to skewed t-distribution.

9 Alternative Investments

9.1 Introduction to commodities and commodity derivatives

Portfolio diversification. Historically low correlation with bonds and stocks. Inflation hedging qualities.

Commodity futures. Net-zero position because for each seller there's a buyer. In contrast, stocks and bonds are net long positions - investors can keep them forever.

Hedging and speculating positions - examples.

	Long Position	Short Position
Hedging	Food manufacturer seeking to hedge the price of corn needed for snack chips	Gold mining company seeking to hedge the future price of gold against potential declines
Speculating	Integrated oil company seeking to capitalize on its knowledge of physical oil markets by making bets on future price movements	Commodity trading adviser (CTA) seeking to earn a profit for clients via a macro-commodity investment fund

Futures prices.

- Static futures curves. Normal futures curve - prices are sloping up from today. Inverted futures curve - prices are falling.
- Dynamic futures curves. Contango - front-month futures prices are above current spot prices, so they are expected to fall closer to maturity. Backwardation - front-month futures prices are below current spot prices, so they are expected to rise.
- Normal futures curve usually associated with contango, inverted with backwardation but not always.
- If spot price higher than futures price, backwardation; opposite is contango. Futures minus spot price spread = **basis spread**.
- Near-term futures contract price versus long-term futures price. Price spread = **calendar spread**. Calendar spread = earlier price - later price.

Theories of futures returns.

- Insurance theory. Keynes' 'normal backwardation': Producers (long position) sell for less than current spot price as insurance to induce speculators to take short position. So, commodity futures prices should normally be in backwardation. Keynes' theory lacked evidence.

- Hedging pressure. Producers want to sell and consumers want to buy, each to get more certainty. Should normally then see a flat commodity futures curve, but if more producers want to sell than consumers buy then backwardation and in vice versa case it is contango.
- Theory of storage. If there is excess inventory, stored and storage costs increase price, so contango. If just-in-time inventory, backwardation. Convenience yield: benefit from extra physical commodity availability upon convenience, reflected in inventory level. If inventories high, low convenience yield; if inventories low, high convenience yield. Futures price = spot price + direct storage costs (e.g. rent + insurance) - convenience yield. It is minus convenience yield because futures prices are paper contracts; they don't get the benefit of convenience yields which are physical supplies. Moreover, if there're low inventories, spot prices will be high anyways, offsetting the convenience yield.

Convenience yield (CY). A convenience yield is the premium gained from holding a physical asset instead of its derivative contract, often arising from low storage costs and high demand for the underlying product. Typically arises when storage is low. Usually calculate by comparing current spot price x discount rate, versus the futures price, divided by the futures price. CY can rise if oversupply - e.g. US oil shale production - because futures contracts become less important; just important to have the physical oil.

Futures returns. **Total futures price returns = spot return + roll return + collateral return.** Collateral return usually the biggest component of returns. However, total futures price risk (standard deviations) are highest for spot returns and low for roll and collateral returns. Spot return has highest correlation to financial market; roll and collateral return correlations to financial market are near 0. If using a commodity futures price index, add a re-balance return. **Price return.** If buy 3-month contract at low price (e.g. 10 USD) and then 3 months later the price is higher (e.g. 15 USD), then this is the price return $((15 - 10)/10 = 50\%)$. It is NOT the roll return.

Roll return. If buy 3-month contract at low price (e.g. 10 USD) and then 3 months later the price is higher (e.g. 15 USD), then there is a positive price return. But if the new 3-month contract price is now very high (e.g. 25 USD), then the roll return is $(25 - 15)/15 = 67\%$.

- Price return (spot yield). Futures price return (i.e. change in futures prices, generally the front-month prices), not physical commodity price return (which are not standardized).
- Roll return (roll yield) - not a 'real' return; it's an accounting return. Commodity-specific. Usually significant in a single period, but over multiple periods not very significant. If backwardated, futures contract holder can roll by selling current contracts at high price and then buying numerically more future contracts, keeping investment the same. Roll return =

$((\text{near-term futures price}) - (\text{further-term futures price})) / (\text{near-term futures price}) \times \% \text{ of position in futures being rolled (i.e. portfolio fraction being rolled over)}$. Note there is only very weak positive correlation between positive roll return and positive total returns - Ilmanen tried to argue roll return is approximated by risk premium but this argument is hard to make.

- Collateral return (collateral yield). Yield for bonds and cash used in trading margins. Use the entire notional value of the futures (i.e. effectively no leverage). Often just use government bond rates.

Commodity swaps. Excess return swaps (e.g. refiner goes long a swap paying an amount in cash when oil price that month is higher than X USD in exchange for a fixed price of Y USD; this is effectively long a series of call options), total return swaps, basis swaps (e.g. difference between average daily price of Brent and heavy crude oil for delivery in GOM), variance swaps (compare observed and actual variance), volatility swaps (compare observed and expected volatility; volatility = square root of variance).

Commodity indexes. Rebalancing frequency matters more for volatile markets. Rebalancing with floating weight schemes, e.g. production, will lead to gradual rebalancing or automatic rebalancing as rising production (= of more economically important commodities) will mean rising weights; these indices will outperform fixed-weight indices.

9.2 Overview of types of real estate investment

Core real estate investments more like bonds. Opportunistic real estate investments more like equities.

Part of real estate value comes from net operating income $\text{NOI} = \text{effective gross income} - \text{operating expenses} - \text{property maintenance}$. It is like EBITDA for a **single property** (as opposed to operating income for a property which is like EBIT).

- Effective gross income = gross income (e.g. average lease regardless of real estate cycle) - vacancies, concessions. Include pass-through costs (e.g. property tax and insurance) that the tenants pay for, and then subtract them as operating expenses. Include servicing and repairs costs if they are charged to the tenants and then subtract these costs as operating expenses.
- Operating expenses: servicing and repairs, insurance, legal, security, etc.
- Property maintenance: plumbing, electrical, lawncare, etc. It is separate from (1) repairs and servicing (= operating expenses), and (2) capex for major works.

- Vacancy adjustment. If there is less than 100% tenancy (i.e. vacancies), just subtract off revenue - often assume that operating expenses and property maintenance don't scale.

Evaluate real estate investment.

- Profitability, NOI. First-year ROE = **Equity dividend rate** = (pre-tax CF) divided by (Property purchase price - mortgage loan), where pre-tax CF = NOI - debt service.
- Degree of leverage. Note: real estate investors usually can get long-term loans because the real estate is collateral. **Loan-to-value** (LTV; a leverage) ratio = mortgage debt outstanding divided by current property value. Low LTV = high equity to debt. **Debt service coverage** (DSC) ratio = NOI divided by debt service, for commercial real estate. Debt service = mortgage payment (principal repayment + interest).
- Compare operating expenses versus effective gross income or some other income measure.

Depreciation. Depreciation base = real estate original construction or acquisition cost + improvements - land cost. Note: depreciation base includes architect, legal, tax, permits, accounting, contractor profits, etc.

Taxes. Taxes = $t \times (\text{NOI} - \text{interest expenses} - \text{depreciation})$. **After-tax cash-flow** = pre-tax CF - taxes = (NOI - debt service) - taxes.

Real estate companies. Real estate operating companies (REOCs) own, operate, actively manage (i.e. develop, operate businesses inside RE) commercial real estate with few restrictions and are taxed. REOCs can reinvest profits. REITs restricted to owning and operating rentals and purchasing mortgages and pass on nearly all earnings (generally 90% or more) to avoid corporate tax.

REITs. Often use as earnings **FFO = funds from operations** = **NI + D + A - net gains from property sales**, where NI = includes net gains from property sales (which are non-recurring revenue). It is like 'core' operating CF or 'smoothed' CF. REITs distribute most earnings as dividends, so need cash for this, and so operating CF is best.

Rates. Adjustable rate = fixed + variable.

Rental income. Can be fixed, step-up clauses (pre-specified rental increases), indexed rents, overage rent clauses (triggered if sales exceeds targets).

Risks. Prepayment risk = borrowers have right to pay earlier, exposing investors to reinvestment risk. Rollover risk - real estate investment holding period longer than tenancy.

Advantages. Inflation hedge, diversification, tax advantages (e.g. REITs). Private real estate depreciated very quickly in some geographies.

Real estate cycle - recovery, expansion, oversupply, recession. NOI peaks during oversupply and reaches bottom at recovery. Vacancies highest at oversupply + recession, lowest at recovery + expansion.

Real estate valuation approaches: income approach, cost approach, sales comparison approach.

Income approach.

$$V_0 = \sum_i^n \frac{NOI_{t+i}}{(1+r)^i} + \frac{TV}{(1+r)^n}$$

NOI is actual or estimated. May need to normalize or 'stabilize' the NOI.

Direct capitalization method is a perpetuity:

$$V_0 = \frac{NOI}{CapRate}$$

Capitalization rate = CapRate, and V_0 is the property value. Rearranging,

$$CapRate = \frac{NOI}{V_0}$$

which is often compared to a bond yield or to EBITDA-EV. Cap rates might change over the lifetime. Going-in cap rate is first year cap rate - this is only used to discount the first cash flow. Terminal cap rate = required rate of return - growth rate.

Cost approach. Replacement cost = cost of replicating asset and deduct depreciation and other factors that reduce property value. Replacement cost probably higher than market cost during oversupply phase of real estate cycle. Note: cost of 'constructing an identical facility' may or may not include land cost. What is being purchased, e.g. commercial real estate, usually includes land.

Sales comparison approach. Market approach. Find similar properties around same time. Use units of comparison like price per sqm.

Real estate indexes.

Private real estate indexes: Appraisal or transaction based.

Appraisal-based indexes: many indices. Holding period return = (NOI - capex + (end market value - beginning market value) all divided by beginning market value.

Transaction indexes, e.g. repeat sales index, hedonic index. A hedonic index in real estate finance is a sophisticated way to track property prices by adjusting for differences in property characteristics. Instead of just comparing raw sale prices, it uses hedonic regression to isolate the value of individual features—like square footage, number of bedrooms, location, and even proximity to parks or schools. Repeat sales track individual properties through time, hedonic uses regression so doesn't require changes in individual properties.

Appraisal indexes lag transaction indexes. Transaction indexes move first, then appraisal indexes with a lag. Appraisal indexes also lag because properties may not be appraised every year. Appraisal indexes will therefore smooth volatilities and overestimate Sharpe ratios, as well as reduce correlation with other assets. To adjust (unsmooth) appraisal index lag, assume actual return is related to appraised returns (*):

$$R_t^* = aR_t + (1 - a)R_{t-1}^*$$

and solve for R_t :

$$R_t = \frac{R_t^*}{a} + \left(\frac{1 - a}{a} \right) R_{t-1}^*$$

Public real estate indexes: Equity (equity REITs), fixed-income (MBS, covered bonds).

MBS. Face prepayment risk: contraction (earlier repayment than expected) and extension (later repayment than expected).

9.3 Investments in RE through publicly traded securities

Real estate securities.

- REITs are required by law to distribute 90%-100% of their earnings to shareholders, invest at least 75% in RE, and derive at least 75% of income from RE. Some REITs get around these rules but then pay tax on those 'extra' activities. Note REITs have little retained earnings so must raise capital from external sources. REITs that own real estate are equity REITs, and REITs that invest in mortgages are mortgage REITs. REITs can deduct dividends paid from their taxable income, or sometimes are exempt from corporate tax altogether. REITs can be public or private.
- REOCs are ordinary taxable entities. REOCs generally exist where there are no REITs under law, these companies develop a lot of real estate (REITs own but don't develop much), and offer other services like brokerage and property management. REOCs can be public or private. They have more flexibility than REITs.
- MBSs.

- Direct investment in real estate. REOCs and direct investment can lead to max investment and capital gains potential because there are no restrictions on what they can invest in and they can keep their earnings for investments.

Real estate securities valuation. Valuing RE equity: BVPS (book value per share) which is depreciated real estate value (not shareholder equity), NAVPS (net asset value per share). Approaches to calculating: asset value estimates, price multiples (relative values), DCF.

Property accounting. IFRS: (1) cost, which is original purchase price minus accumulated depreciation and any impairment losses, or (2) fair value. GAAP: historical + capex - depreciation.

NAV can be calculated by (1) taking NOI and dividing by cap rate, (2) applying value per sqm of other properties and applying it to the given property, (3) using appraised values. NAV becomes more subjective in a negative and less liquid market (bad economy) with fewer observable transactions, in contrast to P-FFO, P-NAV, NAVPS.

NAVPS = (Market value of assets - market value of liabilities) divided by the number of shares. Uses **market**, not historical, values. To calculate NAVPS, start with NOI, and adjust NOI by **non-cash rent** (difference between average rent reported and actual rent; difference between straight-line rent and is the straight-line rent adjustment), adjustment for acquisitions (i.e. expected revenue due to recent but as-yet unreported acquisitions), to get pro forma NOI. Then add some growth. Then use cap rate to back out the estimated V_0 , then add cash + equivalents, land for future development, AR, prepaid expenses to get gross asset value. Then subtract total debt, other liabilities to get NAV.

NAV exclusions (if in there, remove to get 'hard' economic assets, i.e. the regular NAV): goodwill, deferred financing expenses, DTAs. Replace debt book values with market values, remove DTLs. This means, for e.g. that DTLs are *not* subtracted from NAV.

NAVPS. Compare to equity prices to see if undervalued or overvalued. If **public** REIT or REOC doing well, then should trade above NAVPS because NAVPS is a private sector measure, and greater liquidity + competition of public REIT and REOC should be better.

Relative values approach for REIT stocks. Funds from operations.

Funds from operations (FFO) = (NI + D + A) - (net gains on sale of RE). Depreciation is added because RE investors think generally RE holds value and even appreciates. FFO is a SEC-accepted non-GAAP measure. Adjusted FFO = (FFO - non-cash rent - recurring capex). Non-cash rent adjustment = due

to straight-line rental accounting in NI (removing accruals). So, $\text{AFFO} = \text{FFO} - \text{non-cash adjustment} - \text{recurring maintenance}$. Use P-FFO and P-AFFO instead of P-E ratios. Sometimes use EV-EBITDA or EBITDA-EV ratios; note EBITDA is better than FFO or AFFO because EBITDA includes debt cost while FFO and AFFO are equity-only. FFO is not a cashflow (it doesn't have capex and CF for financing). FFO includes FFO from unconsolidated businesses.

Funds available for distribution (FAD) or cash available for distribution - improvement on FFO.

FFO (more CF-like) generally published in US, NAV (more equity-like, emphasizes long-term investment) generally published in Europe and Asia. AFFO harder to find as published figures.

9.4 Real estate income, cashflows, ratios

Income.

- Effective gross income = gross income (e.g. average lease regardless of real estate cycle) - vacancies, concessions.
- NOI. Net operating income = effective gross income - operating expenses - property maintenance. Excludes interest expenses! Analogous to EBITDA.
- Gross potential rental income = market rent x rentable space. Assume 100% occupancy.
- Gross rental income = current rent x rentable space.
- Net rental income = Gross rental income - vacancies, concessions.

Value. NAV net asset value. Estimated (i.e. future) 12-month NOI divided by cap rate, then add cash + equivalents, AR, subtract debt and other liabilities. Analogous to shareholder equity or value V_0 . But shareholder equity is book value, whereas this is market value. NAV becomes subjective in bad economy with few transactions.

Cashflows. Not considered cashflows but analogous to cashflows because they are net income + depreciation and amortization.

- $\text{FFO} = (\text{NI} + \text{D} + \text{A}) - (\text{net gains on sale of RE})$.
- $\text{AFFO} = \text{FFO} - \text{non-cash adjustment} - \text{recurring maintenance}$. AFFO is more about dividend sustainability. Non-cash examples: stock compensation, amortization of loan issuance or origination costs.
- Pre-tax cashflow = NOI - debt service.

If there is 'land held for future development', P-CF will be higher, as this land doesn't generate cash but has value.

Ratios.

- Cap rate = NOI divided by market value. Like return on market value, like EPS.
- P-FFO, P-AFFO is like P-CF. Likely to fall in bad economy.
- NAVPS = (market value assets - market value liability) divided by shares, is like market value per share, i.e. share price.
- P-NAV = P-NAVS. Like price per market value.

Note: if two REITs have the same market value, then the one with a **lower** cap rate will have lower earnings and cashflow, generally - so that REIT will have lower P-FFO.

9.5 Hedge fund strategies

HF disadvantages and risks. Complex offering memorandum (fund prospectus) needs to be understood by limited partners (the investors), high fees, illiquidity on funds. High leverage, which is dangerous to limited partners, high volatility.

Liquid alternatives - mutual funds, closed-end funds, and ETFs that invest in hedge fund-like strategies. Don't perform as well as HFs. Liquid alts can't use same techniques as HF and liquidity has advantages but so does have having funds locked up and illiquid. HFs usually charge management fee + incentive fee; some liquid alts (e.g. mutual funds for popular retail) cannot charge incentive fees by law for transparency reasons.

HF characteristics. More regulatory oversight. Limits investors, limits HF subscriptions. Flexible investment mandates. Larger investment universe - distressed assets, private securities, life insurance contracts, film royalties, etc. Aggressive investment techniques. High leverage. Liquidity constraints - HF advantage over liquid alts. High fees.

Single-manager fund v multi-manager fund. Latter is multi-strategy or fund-of-funds.

HF strategies

Equity	Event-Driven	Relative Value	Opportunistic	Specialist	Multi-Manager
• Long/Short Equity • Dedicated Short Bias • Equity Market Neutral	• Merger Arbitrage • Distressed Securities	• Fixed Income Arbitrage • Convertible Bond Arbitrage	• Global Macro • Managed Futures	• Volatility Strategies • Reinsurance Strategies	• Multi-Strategy • Fund-of-Funds

Equity strategies: long-short, dedicated short-selling, short-biased, equity market neutral.

Long-short equity. More hedged, less volatile than long-only. Not about market timing, e.g. beta tilts not big historical factor. Mostly about stock-picking + position concentration (e.g. through leverage), not market timing. 30% of HF strategies. Generally net long. If more market-neutral, then need MORE leverage to get meaningful returns (need more leverage because balanced positions mean less volatility so have to amplify returns + buying on long positions). Diverse investment styles, e.g. value-growth, large-small cap, specialized industries, etc. Extract alpha on long and short by stock selection + naturally net long beta.

Specialized industries or sectors. Can get better alpha, but harder to judge, fewer funds like this, harder to re-allocate funds between sectors.

Equity: short-sales. Borrow, sell borrowed equities now high + buy later low. Go for overvalued, e.g. high P-E, companies. Stock-picking ability is important; need deep-dive detailed research. Concretely:

- Borrow securities with loan, sell borrowed securities now high + buy later low.
- Buy put options.
- **Long put calendar spread** (for anticipated negative events): sell put with expiry which expires before the event (+ receive premium) and simultaneously buy put with expiry after (for the short position).
- If cost to borrow shares is too high (i.e. the interest to borrow them), then buy shares and lend them yourself. Additionally, buy a put which turns the combination into a **protective put** (limit downside while allowing upside).
- **Synthetic short** position - sell call options to finance purchase of puts. Lose if price rises, win if price falls. (Note: opposite of synthetic short is **synthetic long**, where puts are sold to finance long calls). In a synthetic short or long, the strike price of the put and calls is the same. In short-risk reversals and long-risk reversals, the strike prices are apart, and the

investor is OTM on both put and call, and the spread is so that, e.g. investor can buy the call using the proceeds from selling the put.

Advantages:

- Source of negative correlation with other investments.
- Provide upside in a falling market.
- Typically don't need much leverage because (1) there is enough volatility (i.e. it's not balanced like long-short and market neutral strategies) and (2) short positions have natural built-in leverage, which is automatic collateral so don't need huge loans to buy securities outright.

Disadvantages:

- Must post collateral to securities lender + interest on securities loan.
- Might have to give back securities without warning.
- Might be blocked from company research if active short-seller.
- Limits on short-selling (to prevent market crashes), e.g. in US market has to be rising. Limits can be introduced on short notice.
- Typically lower returns than other HF strategies (but negative correlation benefit).

Short-selling strategies.

- **Dedicated short-selling.** HF strategy where only short sales used (ofc on overpriced stocks) - sometimes reduce short-selling by holding cash.
- **Short-biased.** Not as extreme as dedicated short-selling - also take some long positions. Secular upward market trends make short-selling in general hard. Seek out failing business models + fraud for short-sales.
- **Activist short-selling.** Take short position + then publish why, causing stock price decline. Not market manipulation if (1) don't charge for information, (2) no wrong information, (3) best interests of limited partner investors.

Equity market neutral (EMN). Uses relative value strategies. Best when non-trending or declining markets - because they don't rely on upward trends like long-short equities; they just rely on relative movements. Aim for 0 exposure to market by making net beta = 0, while capturing alpha. Primarily stock-picking, secondarily market-timing. Typically very quantitative, very diverse holdings, need big leverage. Need a lot of leverage because less volatility + long positions.

- Pairs trading, e.g. securities that are usually highly correlated are found to be in a state where one is overvalued and other is undervalued. Gain when they correct + converge.

- Stub trading. Parent is over or undervalued compared to subsidiary; buy-sell based on parent ownership percentage of subsidiaries.
- Multi-class trading. Exploit under or overvalued shares of different classes within same company.
- Quantitative market neutral trading. Frequent trading by computers. When frequency increases to point where mean reversion and relative momentum are emphasized, then it's statistical arbitrage trading.

Major risk of equity market neutral strategies. Often brokers allow high leveraging, but if some margin loss, then brokers can force market neutral managers to downsize quickly.

EMN example. If 1 million USD allocated to trade, then use it to buy 1 million USD of undervalued shares + short e.g. 800,000 USD of overvalued shares, depending on beta ratios.

Event-driven (ED) strategies: Merger-arbitrage (moderate-high leverage), distressed securities (moderate low leverage). Very common. Exposed to some equity market beta, e.g. which determines whether a merger is successful (+ high hedge fund fees result in embedded beta).

Merger-arbitrage.

- Soft-catalyst ED approach: anticipate event not yet happened (more volatile); hard-catalyst ED: prices related to event not yet converged (less volatile).
- Cash-for-stock (acquirer pays for target stock) and stock-for-stock deals.
- In M&A cash-for-stock deals, typically buy just target company. In M&A stock-for-stock deals, take position (successful or not successful) on deal and buy-sell target and acquirer shares.
- It is like the HF is selling insurance on the deal (if they take a positive position on the deal). Like riskless bond (if deal successful, then collect insurance premium) + short put option (if deal unsuccessful, then loss). Also like owning call option if white knight.
- Can use credit default swaps, e.g. short CDS on target company credit cost if take position that deal will succeed.

If after merger announcement, target price rises and acquirer target falls, then spread = potential gain (riskless bond). Reversion to pre-announcement prices = potential loss (short put).

Distressed securities. Usually long bias, moderate leverage. HFs have relative advantage: no minimum credit quality required in their investments (unlike

institutional investors), taking out cash from HF by investors is illiquid (unlike liquid alts). In fact, institutional investors may get rid of such securities in fire sales - and HFs can buy them.

Distressed securities - **capital structure arbitrage**. Long securities where they can get recovery in bankruptcy proceedings and short where there's less chance.

Distressed securities - sometimes HFs invest in **fulcrum securities** which are senior unsecured debt which will convert to equities and give them company control.

Relative value strategies: Fixed-income arbitrage, convertible bond arbitrage. Count on mean reversion. High leverage.

High leverage (arbitrage differences are small, fixed-income is highly correlated to market). Includes **yield curve trades** and **carry trades**. Like long-short equity but for debt - across a big range of debt. Debt is a bigger market than equity, but it's less liquid and opaque, which creates opportunities but also risks. Resembles a short put option.

Convertible bond arbitrage. Convertible bond = bond + call option. Often thinly traded, so equity has trapped value - the equity call option is valued at superficially low implied volatilities. Hard to understand. HF opportunity. Subtypes:

- Credit-oriented convertible managers. High leverage because lots of transactions to do this. Buy undervalued CB + short overvalued underlying stock. HF managers which accept credit default risk. Treats convertible bond more like debt investment.
- Long-biased. HF managers which hedge credit risk + underhedge equity exposure.
- Volatility-oriented convertible managers. Overhedge equity risk.
- Time decay. If the market suddenly calms, then short option loses value, as does the convertible bond's embedded call option.
- Market volatility - convertible bonds are illiquid, so market volatility makes them even harder to sell.

CB arbitrage example. If the stock price is overvalued compared to what can be obtained from the convertible bond, either (1) convert the CB to equity and sell equity now to get x USD, or (2) buy the CB and convert, short the stock, and sell the converted equity - for x USD profit. If the stock price rises, then converted equity profit is higher but short position loss; if the stock price falls, then converted equity profit is lower but short position gain; in the end, it's the

same x USD profit.

CB arbitrage complications: (1) Short sale means borrowing security, and lender might want it back on short notice. (2) Market changes could lead to differences in credit spreads which will change value of convertible bond. (3) Embedded option is subject to time decay.

Opportunistic strategies: Global macro, managed futures. Exposed to trends and general directions of markets because they look at the big picture. Both are highly liquid, use high leverage, right-tail skew.

Global macro. Heterogeneous as a group. High leverage using derivatives. Volatile because depend on world events and trends which are 'lumpy'.

Managed futures. Focus on futures, options on futures, forwards, swaps, commodities, currencies. Most in short-term government debt. High leverage because of using these instruments - there's inherent leverage (and therefore also capital efficiency). Can go long or short and change because futures markets are liquid. Positive skewness during market distress, so useful for portfolio diversification.

Managed futures: **time-series momentum** (TSM), **cross-sectional momentum** (CSM). TSM = Time-series momentum evaluates the past performance of each individual asset (or futures contract) in isolation to determine its own trend. CSM = long the assets with greatest momentum and short those with least. CSM = often net zero market exposure; TSM = volatile.

Specialist strategies: (1) Volatility trading, (2) Reinsurance and life insurance.

Volatility trading. Volatility instruments are naturally convex; little leveraging.

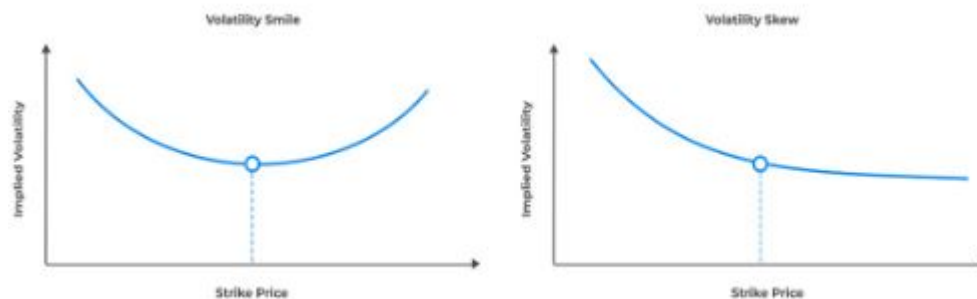
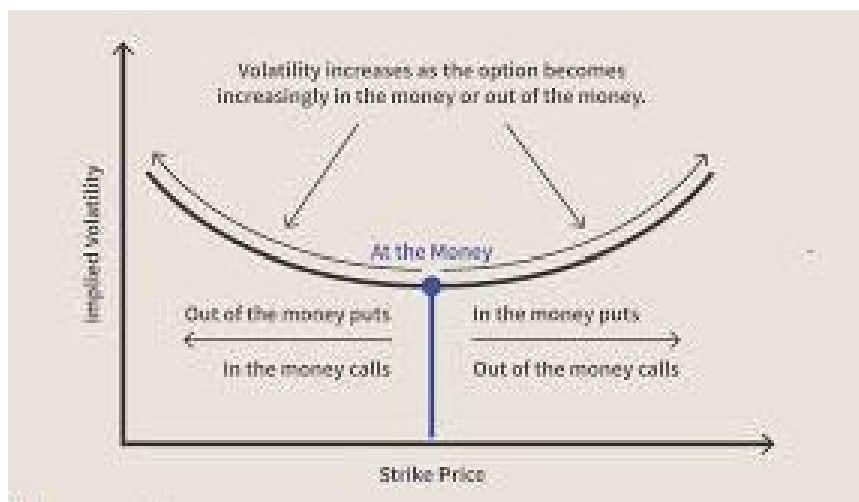
- Volatility trading, e.g. **relative value volatility arbitrage** - buy cheap volatility and sell expensive volatility. Example. Korea has generally low equity pricing volatility, so its equity put options are priced low; US has generally high equity pricing volatility, so its equity put options are priced high. Assume trader thinks Korea equity pricing volatility will be higher in the future and US equity pricing volatility will be lower in the future; then buy Korea puts and sell US puts. Need to adjust (delta-hedge) by buying some Korea equity and selling US equity. Use US sold put premium to buy Korea puts to offset theta decay, and saying that the two markets will move in a certain way because the PM believes they are linked and are unlikely to both move away from the position (i.e. market won't move such that US volatility rises AND Korea volatility falls at the same time).
- Note: equity volatility is negatively correlated to equity returns. So can

be used for diversification (e.g. buy stock and then buy OTM puts on it).

- VIX futures - traders don't prefer; as expiry nears, price falls because of volatility mean reversion.

Notable features of volatility futures.

- Longer-dated options are more sensitive to volatility (vega); short-dated options are more sensitive to price (delta) and price changes (gamma).
- Long volatility strategies are convex because they have right-skews. Long on volatility means black swan events lead to big payoffs.
- Term structure of volatility. Typically upward-sloping. Inverts during crisis.
- Volatility smile - upside and downside protection are valuable, but middle is not. This is for each of call options and put options separately. Volatility skew (downside protection more valuable than upside protection), often seen for options where the underlying is equity.



Ways to implement volatility trading:

- Simple exchange-traded options, including option spreads (e.g. bull, bear, straddles, calendar spreads). But time limits of up to only 2 years. For short-term options, which are high in delta sensitivity, hedge delta and gamma to isolate vega; but if delta hedge too much then overhedge and no vega. During stock price volatility, gamma increases, so need more delta hedging.
- OTC options. More customizable but counterparty and illiquidity risk.
- VIX futures. Pure volatility, so don't have to constantly hedge like for simple exchange-traded options to isolate vega.
- OTC volatility or variance swap. Variance = square of volatility. Pure volatility, unlike standardized options where volatility depends on asset price which must be isolated by delta hedging.

Reinsurance and life settlements. HFs may take different position on reinsurance, life, catastrophes.

Life insurance. Investors in the life settlement market are looking for policies where the fair market value of the policy (what they are willing to pay) is significantly higher than the policy's cash surrender value, but still less than the ultimate death benefit. HF buys the policy from the policyholder themselves. HFs look for (1) low surrender value - value that insurance company would pay to policyholder to quit the policy - which influences the HF's purchase price; (2) low premiums - that the HF must pay; (3) early chance of dying - for earlier payout.

Multi-manager strategies: (1) create own combination, (2) funds of funds, (3) multi-strategy funds (one fund with different strategies).

Fees. FFO investors have to pay double layer of fees (first pay the fee structure of the individual funds in the FFO, then the FFO's own management + incentive fees calculated based on the net gains of the fund), and have **netting risk** - investors have to pay incentive fees on winning funds *without* refunds on losing funds (FFO investors + FFO managers share netting risk). However, multi-strategy funds net the positive and negatives, so investor only pays incentive fees if the net is positive and GP of fund bears netting risk. Note however that some multi-strategy funds have 'pass-through' structure, which is similar to FFO fee structures (FFO investors + GP share netting risk).

Leverage. Multi-strategy HFs use more leverage than FFOs.

HF evaluation. Conditional factor risk model. To evaluate HFs.

$$(\text{Return on HF}_i)_t = \alpha_i + \beta_{i,1}(\text{Factor 1})_t + \beta_{i,2}(\text{Factor 2})_t + \dots + \beta_{i,K}(\text{Factor } K)_t + D_t\beta_{i,1}(\text{Factor 1})_t + D_t\beta_{i,2}(\text{Factor 2})_t + \dots + D_t\beta_{i,K}(\text{Factor } K)_t + (\text{error})_{i,t}, \text{ where}$$

- $(\text{Return on HF}_i)_t$ is the return of hedge fund i in period t ;
- $\beta_{i,1}(\text{Factor 1})_t$ represents the exposure to risk factor 1 (up to risk factor K) for hedge fund i in period t during normal times;
- $D_t\beta_{i,1}(\text{Factor 1})_t$ represents the *incremental* exposure to risk factor 1 (up to risk factor K) for hedge fund i in period t during financial crisis periods, where D_t is a dummy variable that equals 1 during financial crisis periods (i.e., June 2007 to February 2009) and 0 otherwise;
- α_i is the intercept for hedge fund i ; and
- $(\text{error})_{i,t}$ is random error with zero mean and standard deviation of σ_i .

Factors are equity, interest rates, currency, commodities, credit, volatility, etc. Use stepwise regression to eliminate multicollinearity; first regress each factor against each other and if 60% or more correlation, pick whichever has higher R-squared when taking HF returns and regressing on factors. Look at sign of coefficient - if positive, then higher HF returns associated with long positions. The coefficients might switch during market crisis.

Metrics. Sharpe and Sortino ratios. **Sortino ratios** replace standard deviation in Sharpe ratios with downside deviation. So, Sortino ratios measure returns per downside risk unit. **Rho** is a measure of first-order serial autocorrelation; high rho means smooth returns and potential illiquidity. Look at 'normal times' and 'normal + crisis' coefficients. Look at 'maximum drawdowns'.

Results.

- Long-short strategies outperform EMN in total returns, but returns are more volatile so Sharpe ratios are lower.
- Multi-strategy funds have higher Rho compared to FoF due to illiquidity and lockups.
- Standard deviations lowest for managed futures, FOF, macro, EMN. Standard deviations high for event-driven, relative value.
- Opportunistic, merger-arbitrage, EMN have lowest drawdowns. Also lowest Rho - so good liquidity.

Techniques during market stress. If have net long positions, sell equities and market indices, short them, deleverage, buy put options. Short-sales strategies will have negative loadings on equity indices as expected, but they can also have negative loadings on volatility, indicating they might be selling puts on

their short positions to get a volatility premium. During market stress, can't just diversify; have to look at liquidity, volatility, credit, leverage and interactions between them.

General trading and hedging concepts.

- Note on use of leverage by HFs. Often there are opportunities during market distress times. However, this is also when credit lines dry up, increasing credit costs.
- Delta hedging. Delta is sensitivity of option price to underlying. Take opposing sides to reduce directional risk. Example: buy call options + reduce exposure by selling shares. The directional risk is reduced or eliminated, but there's still volatility - if the stock price experiences volatility, the call option price goes up even if the stock price falls on net during the volatile period.
- Gamma. Sensitivity of delta, the second derivative. A convertible bond has positive gamma (whereas a stock has 0 gamma) because as stock price rises, delta increases. Conversely, as stock price falls, delta falls but not at a fast pace because of the convertible bond's debt characteristics. So, the convertible bond has a 'synthetically' long position. This synthetic net long position can be hedged if desired.
- Gamma hedging. Usually use options to control rate of delta change.
- Convexity in trading. Gains more when the underlying (e.g. prices, volatility) gain, and lose less when the underlying declines. Examples: call and put options, bonds, convertible bonds.
- Long beta tilting. Getting more exposure to market beta.
- Difference between carry trade and yield curve trade: yield curve trade is about the term structure changing.
- Short squeeze. Stock price rapidly rises, short traders have to cover positions by buying, which increases the price even more.

10 Ethical and professional standards

10.1 Professional Conduct Program

Board of Governors oversees Professional Conduct Program (PCP). PCP looks at the annual self-disclosure, written complaints, and other sources (e.g. media), to investigate complaint. Then interviews, gathers information and recommends action. If candidate rejects, this goes to Disciplinary review committee (DRC). Throughout process, use CFAI Bylaws.

10.2 Code of ethics and SPCs

Code and SPC apply to individuals.

Firms. CFAI encourages firms to adopt Code and Standards *as part of their code of ethics*. Asset Manager Code applies to firms with 6 principles: loyalty to clients, investment process, trading, compliance, performance, and disclosure.

Use in firms. Using and applying ethical behavior by senior management is the most important in disseminating ethics. Also, important is using it like muscle memory. Simply adopting a code isn't enough.

10.3 Guidance for standards I-VII

I(A): Professionalism.

- Must understand applicable laws and regulations. Don't need detailed knowledge - that's up to compliance and experts.
- If suspect that something unethical is happening by other people in their own firms, still have to dissociate.
- Dissociation is the last step in the process, use intermediate steps first - e.g. contact senior management.
- Must know firm policies and procedures.
- If conflict, take more strict law or regulation, or Code and SPC.
- Must not *knowingly* participate in or assist in violations. Or dissociate.
- Apply law, regulation, Code and SPC of locality or where business is taking place or deemed to have taken place. If reside in strict regulatory country + do business in looser country but home regulations say business of other country's laws apply and those laws are loose, then Code and SPC.
- CFAI encourages reporting violations.
- CFAI suggests having firm-specific Code (or adopt Asset Manager Code of PC), and establish procedures which are regularly reviewed.

I(B): Independence and objectivity

- Reasonable care and judgment for independence and objectivity.
- No offer, solicit, accept gift or consideration reasonably expected to compromise independence and objectivity.
- Can't personally subscribe to IPOs ahead of clients.

- Gifts from clients should be reported to employer but less of a problem. Firm can compare employee who received gift from client versus how that employee has treated other clients.
- If employer is unwilling to provide adverse opinions about corporate client, remove that client from the research universe (a restricted list).
- Corporate issuers shouldn't reimburse for air transport, unless it's modest commercial flight to a destination which is hard to reach. Hotels - if moderate hotels, then probably OK.
- Encourage firms to require prior approval for IPOs, limits on private placements. Encourage firms to have policies reviewing personal investments.
- Encourage firms so that analysts aren't controlled by conflicting interests. Appoint compliance officer.

I(C): Misrepresentation

- Must disclose intended use of external managers, and can't misrepresent their management as own. Need appropriate disclosures.
- Should not shop for higher valuations. Also this adds advisory fees to the client.
- Performance valuations - if no comparison or vague comparison, disclose reasons why.
- Citations. Don't cite and attribute to 'leading' analysts. If citing statistics, must include caveats. If using someone else's model idea, then can take credit for results but must cite model author for idea. If purchase research which is for re-packaging, still cite source.
- Must source and cite.
- A firm can issue reports without providing attribution to prior analysts - the research is the property of the firm.
- Firms can make written lists of qualifications and designate representative.

I(D): Misconduct. Aimed at professional life.

- Examples: Lack of due diligence, bankruptcy if due to fraud or deceit even if in personal life, abusing professional conduct program, taking accidentally-left confidential information to better find more clients.
- Encourage firms to have code, with list of violations, check employee references for good character.

I(E): Competence. Act with and maintain competence.

- Includes managerial competence, even when suddenly promoted - have to get training, go to conferences, etc.

II(A) Material nonpublic information

- This SPC is about information that would impact a security price - it's not about information for where to find more clients, etc. (that's misconduct).
- The less reliable the information, less likely to be material nonpublic information.
- Selective disclosure to some analysts may not be public.
- Have to wait reasonable amount of time for dissemination.
- Mosaic theory. Use nonmaterial and or public sources to get conclusions that might be material nonpublic if communicated confidentially by company to analyst.
- If well-known analyst changes opinion, this is theoretically material nonpublic, and theoretically should be disseminated to public - but presumably analyst used mosaic theory and people can pay for the information, so not a violation.
- If find material information, must make reasonable efforts at public dissemination, usually asking company to make it public. If not, then contact compliance and must not take action on that information.
- Use firewalls, maintain records, personal trading restrictions, firm trading - only be passive in market-making, suspend arbitrage trading.

II(B): Market manipulation. Information-based manipulation and transaction-based manipulation (to give various impressions, or take dominant position in financial instrument to influence it).

- Often occurs together with V(A) Diligence and Reasonable Basis.
- If engage in **pump or liquidity priming**, then must disclose.

III(A): LPC. Loyalty to clients, reasonable care, prudent judgment, clients' interests above own. Legal duties can be difficult to understand and complex; LPC is easier to understand. If have custody over funds, heightened duty.

- Trade execution professionals to get best trade terms for client.
- Duty of loyalty to ultimate beneficiaries of fund.
- If no client (e.g. index fund), then according to mandate. Client can be investing public as a whole.
- If can't avoid conflicts of interest, then disclose.

- Use client brokerage to buy research = soft dollars = soft commissions. Directed brokerage = client specifically directs fund manager to buy research from a specific provider; but must exercise 'best execution'.
- Must vote proxies with LPC
- Regular accounts - submit to each client at least quarterly, itemized statement showing funds, debits, credits, transactions, where the assets are will be moved, separate the assets from other party assets.
- Don't have to seek best execution and price if get written permission from client. Can only direct trades through a specific account as reward for new clients if accounts receive best price and execution and the practice is disclosed to clients.

III(B): Fair dealing. Deal fairly and objectively with all clients (external; not internal to firm).

- Must disseminate research to clients fairly - but not at same time which is logistically impossible.
- Can use different models for different clients - must treat all fairly but not necessarily equally as they are different.
- Can provide different service levels for different fees but must not disadvantage other clients **and must offer to everyone**. Service levels have to be disclosed to all, and different service levels shouldn't be offered selectively.
- Though service levels have to be disclosed to all, extra services, such as invitations to IPO roadshows, don't have to be disclosed - as long as they don't disadvantage other clients.
- Once release recommendation to all clients, can selectively call largest asset size customers even if they aren't paying extra or higher fees.
- Applies to new information, recommendations, etc. as well as changes to them.
- Allocate investment blocks fairly.
- Extent of compliance procedures depend on size and nature of firm.
- Recommend fair dealing by simultaneously notifying firm and clients.
- Can have selection process of notification but has to be based on suitability and known interest.
- Limit the number of people involved before notification, shorten time between decision and dissemination, publish guidelines for pre-dissemination behavior, simultaneous dissemination, maintain client and holdings list (to improve quick fair dealing), written investment allocation procedures, review accounts regularly for fairness.

- IPOs should be distributed to suitable investors on a pro-rate basis for the **entire firm**, not in some way to individual portfolio managers.

III(C): Suitability. Make reasonable inquiries of client before recommendation, reassess and update regularly, determine suitability before recommending, judge in context of **total client portfolio** (not individual securities).

- Execution traders, sell-side analysts don't have anyone to determine suitability for. For PMs, suitability for fund mandate.
- IPS should be updated annually + material changes (e.g. number of children, health; for funds, e.g. changes in liabilities).
- Client trade requests not consistent with IPS - if small, discuss and go. If larger, change IPS or if client won't change IPS consider not serving client anymore.
- Cannot disobey client requests to purchase a specific security.

III(D): Performance presentation. Fair, accurate, complete.

- GIPS best way to meet III(D).
- If presentation brief, make available on request detailed information. Include reference or disclaimer to limited nature of information.
- If third party rater gives a PM a good rating, PM must verify that the third party said that prior to using it for advertisement. Even if we know that the rating is accurate (in a CFA question), PM must verify this (they don't know yet whether it's accurate).

III(E): Confidentiality.

- If report to authorities, usually first discuss a lot within the company, e.g. compliance department.
- Confidentiality in the SPC is about client information AND client trades, not market information.

IV(A): Loyalty. Don't deprive benefits to employer.

- Employers aren't subject to Code and SPC.
- Must notify outside activity could conflict with employer - types of services, duration, and compensation.
- Personal interests and employer are secondary to protecting integrity of capital markets and clients.
- Employee-led buyout of employer is consistent with IV(A), because it depends on permission of employer. Can do this even if information about employer came unsolicited from client.

- If something clearly unethical at employer but perhaps can't figure it out exactly, consult company whistleblowing procedure.
- Publicly disagreeing with employer's actions or views is not a violation, but indirectly using confidential information to do this or indirectly using firm resources for personal advantage is a violation.
- Loyalty to employer includes following compliance procedures and harming employer prospects of business.

IV(B): Additional compensation arrangements. Can't accept gifts or benefits reasonably expected to create a conflict of interest unless written consent from all parties.

- Donations to personal charity, even one you founded, isn't additional compensation, though it could be a conflict of interest.
- Example: extra compensation from firm's client, or on a board of a different company (which might lead to partiality for that other company using employer's resources).
- Must be immediate upon offer, confirmed by offeror, and consent should state the terms of the agreement.
- If part-time hire, then depends on circumstances and what was negotiated in the contract.

IV(C): Responsibility of supervisors. Reasonable efforts to prevent + detect violations.

- Establish record-keeping procedures to prevent violations.
- Includes reasonable supervision of quality of work of subordinates.
- If cannot discharge supervisory responsibility, reject responsibility.
- Depends on size and nature of organization.
- If violation detected, prompt assessment. Can't just rely on employee assurances.
- Give code to clients.

V(A): Diligence and reasonable basis. Reasonable and adequate basis. Not just for work - both the analyst + supervisor - **but also for external fund manager selection.**

- Secondary research = someone else at same firm. Third-party research: outside firm. Must scrutinize both - but can rely on others in firm for scrutiny unless reason for suspicion. Reasonable inquiries into source.

- If using quantitative model output, should reasonably understand the process and model.
- Have procedures for what reasonable and adequate basis is. Have procedures for evaluating external managers.
- External managers need to be vetted - can't just rely on personal relationships.
- Can't just review separate and disparate parts of model; must look at whole on a regular basis.

V(B): Communication with clients and prospective clients. Disclose services and **costs**, processes used, limitations (e.g. liquidity, capacity) of investment processes, limitations of forecasts and statistical methods, changes in methodologies / models / investing styles, identify factors important to investment and communicate that to clients, distinguish fact from opinion in analysis.

- Provide reasonable detail re costs, but not specific dollar amounts - can state percentages. State disbursements, which kind, etc. Can tailor to sophistication of client.
- Generally this standard only affects client-facing members.
- Explain use of external managers if any.
- Disclose use of leverage usually.
- Cannot assume investors know what the 'standard industry practice' is on fee structure.
- Cannot oversimplify complex model or processes - at least provide some basic logic for evaluating risks.
- Quantitative estimates are **opinions**, not fact.
- If internal process for stock-picking (e.g. significant change in who approves securities) changes, then must communicate to clients.
- If error in model or anywhere, must communicate + correction actions taken.
- Explain how to get model inputs and assumptions and model limitations.

V(C): Record retention. Though the standard itself doesn't say it per se, it's also about moving records to a new employer.

- To support opinions - whether to change a rating, or NOT to change a rating.
- Cannot recreate from memory - that's during the time with previous employer (though can use the knowledge - but that knowledge must be recreated using public sources).

- Record-keeping usually a responsibility of the firm, not individuals.

VI(A): Avoid or disclose conflicts. Full and fair disclosures. Conflicts = conflicts versus clients, prospective clients, and employer. Includes appearance of conflict of interest.

- Member beneficially owns securities if direct or indirect pecuniary interest, power to vote or direct the voting, power to dispose or direct disposition of security.
- Spouse owning securities is a beneficial interest.

VI(B): Priority of transactions. Investment transactions for clients and employers have priority over personal ones.

- Establish blackout periods before IPOs, private placements, so that clients can trade first before personal trades. Blackout periods can be as long as each firm - and even each branch of each firm - wants.
- Establish procedures. Disclose personal holdings at beginning of employment and at least annually. Direct personal brokers to provide trade confirmations to employer. Preclearance for trades.
- Disclose policies - not just in boilerplate or hard-to-understand language.

VI(C): Referral fees. Disclose to employer + clients + prospective clients, anything received OR paid for recommendations or referrals of customers.

- A reward payment from the employer to an employee (e.g. bringing clients over) is NOT a referral fee.
- Must do this even if the referral arrangement is such that the investment advisor is to refer the client back to the referrer (e.g. brokerage) for services and that brokerage undertakes 'best price and execution' trades.
- Must also disclose nature of benefit - flat fee, percentage, one-time, ongoing, based on performance, non-cash, etc.
- Update employer quarterly of nature and amount of compensation received.
- Referrals within a company (to a different department) - still have to disclose first time the referral comes in.

VII(A): Conduct as participants in CFAI programs. Can't compromise reputation, integrity, validity of CFAI programs.

- Exams. Can't discuss specific details of exams or even broad topical areas covered or not covered.
- This SPC extends to CFAI events too, e.g. who is included and who is excluded.

VII(B): Reference to CFAI, CFA designation, CFA program

- CFA charterholder designation agreement in PCP agreement signed every year.
- Candidate if enrolled + accepted for exam, and awaiting results.
- Cannot state that they have a partial designation.

GIPS

- There is a GIPS guidance on error correction.

General

- Cannot disobey client requests to purchase a specific security.
- In the US, the principal but not the interest is guaranteed on COD and money market accounts.
- Fiduciary duty broader than duty of loyalty. Fiduciary duty is to act in best interests of client; duty of loyalty is avoiding conflicts of interest and act in good faith.
- High income mutual fund - provides a high level of current income for investors. Not about whether the mutual fund is suitable for high earners.

10.4 Application of the code and standards

I(B) Independence and objectivity, is more general - not about specific investment recommendations or analysis - but rather about professionalism in general.